

Tax Evasion and Productivity

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Abstract

I propose a novel strategy that uses production functions to estimate corporate tax evasion through cost overreporting. Employing a structural production function approach using validation data with correctly reported costs, I investigate potential tax evasion by testing the presence of cost overreporting on a testing data, where the statistically significant differences on the output elasticities of the overreported input in the correct direction are interpreted as evidence of tax evasion. I apply this method to a well-known dataset where I find evidence suggesting that cost overreporting is widespread and quantitatively large. My results indicate that ignoring cost overreporting leads to consistently larger elasticities of intermediate inputs. The bias on the intermediate inputs spreads to the elasticities of labor and capital, whose bias direction varies by industry. Finally, I find significant differences in the productivity distributions. How much cheating there could be (back-of-envelope calculation, be careful)

Keywords: Tax Evasion, Cost Overreporting, Production Function Estimation, Productivity

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TODO

- JMP Structure:
 - Introduction
 - Literature Review
 - Motivating model
 - * Key assumptions:
 - Firms artificially inflate material inputs to reduce sales tax liability (VAT type) and CIT (increasing costs reduces taxable income)
 - Detection probability independent of input choice
 - Corporations do not evade taxes through input overreporting because the marginal economic cost of evasion exceeds the marginal benefit for any non-zero level of evasion
 - Testing for tax evasion through input overreporting
 - Why? Testing is robust to relaxing the the assumption of the probability of detection being independent of input choice
 - * In Colombia: By industry
 - Bootstrap
 - Randomization inference

- * Other countries (Spain, India, Chile)
- Empirical approach: Heterogeneity in tax evasion
 - * Across firms: Who evades more?
 - in COL: Small firms, exporters, importers, by J.O.
 - Other countries (?): Spain, India
 - * Across Time: Exploiting tax policy changes in Colombia
 - Triple difference identification strategy
- Deconvolution approach:
 - * NP deconvolution of densities:
 - tax evasion
 - productivity
 - * Countries: Colombia (main), Chile, Spain, and India
 - * Application:
 - Compare Productivity and PF estimates with naive estimation
 - Back of envelope estimates of tax revenue losses due to evasion
 - Misallocation measure
- Possible extensions:
 - * Estimate detection probability/tax evasion cost function from unused FOC
 - * Non-random auditing from tax authority
 - * More flexible functional forms for production function (Translog)
 - * More flexible process for productivity (ELVIS)
- Other Data: Spain, India, and Chile
 - All include Type of Firm (Juridical Organization)
 - Spain data needs to be bought

- Indian data is publicly available
 - * Need to download and consolidate
- Chile data is ready
- Summarize main points of Lit Review

Introduction

Cost overreporting arises when firms acquire false invoices to claim additional tax deductions on value-added (VAT) and corporate income taxes (CIT). According to the OECD’s document [OECD \(2017\)](#), cost overreporting — also known as “fake invoicing”, “ghost firms”, “invoice mills”, or “missing traders”— permeates internationally.

Reports from Latin America, Eastern Europe, Asia, and Africa claim cost overreporting led to annual tax revenue losses as large as 5.6% of the GDP, for example, in Poland, 2016 (Poland’s Minister of Finance, 2018). Other reports show that cost overreporting led to revenue losses of 0.2% of Chile’s GDP in 2004 (Gonzalez and Velasquez, 2013; Jorrat, 2001; CIAT, 2008); 0.2% of Colombia’s GDP (Portafolio, 2019); and 0.03% of Mexico’s GDP in 2018 (Senado de la Republica, 2019).

Furthermore, ignoring tax evasion leads to biased estimates of productivity. In the proxy variable literature, productivity is measured as the residual of the production function, where the output of a firm is a function of the inputs, capital, labor, and intermediates. A key assumption is that intermediate input demand is strictly monotonic on the productivity of the firm ([Gandhi et al. 2020](#), [Akerberg et al. 2015](#), [Levinsohn & Petrin 2003](#)). In other words, we expect highly productive firms will use fewer inputs to produce a given level of output. Intermediate inputs, however, are also the most likely input to be misreported.

When firms overreport their inputs increasing their costs to claim additional tax deductions, their reported inputs are higher than their actual utilization, resulting in lower productivity estimates.

Despite its relevance, the literature on corporate tax evasion has mostly overlooked cost overreporting. The few studies focusing on this tax evasion strategy rely on exploiting detailed administrative data ([Zumaya et al. 2021](#), [Carrillo et al. 2022](#)). Due to firms' confidentiality concerns, government tax authorities restrict access to kind of data. My approach complements these methods. It is not restricted to confidential tax records and can be applied using more commonly available data such as firm-level surveys.

To the best of my knowledge, no other study has attempted to structurally identify cost overreporting. A fundamental problem is that when it comes to corporate tax evasion, researchers have to account for an additional source of unobserved heterogeneity, productivity. Why? Because cost overreporting might be naively quantified as low productivity. Intuitively, for a given output level, high input utilization by a firm could be explained by either the amount of input the firm overreports to evade taxes or by a negative productivity shock.

To address this gap in the literature, first I formally show that ignoring tax evasion leads to downward biased productivity estimates. I then provide a new estimation strategy using production functions to jointly recover the densities of tax evasion and productivity. The intuition works as follows. In the absence of tax evasion, the first-order conditions of the firms' cost-minimization problem let us recover the common technology, the production function. Consequently, in the presence of cost overreporting, deviations from this common technology identify tax evasion up to the measurement error. Then, from a subset of non-overreporting firms, the strategy identifies the production function parameters and the

density of the output shock. Finally, using non-parametric deconvolution techniques, I jointly recover the distributions of tax evasion and productivity.

Applying the method using firm-level data from Colombia between 1981 and 1991—a commonly used dataset in the production function literature—I find evidence suggesting that firms in four of the top five industries (8 of the top 20) engage in cost overreporting. These firms overreport up to 25% of their costs. My estimates suggest that the tax evading firms in the top 20 revenue industries caused the government of Colombia approximately XXXX in tax revenue losses. [exporters/importers/proprietorships/limited liability companies] are [less/more] likely to engage in cost overreporting. I also find that ignoring cost overreporting leads to consistently larger elasticities of intermediate inputs by [what factor]. The bias on the intermediate inputs spreads to the elasticities of labor and capital, whose bias direction varies by industry [what range?]. Lastly, I find significant differences in the productivity distributions. In particular, true productivity distributions are [how]. The differences between [exporters/corporations] and [importers/limited liability companies] are [what?].

1 A parsimonious model of tax evasion through input overreporting

Price-taking firms maximize expected after-tax profits. Firms choose their inputs $\{K_{it}, L_{it}, M_{it}\}$, capital, labor, and raw materials to produce output Y_{it} given output and input prices $\{P_t, \rho_t, r_{it}, w_{it}\}$, a common production function (Equation 1), and their productivity ω_{it} . This is a characteristic of the production function literature, firms share a common technology, but differ in their productivity.

$$Y_{it} = K_{it}^{\alpha_K} L_{it}^{\alpha_L} M_{it}^{\beta} \exp(\omega_{it} + \varepsilon_{it}) \quad (1)$$

As standard in the literature, firms observe ω_{it} when they choose their inputs. This is the well-known endogeneity problem of simultaneity (Griliches & Mairesse 1995). ε_{it} is an error term. The error term is not part of the firms' information set.

In addition, firms can freely choose their materials in the current period, but capital and labor are either predetermined or dynamic. In practice what this means is that when it comes to choosing materials, firms face static optimization problem.

The characteristic of the materials being flexible is what makes them more susceptible to be overreported. Capital can be audited more easily than materials. For example, the tax authority can verify the existence of capital goods in the firm's premises. In contrast, materials are consumed in the production process and are more difficult to verify.

Likewise, labor leaves a larger paper trail than materials because of payroll taxes and social security contributions. These additional taxes and contributions makes labor more costly to evade than materials. The incentive for firms is to underreport labor to evade payroll taxes and social security contributions, but this is not the focus of this paper.

In the model, firms can choose to overreport their inputs by an amount $e_{it} \equiv M^* - M \geq 0$ to claim additional tax deductions and maximize expected after-tax profits. M^* is the reported level of materials, while M is the true level of materials.

Firms pay a tax rate τ on their revenues after deducting their materials costs. τ is, therefore, equivalent to a value-added tax (VAT) or corporate income tax (CIT), where firms can only claim deductions for their expenses on materials.

The tax authority audits firms at random and always detects misreporting if the firm engages in tax evasion and gets audited (Allingham & Sandmo 1972). The probability of detection is then, $q(e_{it}, a)$; where a represents the exogenous monitoring effort by the tax authority. The probability of detection is continuous and increasing in both of its arguments, $\frac{\partial q}{\partial e_{it}} > 0$ and $\frac{\partial q}{\partial a} > 0$. The more a firm overreports, the more likely it is to be detected. Likewise, the larger the efforts of the tax authority in monitoring suspicious behaviour of firms, the higher the probability of detection. Let's also assume that if the firm does not overreport, $e_{it} = 0$, then it cannot be detected, $q(0, a) = 0$.

The evasion cost $\kappa_i(e_{it})$ is increasing, $\kappa_i(e_{it})' > 0$, and convex in the amount overreported, $\kappa_i''(e_{it}) \geq 0$. In addition, not overreporting is costless, $\kappa_i(0) = 0$. This cost represents the direct cost of overreporting such as acquiring fake invoices and resource cost of maintaining parallel accounting books to keep track of the true and reported costs. It might also represent indirect costs or losses of incurring in tax evasion. If the firms decides to evade, the evading cost is paid whether the firms is detected evading or not.

Finally, if tax evasion is detected, the tax authority imposes a penalty. Detected firms are fined with a rate of ϕ over the value of evaded taxes in addition to the true tax liability.

Firms, then, solve Equation 2

$$\begin{aligned}
\max_{K_{it}, L_{it}, M_{it}, e_{it}} & (1 - q(e_{it}, a)) \left[P_t \mathbb{E}_\varepsilon[Y_{it}] - \rho_t M_{it} - r_t K_{it} - w_t L_{it} \right. \\
& \quad \left. - \tau (P_t \mathbb{E}_\varepsilon[Y_{it}] - \rho_t (M_{it} + e_{it})) - \kappa_i(e_{it}) \right] \\
& + q(e_{it}, a) \left[(1 - \tau) (P_t \mathbb{E}_\varepsilon[Y_{it}] - \rho_t M_{it}) - r_t K_{it} - w_t L_{it} \right. \\
& \quad \left. - \kappa_i(e_{it}) - \phi \rho_t \tau e_{it} \right] \tag{2} \\
\text{s.t. } & Y_{it} = K_{it}^{\alpha_k} L_{it}^{\alpha_L} M_{it}^\beta \exp(\omega_{it} + \varepsilon_{it})
\end{aligned}$$

The firms' problem (Equation 2) can be rewritten as follows,

$$\begin{aligned}
\max_{K_{it}, L_{it}, M_{it}, e_{it}} \quad & \mathbb{E}[\pi_{it}|\Gamma] = (1 - \tau) (P_t \mathbb{E}[Y_{it}] - \rho_t M_{it}) - r_t K_{it} - w_t L_{it} \\
& + [1 - q(e_{it}, a)] (\rho_t \tau e_{it}) - q(e_{it}, a) \phi \rho_t \tau e_{it} - \kappa_i(e_{it}) \\
\text{s.t.} \quad & Y_{it} = K_{it}^{\alpha_K} L_{it}^{\alpha_L} M_{it}^{\beta} \exp(\omega_{it} + \varepsilon_{it})
\end{aligned} \tag{3}$$

An interior solution leads to following first-order conditions,

$$\beta K_{it}^{\alpha_K} L_{it}^{\alpha_L} M_{it}^{\beta-1} \exp(\omega_{it}) \mathcal{E} = \frac{\rho_t}{P_t} \tag{4}$$

$$\tau \rho_t \left(1 - (1 + \phi) \left(q(e_{it}, a) + \frac{\partial q(e_{it}, a)}{\partial e_{it}} e_{it} \right) \right) = \kappa'_i(e_{it}) \tag{5}$$

where $\mathcal{E} = \mathbb{E}[\exp(\varepsilon_{it})]$.

Under random auditing, Equation 4 is the familiar optimality condition, where the left-hand side is the marginal product of materials and the right-hand side is the relative prices of materials.

To better understand the incentives to overreport inputs, we can rewrite Equation 5 in terms of the evasion elasticity of the probability of detection, $\xi_{q,e} = \frac{\partial q(e_{it}, a)}{\partial e_{it}} \frac{e_{it}}{q(e_{it}, a)}$, as follows,

$$\tau \rho_t \left(1 - (1 + \phi) (1 + \xi_{q,e}) q(e_{it}, a) \right) = \kappa'_i(e_{it}) \tag{6}$$

The left-hand side of Equation 6 is the marginal benefit of tax evasion, while, the right-hand side is the marginal cost of evasion. The marginal benefit of evasion decreases with the evasion penalty ϕ , the probability of detection $q(e_{it}, a)$, and the evasion elasticity of

the probability of detection $\xi_{q,e}$. The marginal cost of evasion increases with the level of evasion because $\kappa'(e_{it}) \geq 0$.

Note that there could be firms that can optimally choose not to evade, $e_{it} = 0$. Because no overreporting would be a corner solution, Equation 5 will not hold for these firms. For these firms, the marginal cost of tax evasion is greater than the marginal benefit for any non-zero level of tax evasion. This implies that for these firms, Equation 5 becomes,

$$\tau\rho_t \leq \kappa'_i(0) \quad (7)$$

1.1 Parametrization and normalization

To calibrate the model, we need to parameterize the evasion cost function and the probability of detection function. Also, we would need to normalize some variables for which we do not have data.

To start, we can parametrize the probability of detection with an exponential cdf as follows,

$$q(e_{it}, a) = 1 - \exp(-\lambda_0 a e_{it}) \quad (8)$$

where $\lambda_0 > 0$ is a parameter that governs how quickly the probability of detection increases with the amount overreported and the monitoring effort of the tax authority.

Since we do not have data on the monitoring effort of the tax authority, we can normalize it to $a = 1$. In addition, there was no penalty in Colombia for tax evasion during this period. If you were caught evading taxes, you only had to pay the evaded taxes. Thus, we can set $\phi = 0$.

For the evasion cost function, we can use a linear function. We can allow for Corporations to have a greater marginal cost of evasion than the rest of the firms. Thus, we can write the evasion cost function as follows,

$$\kappa_i(e_{it}) = e_{it}(\gamma_0 + \gamma_1 \mathbb{1}[i \in \text{Corp}]) \quad (9)$$

where $\gamma_0 > 0$ is the baseline marginal cost of evasion, and $\gamma_1 \geq 0$ is an additional marginal cost for Corporations. $\mathbb{1}[i \in \text{Corp}]$ is an indicator function that takes the value of 1 if firm i is a Corporation and 0 otherwise.

1.2 Estimating equations

For estimation purposes, we can simplify Equation 4 by multiplying by M_{it} and dividing by $Y_{it} = K_{it}^{\alpha_K} L_{it}^{\alpha_L} M_{it}^{\beta} \exp(\omega_{it} + \varepsilon_{it})$. Then, by taking logs, this leads to the following expression,

$$\ln \left(\frac{\rho_t M_{it}}{P_t Y_{it}} \right) = \ln(\beta \mathcal{E}) - \varepsilon_{it} \quad (10)$$

For the case of Colombia, what matters is the effective sales tax rate see 9. The effective sales tax rate will vary by firm within an industry and over time depending on several firm characteristics such as size, juridical organization, share of exports and imports among others. Thus, we can use this rich variation and replace the statutory tax rate τ with the effective tax rate τ_{it} .

Then, substituting the parametrizations of the probability of detection (Equation 8) and the evasion cost function (Equation 9) into Equation 5 leads to the following expression,

$$\tau_{it}\rho_t (1 - (1 - \exp(-\lambda_0 e_{it}) + \lambda_0 e_{it} \exp(-\lambda_0 e_{it}))) = \gamma_0$$

Then, by simplifying and taking logs,

$$\ln(\tau_{it}\rho_t) = \ln(\gamma_0) + \lambda_0 e_{it} - \ln(1 - \lambda_0 e_{it})$$

Moreover, we need to consider that what the econometrician observes is the reported intermediate inputs, $M_{it}^* = M_{it} + e_{it}$. However, because Corporations do not overreport, for them, $M_{it}^* = M_{it}$.

Thus, for Corporations, we have two equations,

$$\ln\left(\frac{\rho_t M_{it}^*}{P_t Y_{it}}\right) = \ln(\beta\mathcal{E}) - \varepsilon_{it} \quad (11)$$

$$\tau_{it}\rho_t \leq \gamma_0 + \gamma_1 \quad (12)$$

For the rest of the firms, we have,

$$\ln\left(\frac{\rho_t (M_{it}^* - e_{it})}{P_t Y_{it}}\right) = \ln(\beta\mathcal{E}) - \varepsilon_{it} \quad (13)$$

$$\ln(\tau_{it}\rho_t) = \ln(\gamma_0) + \lambda_0 e_{it} - \ln(1 - \lambda_0 e_{it}) \quad (14)$$

Alternatively, following the measurement error literature, we can rewrite these equations in terms of the true inputs, with a centering condition.

The centering condition would be that the expected value of the tax evasion is zero for Corporations,

$$\mathbb{E}[M_{it}^* - M_{it} | i \in \text{Corp}] = 0 \quad (15)$$

The other equations we can rewrite as,

$$\mathbb{E} \left[\ln \left(\frac{\rho_t M_{it}}{P_t Y_{it}} \right) \right] - \ln(\beta \mathcal{E}) = 0 \quad (16)$$

$$\mathbb{E} [\ln(\tau_{it} \rho_t) - \ln(\gamma_0) - \lambda_0(M_{it}^* - M_{it}) + \ln(1 - \lambda_0(M_{it}^* - M_{it})) | i \notin \text{Corp}] = 0 \quad (17)$$

$$\mathbb{E} [\mathbb{1}[\ln(\tau_{it} \rho_t) - \ln(\gamma_0 + \gamma_1) \leq 0] - 1 | i \in \text{Corp}] = 0 \quad (18)$$

2 Identification Strategy

Because the firms' optimization decisions depend on the fiscal environment, the identification strategy should be motivated by the fiscal environment Γ . In particular, the identification strategy will be as good as how well we can tell apart a subset of firms that have the highest incentive to not evade. For example, in the case of Spain, the firms above the revenue LTU threshold. In the case of Colombia, the corporations.

Assumption 2.1: Non-Evaders

Based on the fiscal environment Γ , the researcher can identify a subset of firms $\theta_i \in \Theta^{NE} \subset \Theta$ that does not evade taxes by overreporting inputs.

For those firms, then $\mathbb{E}[e_{it}|\theta_i \in \Theta^{NE}] = 0$

In addition, I impose the following timing assumption.

Assumption 2.2: Independence

Firms choose overreporting e_{it} *before* the output shock ε_{it}

Assumption 2.2 implies that input overreporting is independent of the current period output shock, $e_{it} \perp \varepsilon_{it}$. In the literature is not rare to assume that the output shock is not part of the information set of the firms, $\varepsilon_{it} \notin \mathcal{I}_t$ (Gandhi et al. 2020). Timing and information set assumptions are not uncommon for identification strategies in production functions and demand estimation (Akerberg et al. 2021, Akerberg 2019).

2.1 Identifying the production function parameters

Assume the production function is Cobb-Douglas, $G(M_{it}, K_{it}, L_{it}) \exp(\omega_{it} + \varepsilon_{it}) = M_{it}^\beta K_{it}^{\alpha_K} L_{it}^{\alpha_L} \exp(\omega_{it} + \varepsilon_{it})$.

The econometrician observes output, capital, labor, and prices of intermediate inputs and output $\{Y_{it}, K_{it}, L_{it}, \rho_t, P_t\}$, but she only observes overreported intermediate inputs, $M_{it}^* = M_{it} \exp(e_{it})$ ¹.

Then, for the case of Colombia, **eq-foc:ind** applies since the type of firms is the juridical organization and the non-evaders are Corporations. By multiplying both sides by the intermediate inputs and dividing by the output, we get

¹Note we can always rewrite $M^* + e = M^* \exp\{a\}$, then $\exp\{a\} = \frac{e}{M^*} + 1$.

$$\begin{aligned}
\ln \left(\frac{\rho_t M_{it}}{P_t Y_{it}} \right) &= \ln \beta + \ln \mathcal{E} - \varepsilon_{it} \\
&\equiv \ln D^{\mathcal{E}} - \varepsilon_{it}
\end{aligned}
\tag{19}$$

$$\ln(M_{it}^*) = \ln(M_{it}) + e_{it}$$

$$\ln \left(\frac{\rho_t M_{it}^*}{P_t Y_{it}} \right) = \ln \beta + \ln \mathcal{E} - \varepsilon_{it} + e_{it}$$

where, $\mathcal{E} \equiv \mathbb{E}[\exp(\varepsilon_{it})|\mathcal{J}_{it}] = \mathbb{E}[\exp(\varepsilon_{it})]$.

We can use Equation 19 and assumption 2.1 to recover the production function parameter β

$$\mathbb{E} \left[\ln \left(\frac{\rho_t M_{it}^*}{P_t Y_{it}} \right) \middle| \Theta^{NE} \right] = \ln D^{\mathcal{E}} \tag{20}$$

The constant $\mathcal{E}$ is also identified (Gandhi et al. 2020),

$$\mathcal{E} = \mathbb{E} \left[\exp \left(\ln D^{\mathcal{E}} - \ln \left(\frac{\rho_t M_{it}^*}{P_t Y_{it}} \right) \right) \middle| \theta^{NE} \right] = \mathbb{E} [\exp(\varepsilon_{it})|\theta^{NE}] = \mathbb{E}[\exp(\varepsilon_{it})] \tag{21}$$

and, thus, the output elasticity of input, β , is also identified,

$$\beta = \exp(\ln D^{\mathcal{E}} - \ln \mathcal{E}). \tag{22}$$

2.2 Identifying Tax Evasion

Having recovered both the flexible input elasticity, β , and the constant $\mathcal{E}$, for all firms, I can form the following variable using observed data.

$$\begin{aligned}\mathcal{V}_{it} &\equiv \ln \left(\frac{\rho_t M_{it}^*}{P_t Y_{it}} \right) - \ln \beta - \ln \mathcal{E} \\ &= -\varepsilon_{it} + e_{it}\end{aligned}$$

Tax evasion therefore can be recovered up to an independent random variable. We can do better however. By assumption 2.2, the tax evasion, ε_{it} , is independent of e_{it} . Note that, from Equation 19, we also recovered $f_\varepsilon(\varepsilon)$ the distribution of ε . By using deconvolution methods, I can recover the tax evasion apart from the output shock.

In particular, from probability theory,

Definition 2.1.

Definition: Convolution

The density of the sum of two **independent** random variables is equal to the **convolution** of the densities of both addends; hence

$$h = f * m = \int f(\mathcal{Z} - \varepsilon) m(\varepsilon) d\varepsilon$$

where f is the density of $\mathcal{Z}$ (Meister, 2009)

2.3 Identifying Productivity

Here, I show how to recover the rest of the parameters of the production function, including productivity and its Markov process. We can do it in several ways, depending on our object of interest.

In the literature, it is not uncommon to assume that productivity follows a Markov process. That is,

$$\omega_{it} = h(\omega_{it-1}) + \eta_{it} \tag{23}$$

Let $\ln X_{it} = x_{it}$. With Equation 23, we can form the following observed variable,

$$\begin{aligned}
\mathcal{W}_{it} &\equiv y_{it} - \beta(m_{it}^* - \mathcal{V}_{it}) \\
&= y_{it} - \beta(m_{it} + e_{it} - e_{it} + \varepsilon_{it}) \\
&= y_{it} - \beta m_{it} - \beta \varepsilon_{it} \\
&= \alpha_K k_{it} + \alpha_L l_{it} + \omega_{it} + (1 - \beta)\varepsilon_{it}
\end{aligned} \tag{24}$$

Here, intuitively, we are trading unobservables, the unobserved tax evasion for the output shock, just to make our lives easier. We could have use use directly m^* since we know the distribution of tax evasion e , which is equivalent to the distribution of $f_{m^*|m}(m^*|m)$. Also note that is it not necessary to learn tax evasion to recover productivity. This is useful if a practitioner is only interested in productivity and not on tax evasion.

By replacing ω_{it} from Equation 24 in Equation 23, we get

$$\mathcal{W}_{it} = (1 - \beta)\varepsilon_{it} + \alpha_K k_{it} + \alpha_L l_{it} + h(\mathcal{W}_{it-1} - (1 - \beta)\varepsilon_{it-1} - \alpha_K k_{it-1} - \alpha_L l_{it-1}) + \eta_{it} \tag{25}$$

If $h(\xi) = \delta_0 + \delta_1 \xi$, to learn α and h , we need to find instruments Z such that,

$$\mathbb{E}[\eta_{it} + (1 - \beta)\varepsilon_{it} + \delta_1(1 - \beta)\varepsilon_{it-1} | Z] = 0. \tag{26}$$

Table 1 displays candidates for Z

Table 1: Orthogonality by Residuals

	Orthogonality
η_{it}	$k_{it}, l_{it}, \varepsilon_{it}, k_{it-1}, l_{it-1}, \varepsilon_{it-1}, \omega_{it-1}, \mathcal{W}_{it-1}, m_{it-1}, m_{it-1}^*$
ε_{it}	$k_{it}, l_{it}, \omega_{it}, m_{it}, m_{it}^*, \eta_{it}, k_{it-1}, l_{it-1}, \varepsilon_{it-1}, \omega_{it-1}, \mathcal{W}_{it-1}, m_{it-1}, m_{it-1}^*$
ε_{it-1}	$k_{it}, l_{it}, \omega_{it}, \varepsilon_{it}, m_{it}, m_{it}^*, \eta_{it}, k_{it-1}, l_{it-1}, \omega_{it-1}, \mathcal{W}_{it}, m_{it-1}, m_{it-1}^*$

The possible instruments are $Z_{it} = \{k_{it}, l_{it}, k_{it-1}, l_{it-1}, m_{it-1}^*, \mathcal{W}_{it-2}\}$.

During estimation, I use labor and capital as instruments for themselves $Z_{it} = \{k_{it}, l_{it}\}$, and the constant and the lag of observed overreported intermediates as instrument for the AR(1) productivity Markov process $h(\cdot)$.

Once the parameters of the production function are identified, we can use the following observed variable to recover ω_{it}

$$\begin{aligned} \widetilde{\mathcal{W}}_{it} &\equiv \mathcal{W}_{it} - \alpha_K k_{it} - \alpha_L l_{it} \\ &= \omega_{it} + (1 - \beta)\varepsilon_{it} \end{aligned} \tag{27}$$

2.4 Identification with two Flexible Inputs

Suppose now that M_{it} and L_{it} are flexible inputs, but K_{it} is not. Furthermore, M_{it} is deductible and L_{it} is not. Under a Cobb-Douglas production function, `?@eq-foc:deduct` and `?@eq-foc:non-deduct` become,

$$\begin{aligned} \ln \left(\frac{\rho_t M_{it}}{P_t Y_{it}} \right) &= \ln \beta + \ln \mathcal{E} - \varepsilon_{it} \\ &= \ln D_M^\mathcal{E} - \varepsilon_{it} \end{aligned} \tag{28}$$

$$\begin{aligned}\ln\left(\frac{w_t L_{it}}{(1-\tau)P_t Y_{it}}\right) &= \ln \alpha_L + \ln \mathcal{E} - \varepsilon_{it} \\ &= \ln D_L^\mathcal{E} - \varepsilon_{it}\end{aligned}\tag{29}$$

Note that firms do not have incentives to overreport non-deductible flexible inputs. So the observed non-deductible flexible inputs are the true inputs consumed in production. For the deductible flexible inputs, we still observe the overreported inputs, $M_{it}^* = M_{it} \exp(e_{it})$.

So, Equation 28 becomes,

$$\ln\left(\frac{\rho_t M_{it}^*}{P_t Y_{it}}\right) = \ln D_M^\mathcal{E} - \varepsilon_{it} + e_{it}\tag{30}$$

The constant $\ln D_L^\mathcal{E}$ is identified by a linear regression from Equation 29. Likewise, using assumption 2.1, we can identify the constant $\ln D_M^\mathcal{E}$ by running a linear regression of Equation 30 for the non-evaders.

To identify the elasticities from the constant $\mathcal{E}$, let $\hat{\varepsilon}_{it}^M$ and $\hat{\varepsilon}_{it}^L$ be the residuals of the regressions of Equation 30 and Equation 29, respectively. Then, we can form the following observed variable,

$$\varepsilon_{it} = \begin{cases} \lambda_{it} \hat{\varepsilon}_{it}^M + (1 - \lambda_{it}) \hat{\varepsilon}_{it}^L & \text{if } i \text{ is a Non-evader} \\ \hat{\varepsilon}_{it}^L & \text{otherwise} \end{cases}\tag{31}$$

where λ_{it} is a weight that can be chosen by the practitioner. For example, it can be set to the share of the cost of both flexible inputs in the total cost of production, $\lambda_{it} = \frac{\rho_t M_{it}^*}{\rho_t M_{it}^* + w_t L_{it}}$.

In this way, we would have a time-firm specific weight that captures the relative importance of each flexible input in the production process. alternatively, we can have time or firm fixed weights, $\lambda_t = N^{-1} \sum_i \lambda_{it}$ or $\lambda_i = T^{-1} \sum_t \lambda_{it}$, respectively.

Another option is to use the optimal weight to minimize the variance of ε_{it} ,

$$\lambda = \arg \min_{\lambda} \mathbb{E} [\varepsilon^2 | \Theta^{NE}]$$

$$\begin{aligned} \mathbb{E} [\varepsilon^2 | \Theta^{NE}] &= \lambda^2 \mathbb{E} [(\hat{\varepsilon}_{it}^M)^2 | \Theta^{NE}] + (1 - \lambda)^2 \mathbb{E} [(\hat{\varepsilon}_{it}^L)^2 | \Theta^{NE}] \\ &\quad + 2\lambda(1 - \lambda) \mathbb{E} [\hat{\varepsilon}_{it}^M \hat{\varepsilon}_{it}^L | \Theta^{NE}] \end{aligned}$$

Deriving the first order condition, we get

$$2\lambda \mathbb{E} [(\hat{\varepsilon}_{it}^M)^2 | \Theta^{NE}] + 2(1 - 2\lambda) \mathbb{E} [\hat{\varepsilon}_{it}^M \hat{\varepsilon}_{it}^L | \Theta^{NE}] - 2(1 - \lambda) \mathbb{E} [(\hat{\varepsilon}_{it}^L)^2 | \Theta^{NE}] = 0$$

Finally, solving for λ gives us the optimal weight,

$$\lambda^* = \frac{\mathbb{E} [(\hat{\varepsilon}_{it}^L)^2 | \Theta^{NE}] - \mathbb{E} [\hat{\varepsilon}_{it}^M \hat{\varepsilon}_{it}^L | \Theta^{NE}]}{\mathbb{E} [(\hat{\varepsilon}_{it}^M)^2 | \Theta^{NE}] + \mathbb{E} [(\hat{\varepsilon}_{it}^L)^2 | \Theta^{NE}] - 2\mathbb{E} [\hat{\varepsilon}_{it}^M \hat{\varepsilon}_{it}^L | \Theta^{NE}]}. \quad (32)$$

This might work only if the covariance of the residuals is less than the variance of any residual. In theory, the covariance cannot be larger than the variance of either residual. In that case, it means the both residuals are good estimates of the output shock ε_{it} and we can use either of them to recover the output shock ε_{it} .

Then, we can recover $\mathcal{E} = \mathbb{E}[\exp(\varepsilon_{it})]$ and the elasticities $\beta = \exp(\ln D_M^\mathcal{E} - \ln \mathcal{E})$ and $\alpha_L = \exp(\ln D_L^\mathcal{E} - \ln \mathcal{E})$.

2.5 Identifying Tax Evasion with Two Flexible Inputs

$$\begin{aligned}
\mathcal{V}_{it} &\equiv \ln \left(\frac{\rho_t M_{it}^*}{P_t Y_{it}} \right) - \ln D_M^\mathcal{E} + \varepsilon_{it} \\
&= \ln \left(\frac{\rho_t M_{it}}{P_t Y_{it}} \right) - \ln D_M^\mathcal{E} + \varepsilon_{it} + e_{it} \\
&= -\varepsilon_{it} + \varepsilon_{it} + e_{it} \\
&= e_{it}
\end{aligned}$$

In the case of the two flexible inputs, we can form a firm-time specific variable that captures the tax evasion by overreporting the deductible flexible input, $\mathcal{V}_{it} = e_{it}$.

2.6 Identifying Productivity with Two Flexible Inputs

To recover α_K , we can now form the following observed variable,

$$\begin{aligned}
\mathcal{Y}_{it} &\equiv y_{it} - \beta(m_{it}^* - e_{it}) - \alpha_L l_{it} - \varepsilon_{it} \\
&= y_{it} - \beta m_{it} - \alpha_L l_{it} - \varepsilon_{it} \\
&= \alpha_K k_{it} + \omega_{it}
\end{aligned}$$

Then, using ortogonality of η_{it} ,

$$\mathbb{E}[\mathcal{Y}_{it}|Z_{it}] = \alpha_K k_{it} + \mathbb{E}[h(\mathcal{Y}_{it-1} - \alpha_K k_{it-1})|Z_{it}]$$

where $Z_{it} = \{k_{it}, k_{it-1}, \mathcal{Y}_{it-1}\}$.

2.7 Translog Production Function

To identify tax evasion and relaxing the assumption of a CD production function, we would need a flexible input for which firms have no incentives to overreport. Firms might face

flexible inputs that they cannot deduct from their VAT or CIT, for example. If this is the case, then firms would have no incentives to overreport non-deductible flexible inputs.

Assume now that L_{it} is a non-deductible flexible input and, without loss of generality, there are only two inputs (L_{it}, M_{it}) . Let's assume the production function is now *translog*. We have,

$$\ln G(l, m) = \beta_0 m_{it} + \beta_1 m_{it} l_{it} + \beta_2 m_{it} m_{it} + \beta_3 l_{it} + \beta_4 l_{it} l_{it}$$

Then, equation [?@eq-foc:non-deduct](#) becomes

$$\begin{aligned} s_{it}^L &= \ln(\beta_3 + 2\beta_4 l_{it} + \beta_1(m_{it}^* - e_{it})) + \ln \mathcal{E} - \varepsilon_{it} \\ &\equiv \ln D^\mathcal{E}(l_{it}, m_{it}^* - e_{it}) - \varepsilon_{it} \end{aligned} \tag{33}$$

where $s_{it}^L \equiv \ln\left(\frac{w_t L_{it}}{(1-\tau)P_t Y_{it}}\right)$.

Note that by assumption [2.1](#), $D^\mathcal{E}$ and the density of ε are still identified.

2.7.1 Testing for Tax Evasion

We can design a test for tax evasion by forming the analog of Equation [23](#),

$$\begin{aligned} \mathcal{V}_{it}^L &= \ln D^\mathcal{E}(l_{it}, m_{it}^*) - s_{it}^L \\ &= \ln D^\mathcal{E}(l_{it}, m_{it} + e_{it}) - s_{it}^L \\ &= \ln D^\mathcal{E}(l_{it}, m_{it} + e_{it}) - \ln D^\mathcal{E}(l_{it}, m_{it}) + \varepsilon_{it} \\ &= \ln\left(\frac{D^\mathcal{E}(l_{it}, m_{it} + e_{it})}{D^\mathcal{E}(l_{it}, m_{it})}\right) + \varepsilon_{it} \end{aligned}$$

Note that $\ln \left(\frac{D^\mathcal{E}(l, m+e)}{D^\mathcal{E}(l, m)} \right) \geq 0$ is always positive because $e_{it} \geq 0$ and $D^\mathcal{E}$ is strictly increasing in m . Thus, $\frac{D^\mathcal{E}(l, m+e)}{D^\mathcal{E}(l, m)} \geq 1$.

2.7.2 Identifying Tax Evasion

Once we recovered the parameters of Equation 33, we can invert it to solve for the true inputs m^* as a function of observed data and the output. Then, the difference between the observed and true inputs is by definition the tax evasion by overreporting.

$$\begin{aligned} m_{it}^* &= m_{it} + e_{it} \\ &= D^{-1}(\exp(s_{it}^L - \ln \mathcal{E} + \varepsilon_{it}), l_{it}) + e_{it} \end{aligned}$$

Again, having recovered the distribution of the output shock $f_\varepsilon(\varepsilon)$, we can use deconvolution techniques to learn the distribution of e . In particular, to learn the p.d.f of tax evasion $f_e(e)$, we can deconvolute it by

$$\int f_e(m_{it}^* - D^{-1}(\exp(s_{it}^L - \ln \mathcal{E} - \varepsilon_{it}), l_{it})) f_\varepsilon(\varepsilon) d\varepsilon$$

2.8 Non-Parametric Identification of Tax Evasion

The previous result also suggests a non-parametric identification strategy, as long as $D^\mathcal{E}$ is monotonic in m . This identification strategy is analogous to [Hu et al. \(2022\)](#), where the authors also require monotonicity and independence to recover a nonparametric function of m^* with nonclassical measurement error.

In our case, intuitively, if we know the density of ε and the function $D^\mathcal{E}$, the variation left is due to e , which can be recovered as long as we can vary $D^\mathcal{E}$ by moving e .

To see why the non-deductible flexible input is needed to identify tax evasion consider the following. Suppose that only the input M is flexible and deductible.

$$\ln \left(\frac{\rho M}{PY} \right) = \ln D^{\mathcal{E}}(K, L, M) - \varepsilon$$

$D^{\mathcal{E}}(K, L, M)$ is still identified by assumption 2.1, however, when we form the analogous of Equation 23, we now have

$$\begin{aligned} \ln \left(\frac{\rho M^*}{PY} \right) - \ln D^{\mathcal{E}}(K, L, M^*) &= \\ &= \ln \left(\frac{\rho(M+e)}{PY} \right) - \ln D^{\mathcal{E}}(K, L, M+e) \\ &= \ln \left(\frac{\rho(M+e)}{PY} \right) - \ln D^{\mathcal{E}}(K, L, M+e) \\ &\quad + \left[\ln \left(\frac{\rho M}{PY} \right) - \ln D^{\mathcal{E}}(K, L, M) \right] \\ &\quad - \left[\ln \left(\frac{\rho M}{PY} \right) - \ln D^{\mathcal{E}}(K, L, M) \right] \\ &= \ln \left(\frac{\rho M}{PY} \right) - \ln D^{\mathcal{E}}(K, L, M) \\ &\quad + \left[\ln \left(\frac{\rho(M+e)}{PY} \right) - \ln \left(\frac{\rho M}{PY} \right) \right] \\ &\quad - [\ln D^{\mathcal{E}}(K, L, M+e) - \ln D^{\mathcal{E}}(K, L, M)] \\ &= -\varepsilon \\ &\quad + [\ln D^{\mathcal{E}}(K, L, M+e) - \varepsilon - \ln D^{\mathcal{E}}(K, L, M) + \varepsilon] \\ &\quad - [\ln D^{\mathcal{E}}(K, L, M+e) - \ln D^{\mathcal{E}}(K, L, M)] \\ &= -\varepsilon(K, L, M) \end{aligned}$$

Now, we are not able to separate the variation of ε from e .

3 Colombia 1981-1991

3.1 Colombian Corporate Tax System

The relevant corporate taxes for input overreporting in Colombia during this period are the Corporate Income Tax (CIT) and the Sales Tax. The Sales Tax gradually transformed into a kind of Value-Added Tax (VAT). Also relevant for the CIT are the different juridical organizations that exist in Colombia.

This period was characterized by high levels of overall tax evasion ([Torres & Sourdis 1994](#)). The fiscal rules had a system of penalties and interest that encouraged false and delinquent returns ([Mclure 1989](#)). The fiscal authority was characterized by having an inefficient auditing system, being overburdened, and legal loopholes ([Perry & de Triana 1990](#)).

3.1.1 Juridical Organizations

In Colombia, there are five types of juridical organizations: Corporations, Partnerships, Limited Liability Companies, and Individual Proprietorships.

Corporations (*sociedad anónima*) are the typical associations of capital. They are the counterpart of the US corporation. The capital of a corporation is provided by the shareholders (no less than 5) and is divided into tradable shares of equal value. Shareholders' liability is limited to the capital contributed. Corporations are subject to the Superintendent of Corporations and are closely supervised, being required to have an auditor.

Joint Stock Companies (*sociedad en comandita por acciones*) comprises two or more managing partners who are jointly and severally liable, and five or more limited partners whose liability is limited to their respective contributions. Joint Stock Companies are taxed as Corporations. Its shares are tradable, like the shares of Corporations.

Partnerships are associations of two or more persons. Partners are jointly and severally liable for the partnership's operations. Partnerships include general partnerships (*sociedad colectiva*), de Facto partnerships (*sociedades de hecho*), and ordinary limited partnerships (*sociedad en comandita simple*).

A limited liability company (*sociedad de responsabilidad limitada*) is an association of two or more persons—not exceeding 20 (Fiscal Survey) or 25 (1992 *EAM* survey documents)—, whose shares cannot be traded. The personal liability of the partners is limited to the capital contributed. The Limited Liability Company is quite important in Colombia (Fiscal Survey).

Finally, proprietorships are individuals (natural persons) who allocate part of their assets to conduct commercial activities.

There are other juridical organizations in the data that will be excluded from the final analysis. These organizations are non-profit, like cooperatives and community enterprises, or state industrial enterprises, the proceeds of which come from taxes, fees, or special contributions.

3.1.2 Corporate Income tax

The juridical organizations were subject to different Corporate Income Tax rates. Corporations were taxed at a fixed rate of 40%, while Partnerships and Limited Liability companies at 20%. Individual proprietors were subject to the graduated Individual Tax Schedule consisting of 56 rates, ranging from 0.50 to 51 percent.

Corporations were taxed on their distributed dividends, while partnerships and limited liability companies were taxed on their profits, whether or not distributed. Owners of juridical organizations were double taxed, at the firm and the individual level, whereas the

income of proprietorships was taxed only once, at the individual level.

Since 1974, individuals and juridical organizations, except for limited liability companies, were subject to the minimum presumptive income. Rent (income and profits) was presumed to be no less than 8 percent of net wealth (assets less depreciation, real estate, livestock, securities).

Certain industries like airlines, publishing, and reforestation sectors, and various activities in selected regions (primarily “frontier” and other less developed ones) had their income tax exempted, limited, or reduced.

Table 2: Juridical Organizations in Colombia (1980s), A Summary

Organization	Corporate			
	In-			
	come			
	Tax	Liability	Capital	Owners
Corporation	40%	Limited to	Tradable	$N \geq 5$
	(on	capital	capital	
	dis-	participation	shares	
	tributed			
	divi-			
	dends)			
Limited Co.	20%	Limited to	Non-	$2 \leq N \leq$
	(on	capital	tradable	20(25)
	prof-	participation	capital	
	its)		shares	

Corporate				
In-				
come				
Organization	Tax	Liability	Capital	Owners
Partnership	20%	Full	Not a	$N \geq 2$
	(on		capital	
	prof-		association	
	its)			
Proprietorship	Individual	Full	Owner	$N = 1$
	In-			
	come			
	Tax			

3.1.3 Sales taxes

Sales taxes were originally targeted at the manufacturing sector on finished goods and imports. Since 1974, manufacturers were allowed to credit taxes paid on any purchase made by the firm, except the acquisition of capital goods (Perry & de Triana 1990). The credits worked through a system of refunds. Consequently, the tax became a kind of value-added tax (VAT).

The basic rate was 15 percent. There was also a preferential rate of 6 percent for certain industries like clothing, footwear, and major inputs used for building popular housing, and a rate of 35 percent for luxury goods. Exports, foodstuff, drugs, and textbooks were excluded from the beginning. Also excluded were inputs, transportation equipment, agricultural machinery, and equipment.

3.1.4 Discussion

From Colombia's tax system, we can conclude that corporations are the least likely to evade taxes by misreporting because of at least three reasons. One, the Superintendent of Corporations closely monitored corporations by requiring them to have an auditor. In the model, this implies that the probability of detection is higher for them. Second, free tradable shares impose an incentive to be profitable because the market value of the shares might be negatively affected otherwise. In other words, if a corporation fakely reports lower profits, the value of its shares would likely decrease scaring away shareholders and potential investors. Joint stock companies have freely tradable shares too. Three, corporations pay CIT on distributed dividends, not on profits as Partnerships and LLCs did. Corporations have an additional margin regarding the income tax they pay because they decide when to pay dividends. This might generate other types of incentives that might be influenced by the corporation's policy regarding their dividends and the demands of their shareholders. On the other hand, Proprietorships and LLCs are subject to the incentive to evade CIT by artificially reducing their profits.

Moreover, Proprietorships, Partnerships, and Limited Companies had incentives to overreport inputs to evade VAT and CIT. The incentives to evade varied across industries because the sales tax rate differed between industries. The incentives to evade also varied within industry sectors because juridical organizations within an industry were subject to different CIT rates. There were additional sources of variation depending on the firm's location due to local exemptions and sales composition (inputs to other firms, to consumers, to the foreign market).

Lastly, Individual proprietorships were likely to bunch at the individual income thresholds because they were subject to individual income tax which was increasing by brackets.

4 Data

The Colombian data is a well-known firm-level panel data set that has been used in the estimation of production functions and productivity before. The Colombian dataset comes from the Annual Survey of Manufacturing (EAM) and contains information about manufacturing firms with more than 10 employees from 1981 to 1991.

Besides the information on output, intermediates, capital, and labor, the dataset includes the type of juridical organization and the sales taxes. Table 3 offers some summary statistics.

4.1 Top Industries by Revenue in Colombia

4.2 Corporations by Industry

Looking into industry sectors.

5 Implementation

We are interested in the distribution of tax evasion e but it cannot be observed. What is observed is the contaminated version $\mathcal{V}$ Equation 23. Evasion e and the output shock ε are independent [2.2] with probability density distributions f_e and f_ε . Then, from Definition 2.1

$$f_{\mathcal{V}}(\mathcal{V}) = \int f_e(\mathcal{V} + \varepsilon) f_\varepsilon(\varepsilon) d\varepsilon$$

where $f_{\mathcal{V}}$ denotes the density of $\mathcal{V}$.

Table 3: Summary Statistics, Manufacturing Firms in Colombia (1981-1991)

	Missing (%)	Mean	SD	Q1	Median	Q3
Share of Revenues						
Sales Taxes	0	0.066	0.051	0.006	0.069	0.100
Skilled Labor (Wages)	0	0.043	1.620	0.009	0.021	0.046
Unskilled Labor (Wages)	0	0.086	0.106	0.019	0.051	0.119
Capital	0	0.495	7.555	0.126	0.261	0.499
Intermediates						
Materials (M)	0	0.457	0.191	0.333	0.457	0.582
Electricity (E)	0	0.018	0.028	0.005	0.010	0.020
Fuels (F)	0	0.011	0.025	0.001	0.003	0.010
Repair & Maintenance (R&M)	0	0.007	0.013	0.001	0.003	0.009
Services (S)	0	0.103	0.167	0.052	0.084	0.130
Deductible Inter. (M+E+F+R&M)	0	0.493	0.180	0.371	0.488	0.610
Non-Deductible Inter. (S)	0	0.103	0.167	0.052	0.084	0.130
J. Org.	N	%				
Proprietorship	5343	12.877				
Ltd. Co.	25457	61.353				
Corporation	8761	21.114				
Partnership	1425	3.434				

Table 4: Top 10 industries in Colombia by revenues

description	sic_3	n_sic	n_Corp	Market Share
Food Manufacturing	311	912	150	23.5
Beverage Industries	313	130	74	9.5
Other Chemical Products	352	301	101	8.5
Industrial Chemicals	351	117	69	7.5
Textiles	321	476	71	7.5
Transport Equipment	384	238	42	6.5
Food Manufacturing	312	193	41	5.5
Paper and Paper Products	341	153	41	5.5
Fabricated Metal Products, Except Machinery and Equipment	381	633	85	3.5
Electrical Machinery Apparatus, Appliances and Supplies	383	213	57	3.5

Table 5: Corporations by Industry. Top 20 Industries in Colombia by number of firms.

Industry
Food Manufacturing
Wearing Apparel, Except Footwear
Fabricated Metal Products, Except Machinery and Equipment
Textiles
Machinery Except Electrical
Printing, Publishing and Allied Industries
Other Chemical Products
Other Non-Metallic Mineral Products
Plastic Products Not Elsewhere Classified
Transport Equipment
Electrical Machinery Apparatus, Appliances and Supplies
Food Manufacturing
Furniture and Fixtures, Except Primarily of Metal
Footwear, Except Vulcanized or Moulded Rubber or Plastic Footwear
Other Manufacturing Industries
Wood and Wood and Cork Products, Except Furniture
Paper and Paper Products
Beverage Industries
Industrial Chemicals
Leather and Products of Leather, Leather Substitutes and Fur, Except Footwear and Wearing Apparel

5.1 Parametric Deconvolution

Assume a functional form for $f_\varepsilon(\cdot; \gamma)$ that depends on known parameters γ . Assume a known functional form for the density $f_e(\cdot; \lambda)$ that depends on unknown parameters λ . We can estimate parameters λ by

$$\hat{\lambda} = \arg \max_{\lambda} \sum_{i=1}^n \log \left(\int f_e(\mathcal{V}_i + \varepsilon; \lambda) f_\varepsilon(\varepsilon; \gamma) d\varepsilon \right) \quad (34)$$

Properties of MLE with unobserved scalar heterogeneity have been derived elsewhere before (Chen 2007, Yi 2021).

5.2 Non-Parametric Deconvolution by Splines

Following (n.d.), consider the following logspline model:

$$f_{e|\Theta}(e) = \exp(s(e; \theta) - C(\theta)), \quad L < e < R$$

where, $[L, R]$ define the support of f_e ,

$$s(e; \theta) = \sum_{j=1}^{K-1} \theta_j B_j(e),$$

$\{B_j(E), j = 1, 2, \dots, K\}$ is a sequence of cubic B-spline basis functions, and

$$C(\theta) = \log \left(\int_L^R \exp(s(e; \theta)) de \right) < \infty.$$

Then $f_{e|\Theta}(e)$ is a positive density function on (L, R) . The logspline model assumes that f_e is a twice continuously differentiable function. The last basis function is not used for

identifiability purposes, given that by the property of B-splines, $\sum_{j=1}^K B_j(e) = 1$ for all $e \in (L, R)$.

The log likelihood of the observed variable $\mathcal{V}$ is

$$\begin{aligned} l_{\mathcal{V}}() &= \sum_{i=1}^n \log \left(\int f_{\varepsilon}(e - \mathcal{V}_i) f_{e|\theta}(e) de \right) \\ &= \sum_{i=1}^n \log \left(\int f_{\varepsilon}(e - \mathcal{V}_i) \exp(s(e; \theta) - C(\theta)) de \right) \\ &= \sum_{i=1}^n \log \left(\int f_{\varepsilon}(e - \mathcal{V}_i) \exp(s(e; \theta)) de \right) - nC(\theta) \end{aligned}$$

The usual maximum likelihood estimate $\hat{\theta}$ is the maximizer of $l_{\mathcal{V}}(\theta)$.

To avoid overfitting, I add a penalty term to the log-likelihood function. The penalty term is a linear combination of the squared second derivatives of the B-spline basis functions. The second derivative of the B-splines is approximated by finite differences ([Eilers & Marx 1996](#)).

The penalty term is given by

$$\text{Penalty}(\theta) = \lambda \cdot \|\mathbf{D}\theta\|^2$$

Intuitively, we want to penalize excessive curvature in the log-density $\log f_e(e) \approx s(e; \theta) = \sum_j \theta_j B_j(e)$, where B_j are B-spline basis functions.

The second derivative of a smooth function is a measure of curvature. Since we don't have an analytical form for $s''(e)$, we can approximate it using finite differences on the coefficients θ .

Given a vector of spline coefficients $\theta = (\theta_1, \theta_2, \dots, \theta_k)$, define the second-order differences

as:

$$\Delta^2\theta_j = \theta_j - 2\theta_{j+1} + \theta_{j+2}$$

These values approximate the second derivative of the spline function at positions along the domain.

We can write all second differences as a matrix operation:

Let $D \in \mathbb{R}^{(k-2) \times k}$, then:

$$D\theta = \begin{bmatrix} 1 & -2 & 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & -2 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & -2 & 1 & \dots & 0 \\ \vdots & & & \ddots & & & \vdots \\ 0 & \dots & 0 & 1 & -2 & 1 & \end{bmatrix} \begin{bmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \\ \vdots \\ \theta_k \end{bmatrix}$$

So $D\theta$ is a vector of second differences:

$$[\theta_1 - 2\theta_2 + \theta_3, \theta_2 - 2\theta_3 + \theta_4, \dots]$$

The roughness penalty becomes:

$$\text{Penalty} = \lambda \cdot \|D\theta\|^2 = \lambda \cdot \sum_{j=1}^{k-2} (\Delta^2\theta_j)^2$$

where λ is a smoothing parameter that can be chosen based on BIC/AIC or cross-validation.

The final log-likelihood function to be maximized is:

$$l_{\mathcal{V}}(\theta) = \sum_{i=1}^n \log \left(\int f_{\varepsilon}(e - \mathcal{V}_i) \exp(s(e; \theta)) de \right) - nC(\theta) - \lambda \cdot \|D\theta\|^2$$

5.2.1 Simulations

Before deconvoluting with the actual data, I run some simulations to check the performance of my implmenetation in R.

I simulated a thousand observations of $\varepsilon \sim \mathcal{N}(0, 0.35)$ and $\log(e) \sim \mathcal{N}(-1.8, 0.95)$, and then I generated $\mathcal{V} = e - \varepsilon$. Figure 1 shows the histogram of the simulated data.

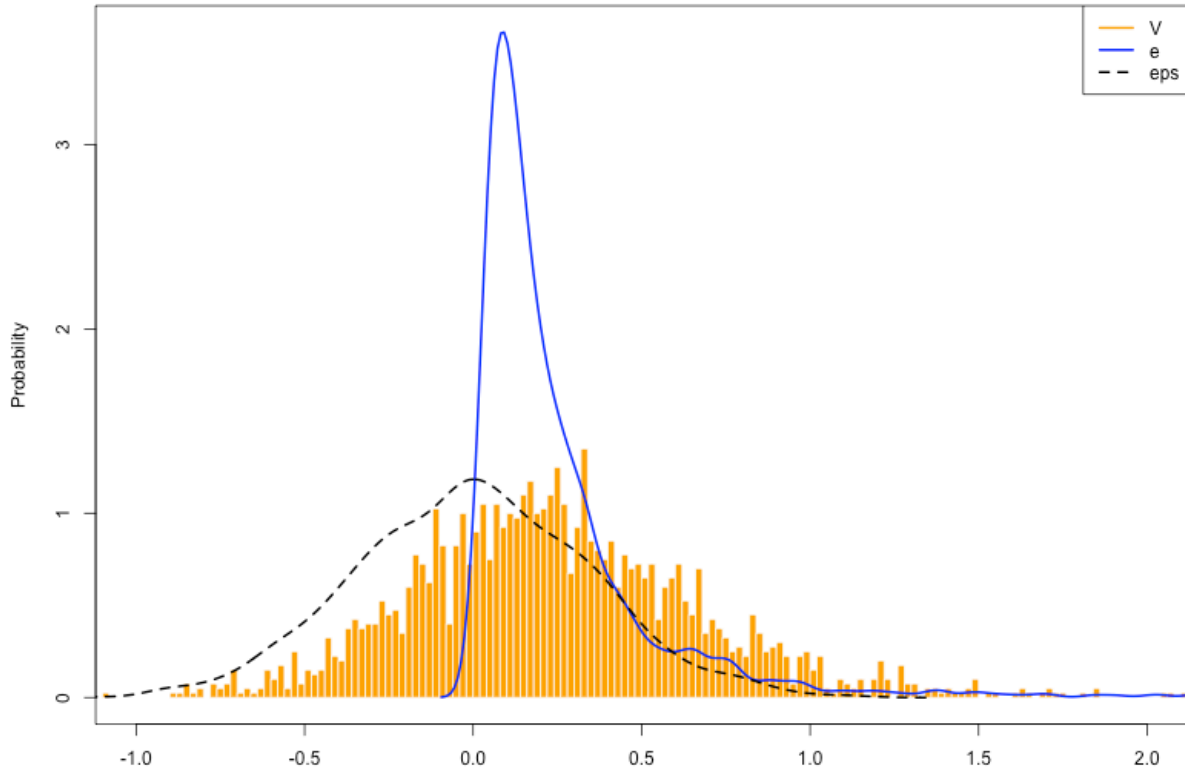


Figure 1: Simulated data

I then used the logspline model to estimate the parameters of the distribution of e . I used a B-spline basis with 10 knots. The knots were placed at equally spaced percentiles of the non-negative V observations. For the integral, I used Gauss Legendre quadrature. To ensure good coverage of the high probability region of the density, I used a recursive function to form a grid placing a higher number of nodes in the high probability region. I set the penalty parameter λ to 0.1. The estimated density is shown in Figure 2.

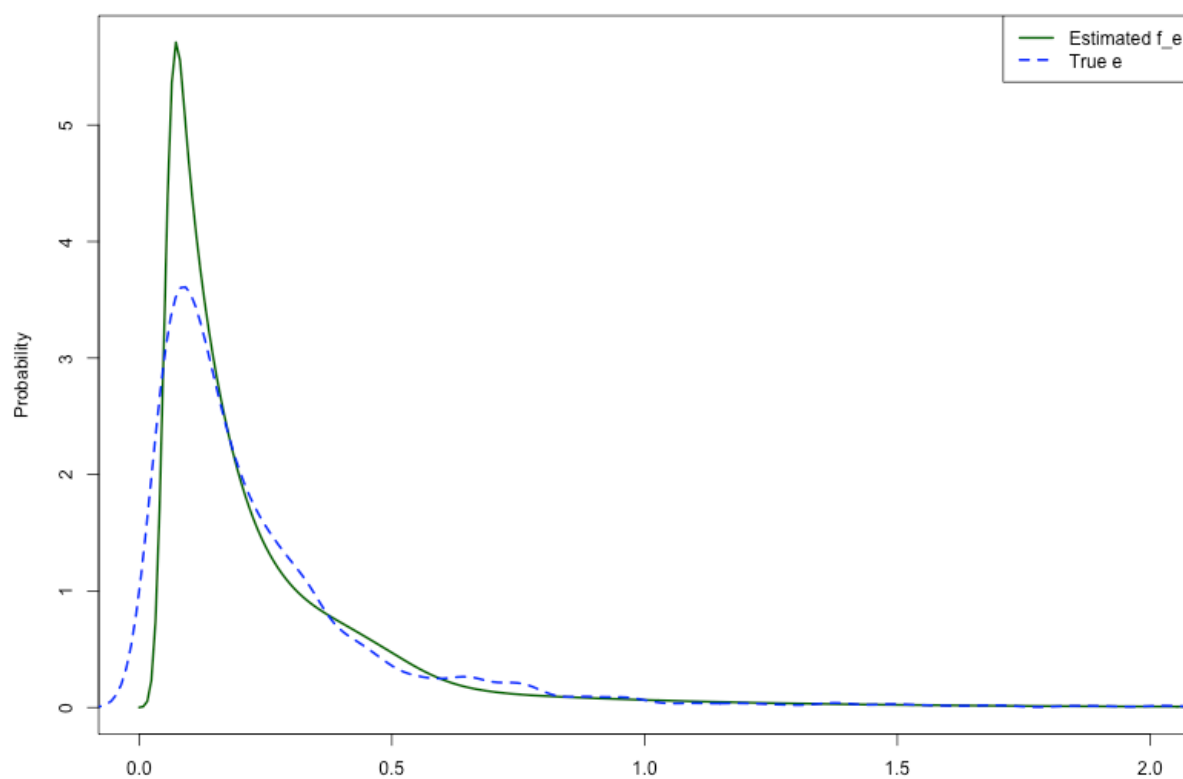


Figure 2: LogSpline Deconvolution

6 Results

6.1 Testing for the Presence of Tax Evasion Through Overreporting

Equation 23 suggest a way to test the presence of tax evasion through cost overreporting. Let $\mathbb{E}[\mathcal{V}_{it}] \equiv \mu_{\mathcal{V}}$ and define the null hypothesis as the absence of cost overreporting and the alternative hypothesis as the presence of cost overreporting. Consequently, to verify for the presence of tax evasion by overreporting, we can use a one-sided test where under the alternative hypothesis $H_1 : \mu_{\mathcal{V}} > 0$, and under the null, $H_0 : \mu_{\mathcal{V}} \leq 0$.

I use a bootstrap hypothesis test to verify the presence of tax evasion. Table 6 displays the estimated elasticities of intermediates for corporations and non-corporations for different industries. The table also displays the estimated value of the test statistic for the null hypothesis of no tax evasion.

I test the three least likely and the three most likely industries to evade taxes. The three least likely industries are exempt of sales taxes or lost exemption during the sample period. Exempt industries include 311 and 312 (Food Products), while industry 384 (Transport Equipment) lost exemption in 1984. Exempt industries are less likely to evade taxes because the sales tax rate is zero, so there is no incentive to overreport costs.

The three most likely industries have low purchase sales tax rates. Lower purchase sales tax rate imply higher incentives to overreport costs to claim additional deductions to the sales taxes owed to the government. These industries include 313 (Beverages), 331 (Wood Products except Furniture), and 369 (Non-Metallic Mineral Products).

?@tbl-evasion-test-CD shows that the null hypothesis of no tax evasion is rejected at

Table 6: Intermediate elasticities for Corporations (Non-Evaders) and non-Corporations (Evaders) for different industries. Estimates were obtained using GNR (2020) method assuming a CD functional form. Intermediates are defined as raw materials. ε is defined as **measurement error**. The Tax Evasion Test displays the coefficient of the statistical test, where the null hypothesis states that there is no tax evasion through cost overreporting. Among the industry characteristics, Sales is the industry's market share of the total sales in Colombia between 1981 and 1991. Sales Tax Rate is the average sales tax reported rate. Exports is the average share of exports of the industry total sales, and Exporters is the share of firms exporting 10 percent or more of their total sales. Bias-corrected bootstrap confidence intervals at 95% significance level are displayed in brackets. *, **, *** denote significance at the 10%, 5%, and 1% levels (one-sided).

SIC	Elasticities		Test				
	Corps.	Others	Tax Ev.	N	N (Corps.)	Sales (Mkt %)	Sales (τ_0)
2 Least Likely Industries							
311	0.6	0.61	0.02*	888	147	23.6	0.2
	[0.57, 0.62]	[0.59, 0.62]	[-0.01, 0.08]				
312	0.58	0.57	-0.01	188	41	5.3	0.2
	[0.49, 0.64]	[0.51, 0.63]	[-0.15, 0.14]				
3 Most Likely Industries							
313	0.29	0.35	0.18***	130	74	9.7	8.4
	[0.26, 0.3]	[0.32, 0.39]	[0.05, 0.32]				
369	0.24	0.3	0.2***	274	58	3.4	5.5
	[0.19, 0.27]	[0.28, 0.33]	[0.06, 0.43]				
331	0.32	0.41	0.24***	172	13	0.6	5.6
	[0.24, 0.37]	[0.38, 0.44]	[0.04, 0.47]				

Table 7: Intermediate elasticities for Corporations (Non-Evaders) and non-Corporations (Evaders) for different industries. Estimates were obtained using GNR (2020) method assuming a CD functional form. Intermediates are defined as raw materials. ε is defined as **measurement error**. The Tax Evasion Test displays the coefficient of the statistical test, where the null hypothesis states that there is no tax evasion through cost overreporting. Among the industry characteristics, Sales is the industry's market share of the total sales in Colombia between 1981 and 1991. Sales Tax Rate is the average sales tax reported rate. Exports is the average share of exports of the industry total sales, and Exporters is the share of firms exporting 10 percent or more of their total sales. Bias-corrected bootstrap confidence intervals at 95% significance level are displayed in brackets. *, **, *** denote significance at the 10%, 5%, and 1% levels (one-sided).

SIC	Elasticities		Test					
	Corps.	Others	Tax Ev.	N	N (Corps.)	Sales (Mkt %)	Sales (τ_0)	P
311	0.6	0.61	0.02*	888	147	23.6	0.2	
	[0.57, 0.62]	[0.59, 0.62]	[-0.01, 0.08]					
313	0.29	0.35	0.18***	130	74	9.7	8.4	
	[0.26, 0.3]	[0.32, 0.39]	[0.05, 0.32]					
352	0.41	0.44	0.06**	300	100	8.5	6.8	
	[0.38, 0.43]	[0.42, 0.47]	[-0.01, 0.16]					
351	0.34	0.37	0.09	116	68	7.5	7.0	
	[0.28, 0.37]	[0.29, 0.43]	[-0.13, 0.33]					
321	0.39	0.44	0.13***	470	71	7.4	8.0	
	[0.36, 0.42]	[0.42, 0.45]	[0.09, 0.24]					

the 1% significance level for twelve of the top twenty manufacturing industries.

In particular, there is no evidence of tax evasion for most industries in which products or raw materials are exempt of sales taxes, such as 312 (Other Food Products), 382 (Non-Electrical Machinery), 384 (Transport Equipment), 323 (Leather Products), and 369 (Non-Metallic Mineral Products); there is also no evidence of materials overreporting for 356 (Plastic Products) industry, whose main raw materials are likely to be specialized and supplied by few local and international suppliers.

Among the industries with exempted products, 311 (Food Products) and 383 (Electrical Machinery) there is evidence of tax evasion but as we'll see later the average overreporting is low compared to other industries.

As expected, the evidence is stronger particularly for the other industries that do not fall in the previous categories. Namely, there is evidence of tax evasion for industries 313 (Beverages); for 321 (Textiles), 322 (Wearing Apparel), and 324 (Footwear); 331 (Wood Products except Furniture) and 332 (Wood Furniture); for industry 341 (Paper) and 342 (Publishing); 351 (Industrial Chemicals) and 352 (Other Chemicals); 381 (Metal Products Except Machinery), and 390 (Other Manufacturing Industries).

6.2 Deconvoluting Tax Evasion Using Moments

One simple way to start with the deconvolution is using moments. In particular, for every n -th moment $\mathbb{E}[\varepsilon_{it}^n | \Theta^{NE}] = \mathbb{E}[\varepsilon_{it}^n | t] = \mathbb{E}[\varepsilon_{it}^n]$

Therefore, any moment of the tax evasion e_{it} distribution $\forall t \in T$ can be estimated in theory.

Table 8: Average tax evasion by Industry. Estimates show the average tax evasion from the output shock in Equation 23. LCI and UCI are the bias-corrected bootstrap confidence intervals at the 10% significance level with 250 bootstrap replicates. Intermediates are defined as raw materials.

SIC	Inter	Mean	LCI	UCI
331	Materials	0.2177	0.0973	0.3764
322	Materials	0.1955	0.1007	0.3007
369	Materials	0.1446	0.0580	0.2702
313	Materials	0.0659	0.0312	0.1091
321	Materials	0.1094	0.0793	0.1793

Namely, from Equation 23, we can estimate the average tax evasion by

$$\begin{aligned}
\mathbb{E}[e_{it}|t] &= \mathbb{E}[\mathcal{V}_{it}|t] + \mathbb{E}[\varepsilon_{it}] \\
&= \mathbb{E}[\mathcal{V}_{it}|t] + \mu_{\varepsilon}
\end{aligned}$$

Note that we learned the distribution $f_{\varepsilon}(\varepsilon)$ of ε from the first stage, so μ_{ε} and σ_{ε} are known.

Table 8 displays the estimated average of the log fraction that firms increase their costs of raw materials to evade taxes by claiming a greater deductible amount of their owed sales taxes.

The top five tax evading industries, 322 (Wearing Apparel), 342 (Publishing), 313 (Beverages), 351 (Industrial Chemicals), and 331 (Wood Products), display an average tax evasion e greater than 12%, which is non-trivial.

Although deconvolution by moments is the simplest method, it displays the undesirable

Table 9: Semiparametric Deconvolution of Tax Evasion using Penalized B-Splines. I used a kernel density estimator to fit the error density on the residuals of the first stage conditional on firms being Non-Evaders.

sic_3	mean	sd	skewness
331	0.191	0.080	0.782
322	0.185	0.037	0.117
369	0.077	0.027	22.485
313	0.077	0.096	7.320
321	0.127	0.044	0.566

characteristic that estimate of variances frequently result with a negative sign. In the next section, I address this problem by using parametric MLE.

6.3 Non-Parametric Deconvolution using Penalized B-Splines

6.4 Production Function

?@tbl-CD-ar1-comparision displays the results of estimating the production function parameters following the identification strategy described above. I use an AR(1) model for the Markov process of productivity. I employed $Z = \{k_{it}, l_{it}\}$ as instruments for themselves. One additional instrument to control for the endogeneity between ε_{it-1} and $\mathcal{W}_{it-1}$ is needed in the first stage of ‘TE+GNR’. The instrument used is specified in the table **?@tbl-CD-ar1-comparision**.

To compare, I also include the uncorrected estimates using GNR and a simple OLS. In the case of GNR estimation method, the Markov process of productivity is a third degree

polynomial, and capital and labor, the instruments.

I focus on the top 5 industries in terms of revenue: 1) 311 - Food Products, 2) 313 Beverages, 3) 352 Other Chemical Products. 4) 351 Industrial Chemicals, and 5) 321 Textiles.

From Table 11, we can conclude that m_{t-1}^* is a better than $\mathcal{W}_{t-2}$, k_{t-1} and l_{t-1} . The following discussion will be focusing on the results relating using m_{t-1}^* as the instrument.

First, because tax evasion artificially inflates the intermediates's cost share of revenues, lower output elasticities of intermediates than traditional methods are expected. In table **?@tbl-CD-ar1-comparision** shows that correcting for tax evasion results in smaller intermediates' output elasticity with respect with OLS. With respect to GNR, all estimates are statistically lower except one. The output elasticity of intermediates of the Textile industry (321) is significantly greater than the GNR result.

Three of the five intermediate elasticities resulted as not statistically different from zero. These zeros are not exclusive to the method correcting by tax evasion, as we can see that GNR for Cobb-Douglas approach zero for some industries. These zeros could be the result of the functional form. A Cobb-Douglas specification leaves out the cross-elasticities which could lead to omitted variables bias. In other words, we effectively ignore complementarity or substitutability between the inputs. This CD omitted variable bias could be accentuated because we defined raw materials as intermediates leaving out others, such as energy, fuels, and services.

The elasticities of capital and labor vary across industries, with some being lower, some higher, and others quite similar. This is expected. The direction of the bias in the elasticities of capital and labor depends on whether these inputs are complements or substitutes of intermediate inputs. This relationship will vary by industry. Put differently, for different

Table 10: Cobb-Douglas production function parameters estimates, correcting intermediates —raw materials— for tax evasion vs. naive estimation and OLS. h is an AR(1) process for ‘Tax Evasion + GNR’ but a third degree polynomial for ‘GNR’. $Z = \{k_{it}, l_{it}\}$ for GNR. One additional instrument to GNR’s Z control for the endogeneity between ε_{it-1} and $\mathcal{W}_{it-1}$ in the first stage of ‘TE+GNR’. The instrument used is specified in the table. ε_{it-1} is defined as measurement error.

Inds.	Input	Tax Evasion + GNR				GNR	OLS
		k_{it-1}	l_{it-1}	m^*_{it-1}	$\mathcal{W}_{it-2}$		
331	m	0.3209	0.3209	0.3209	0.3209	0.3989	0.5593
	k	0.0373	0.0000	0.1112	0.0693	0.0962	0.0636
	l	0.5596	0.3253	0.5941	0.5587	0.4671	0.4311
322	m	0.3469	0.3469	0.3469	0.3469	0.4218	0.5669
	k	0.0865	0.0520	0.1612	0.1781	0.1759	0.0477
	l	0.2968	0.2837	0.3871	0.3900	0.3302	0.4119
369	m	0.2441	0.2441	0.2441	0.2441	0.2821	0.4974
	k	0.0497	0.3166	1.0000	0.3290	0.2457	0.1253
	l	0.5218	0.3419	0.7503	0.5528	0.5611	0.5206
313	m	0.2894	0.2894	0.2894	0.2894	0.3091	0.7497
	k	0.0155	1.0000	0.4489	0.0000	0.3270	0.1171
	l	0.0276	0.1531	0.1197	0.2100	0.2834	0.2702
321	m	0.3869	0.3869	0.3869	0.3869	0.4316	0.6616
	k	0.0662	0.0000	0.3218	0.3052	0.2414	0.0966
	l	0.1257	0.2640	0.2838	0.3213	0.3163	0.2796

Table 11: First stage ‘Weak Instruments’ F-test of the 2SLS-IV estimation. The null hypothesis is that the coefficients are equal to zero, in other words, uncorrelated with the endogenous variable $\mathcal{W}_{it-1}$.

Ins.	Inds.	df1	df2	statistic	p-value	stars
k_{t-1}	331	1	785	121.4029	0	***
	322	1	3581	1419.5262	0	***
	369	1	1235	1764.8363	0	***
	313	1	985	2758.2637	0	***
	321	1	2339	3714.9184	0	***
l_{t-1}	331	1	785	778.9743	0	***
	322	1	3581	3840.2193	0	***
	369	1	1235	308.5926	0	***
	313	1	985	435.1505	0	***
	321	1	2339	4176.4138	0	***
m^*_{t-1}	331	1	785	416.9415	0	***
	322	1	3581	4963.1390	0	***
	369	1	1235	1048.3330	0	***
	313	1	985	458.2628	0	***
	321	1	2339	420.5456	0	***
$\mathcal{W}_{t-2}$	331	1	636	1961.0846	0	***
	322	1	2949	8673.2398	0	***
	369	1	1004	3778.8152	0	***
	313	1	865	24073.3213	0	***
	321	1	1959	7279.2133	0	***

industries, the cross-elasticity between intermediates and capital, and intermediates and labor will be different in direction and magnitude. Since these cross-elasticities are not included in the CD production function, the elasticities of capital and labor should reflect different degrees of these variations once intermediates are corrected for overreporting.

6.5 Productivity

6.5.1 Non-Parametric Productivity Density

Nothing impedes us from estimating a non-parametric distribution of ω . I deconvolute ω using penalized B-Splines from Equation 27. First, I estimate the density of ε using a kernel density estimator.

To compare these productivity estimates to uncorrected ones, I simulate productivity using the B-Splines. To simulate 1000 independent draws from ω , I numerically invert the B-Splines. First, I draw 1000 $u^S \sim U(0,1)$. Then I find the $\omega^S = \arg \min \left(\int_a^{\omega^S} s(t, \theta) dt - u^S \right)^2$. Then, I computed productivity, $\varphi_{it}^S = \exp\{\omega_{it}^S\}$. ε is measurement error, so not part of productivity.

6.6 Over-time tax evasion patterns

Incentives to evade taxes may change over time due to changes in fiscal policy. In particular, there were two major tax reforms in Colombia in 1983 and in 1986.

6.6.1 Fiscal Reforms of 1983 and 1986

6.6.1.1 1983 Reform

6.6.1.1.1 Sales Tax Changes

Table 12: Non-Parametric Deconvolution of ω using Penalized B-Splines. The error density function is estimated using a kernel density estimator fitted on the residuals of the first stage conditional on firms being Non-Evaders.

sic_3	instrument	mean	sd	skewness
313	$\mathcal{W}_{t-2}$	8.410	0.926	-0.496
	m_{t-1}	3.799	0.454	-0.088
321	$\mathcal{W}_{t-2}$	2.885	0.378	-0.069
	m_{t-1}	2.885	0.391	-0.127
322	$\mathcal{W}_{t-2}$	4.102	0.297	-0.481
	m_{t-1}	4.253	0.303	-0.455
331	$\mathcal{W}_{t-2}$	4.616	0.269	-0.179
	m_{t-1}	4.140	0.256	-0.724
369	$\mathcal{W}_{t-2}$	3.017	0.305	-0.489
	m_{t-1}	-3.984	1.929	0.542

Table 13: Summary statistics of the simulated productivity using the non-parametric estimates of ω and ε . Both functions were numerically inverted to obtain simulated draws of ω . 1000 draws were used. Productivity is computed as $\varphi_{it}^S = \exp \omega_{it}^S$. ε is measurement error, so not part of productivity.

sic_3	Method	Mean	SD	Q1	Median	Q3
313	GNR	60.8	30.7	43.3	55.3	69.6
	TE: $\mathcal{W}_{t-2}$	6253.5	5744.5	2697.1	4867.0	7690.4
	TE: m^*_{t-1}	48.2	25.1	35.6	44.9	54.9
321	GNR	21.0	11.2	13.8	18.7	25.5
	TE: $\mathcal{W}_{t-2}$	18.9	8.7	14.8	17.4	20.8
	TE: m^*_{t-1}	18.9	8.8	14.8	17.5	21.0
322	GNR	34.9	16.4	24.0	33.4	43.2
	TE: $\mathcal{W}_{t-2}$	60.9	17.3	51.6	60.0	68.8
	TE: m^*_{t-1}	71.0	20.8	59.7	69.8	80.3
331	GNR	51.3	26.1	33.2	46.4	62.7
	TE: $\mathcal{W}_{t-2}$	103.1	31.4	85.9	99.6	115.2
	TE: m^*_{t-1}	63.8	17.8	55.4	63.0	70.9
369	GNR	34.5	26.9	17.0	29.2	45.5
	TE: $\mathcal{W}_{t-2}$	20.3	4.6	18.1	20.0	22.0
	TE: m^*_{t-1}	0.2	1.2	0.0	0.0	0.0

In 1983, the government tried to homogenize the sales tax rates across industries which led to some industries facing higher sales tax rates while others faced lower rates or were exempted. The 1983 reform relatively unified the sales tax rates by combining previously taxed goods at 6 and 15 percent into 10 percent.

The number of the products and services that were levied expanded. The sales tax was extended to the retail level, with a *simplified system* being made available to small retailers to ease compliance costs and the administrative burden.

6.6.1.1.2 Income Tax Changes

However, these were not the only changes in fiscal policy. The 1983 reform also adjusted the income tax rates for limited liability companies and individuals. For individuals, the income tax rate increased by 8% in most scales, while the maximum was reduced from 56 to 49%. For limited liability companies, the CIT was reduced from 20 to 18%.

Lastly, 1983 saw the extension of the presumptive income tax to limited liability companies, which before were exempted. This meant that LLCs could not declare an income lower than 2 percent of their capital stock.

6.6.1.2 1986 Reform

In 1986, there were further changes to the income tax rates for individuals, limited liability companies, and corporations. For individuals, the highest income tax rate was reduced from 49% to 30%. The CIT for LLCs and Corporations was homogenized at 30%. Before the reform, Corporations faced a 40% CIT rate, while LLCs faced an 18% CIT rate.

6.6.2 Theoretical implications

According to our model, both the sales tax rate and the corporate income tax rate affect the incentives to evade taxes by overreporting inputs.

An increase in the sales tax rate of the corporate income tax rate would increase the incentives to evade taxes by overreporting inputs. Conversely, a decrease in either the sales tax rate or the corporate income tax rate would decrease the incentives to evade taxes. Therefore, it is ambiguous what to expect from these fiscal reforms.

1983 Fiscal Policy:

- Changes *across* industries: Changes to Sales tax rate, some increased, some decreased depending on the industry, still others remained exempted;
- Changes *within* industries: Changes to CIT of Proprietorship (individuals income tax) and LLCs;
 - Some individuals got an increase, others a decrease. Most individuals 8 percent increase, top was reduced from 56 to 49.
 - LLCs received a decrease from 20 to 18 percent of their income. However, they were also subject to the presumptive income tax starting in 1983.

6.7 Empirical approach

According to our model, Corporations are the truth reporters. In other words, Corporations do not overreport inputs to evade taxes—they might have other more-sophisticated strategies to avoid taxes, though. So for Corporations i in industry j at year t , the FOC implies,

$$s_{ijt}^C = \gamma_j + \varepsilon_{ijt}$$

CIT and ST generate tax evasion incentives for non-Corporations NonCorp = {Proprietorship, LLC, Other} in taxed industries. For a non-corporation firm i in taxed industry j and year t , thus we have,

$$s_{ijt}^{NC} = \gamma_j + e_{ijt}^{ST} + e_{ijt}^{CIT} + \varepsilon_{ijt} \quad j \in \text{Taxed}$$

where e_{ijt}^{ST} and e_{ijt}^{CIT} are the overreporting due to sales tax and corporate income tax, respectively. Since the fiscal reform of 1983 affected both the ST and CIT, these terms will vary accordingly.

Finally, because firms in exempt of ST industries are subject only to CIT, we have,

$$s_{ijt}^{NC} = \gamma_j + e_{ijt}^{CIT} + \varepsilon_{ijt} \quad j \in \text{Exempt}$$

Then, let D_{NC} be a dummy indicating whether the firm is not a Corporation. Furthermore, let D_{NE} be a dummy indicating whether the industry is taxed (not-exempted of ST), and D_E be a dummy indicating whether the industry is exempted, such that $D_E + D_{NE} = 1$. Thus, for firms in taxed industries,

$$\begin{aligned} (1 - D_{NC})s_{ijt} + D_{NC}s_{ijt} &= (1 - D_{NC})(\gamma_j + \varepsilon_{ijt}) \\ &\quad + D_{NC}(\gamma_j + e_{ijt}^{ST} + e_{ijt}^{CIT} + \varepsilon_{ijt}) \\ s_{ijt} &= \gamma_j + D_{NC}(e_{ijt}^{ST} + e_{ijt}^{CIT}) + \varepsilon_{ijt} \\ s_{ijt} &= \gamma_j + D_{NC}(e_{ijt}^{ST} + e_{ijt}^{CIT}) + \varepsilon_{ijt} \quad j \in \text{Non-Exempt} \end{aligned}$$

Likewise, for firms in exempt industries,

$$(1 - D_{NC})s_{ijt} + D_{NC}s_{ij't} = \gamma_j + D_{NC}e_{ijt}^{CIT} + \varepsilon_{ijt}$$

$$s_{ijt} = \gamma_j + D_{NC}e_{ijt}^{CIT} \quad j \in \text{Exempt}$$

Then, combining both equations, we have,

$$D_E s_{ijt} + D_{NE} s_{ijt} = D_E (\gamma_j + D_{NC} e_{ijt}^{CIT}) + D_{NE} (\gamma_j + D_{NC} (e_{ijt}^{ST} + e_{ijt}^{CIT})) + \varepsilon_{ijt}$$

$$(D_E + D_{NE}) s_{ijt} = (D_E + D_{NE}) \gamma_j + D_E D_{NC} e_{ijt}^{CIT} \tag{35}$$

$$+ D_{NE} D_{NC} (e_{ijt}^{ST} + e_{ijt}^{CIT}) + (D_E + D_{NE}) \varepsilon_{ijt}$$

$$s_{ijt} = \gamma_j + D_E D_{NC} e_{ijt}^{CIT} + D_{NE} D_{NC} (e_{ijt}^{ST} + e_{ijt}^{CIT}) + \varepsilon_{ijt}$$

So, by taking the expectation,

$$E[s_{ijt} | D_E, D_{NE}, D_{NC}] = \gamma_j + \mu_E^{CIT} + \mu_{NE}^{ST+CIT},$$

we can learn the average overreporting by firms in exempt and taxed industries.

We can also take the expectation over time to learn how overreporting changes from year to year due to fiscal policy changes in sales tax rates and income tax rates.

$$E[s_{ijt} | D_E, D_{NE}, D_{NC}, t] = \gamma_j + \mu_{E,t}^{CIT} + \mu_{NE,t}^{CIT+ST}$$

In this approach, $\mu_{E,t}^{CIT}$ captures the average overreporting in exempt industries at year t . $\mu_{NE,t}^{CIT+ST}$ captures the average overreporting due to ST and CIT for firms in taxed industries at year t . Note that in this approach we cannot separate the effect of ST and CIT for firms in taxed industries.

In the case of the fiscal policy changes, we are interested in the difference in average tax evasion with respect to the year before the fiscal reform of 1983. Thus, we have two periods, t' after the reform and t before the reform.

$$E[s_{ijt}|D_E, D_{NE}, D_{NC}, t'] - E[s_{ijt}|D_E, D_{NE}, D_{NC}, t] = \Delta_E^{CIT} + \Delta_{NE}^{CIT+ST}$$

Where $\Delta_E^{CIT} = \mu_{E,t'}^{CIT} - \mu_{E,t}^{CIT}$, $\Delta_{NE}^{CIT+ST} = \mu_{NE,t'}^{CIT+ST} - \mu_{NE,t}^{CIT+ST}$.

Since most taxed industries receive an increase in the ST rate of 4 percentage points, we expect a significant increase in tax evasion for firms in taxed industries after the fiscal reform of 1983.

With respect to CIT, the expected change is ambiguous for both exempt and taxed industries. In both industries, Proprietorships faced an average increase of eight percent in their income tax rates, while LLCs faced a decrease in their CIT from 20 to 18 percent. Consequently, Proprietorships faced an increase in their incentives to evade taxes, while LLCs a decrease. Thus, the net effect on tax evasion due to changes in CIT depends on the relative weight of Proprietorships and LLCs in each industry and the magnitude of the changes in their income tax rates.

Alternatively, we could be interested in separating the effect of changes in CIT from changes in ST on tax evasion. In this case, we can rearrange the equations for firms in exempt and taxed industries to obtain the following,

$$\begin{aligned} (1 - D_{NE})s_{ijt} + D_{NE}s_{ijt} &= (1 - D_{NE}) (\gamma_j + D_{NC}e_{ijt}^{CIT} + \varepsilon_{ijt}) \\ &\quad + D_{NE} (\gamma_j + D_{NC}(e_{ijt}^{ST} + e_{ijt}^{CIT}) + \varepsilon_{ijt}) \quad (36) \\ s_{ijt} &= \gamma_j + D_{NC}e_{ijt}^{CIT} + D_{NC}D_{NE}e_{ijt}^{ST} + \varepsilon_{ijt} \end{aligned}$$

Since CIT rates depend on the juridical organization and not on the industry, we assume that CIT generates the same incentives to evade taxes across industries, whether they are exempt of ST or not.

(Because we aggregate different juridical organizations in exempt and taxed industries into non-Corporations, this assumption might break if the composition of juridical organizations is non-trivially different between exempt and taxed industries.)

Then, taking expectations and conditioning on time,

$$E[s_{ijt}|D_{NE}, D_{NC}] = \gamma_j + \mu_t^{ST} + \mu_t^{CIT}.$$

Finally, taking differences before and after the fiscal reform of 1983,

$$E[s_{ijt}^k|t'] - E[s_{ijt}^k|t] = \Delta^{CIT} + \Delta^{ST}$$

In regression form:

1. Equation 35 becomes,

$$\begin{aligned} s_{ijt} = & D_{NC} D_E \mathbf{D}_t \Gamma'_{0,t} \\ & + D_{NC} D_{NE} \mathbf{D}_t \Gamma'_{1,t} + \gamma_j + \varepsilon_{ijt} \end{aligned} \tag{37}$$

2. Equation 36 becomes,

$$\begin{aligned} s_{ijt} = & D_{NC} \mathbf{D}_t \Gamma'_{0,t} \\ & + D_{NC} D_{NE} \mathbf{D}_t \Gamma'_{1,t} + \gamma_j + \varepsilon_{ijt} \end{aligned} \tag{38}$$

Where D_{NC} is a dummy indicating whether the firm is not a Corporation, D_{NE} a dummy indicating whether the industry is taxed (not exempt), and $\mathbf{D}_t$ a $1 \times T$ vector of year dummies. Finally, γ_j are industry fixed effects, and ε_{ijt} is the error term. Consistent with the empirical approach, the industry fixed effects capture the true materials elasticity of outputs, while the coefficients of the interaction terms capture the overreporting due to changes in CIT and ST.

$\Gamma_{0,t}$ and $\Gamma_{1,t}$ are $1 \times T$ vectors of coefficients. In the first specification (Equation 37), $\Gamma_{0,t}$ captures the change in tax evasion due to changes in CIT for firms in exempt industries, while $\Gamma_{1,t}$ captures the change in tax evasion due to changes in sales tax rates and CIT for firms in taxed industries.

In the second specification (Equation 38), $\Gamma_{0,t}$ captures the change in tax evasion due to changes in CIT for all non-Corporations, while $\Gamma_{1,t}$ captures the change in tax evasion due to changes in sales tax rates for non-Corporations in taxed industries.

To estimate the difference in tax evasion after the 1983 fiscal reform, I use the following regression specification,

1. First approach, with the effect of changes in CIT and ST combined,

$$\begin{aligned}
s_{ijt} = & \beta_0 D_{NC} + \beta_1 D_{NC} D_E + \beta_2 D_{NC} D_{NE} \\
& + D_{NC} D_E \tilde{\mathbf{D}}_t \tilde{\Gamma}'_{0,t} \\
& + D_{NC} D_{NE} \tilde{\mathbf{D}}_t \tilde{\Gamma}'_{1,t} + \gamma_j + \varepsilon_{ijt}
\end{aligned} \tag{39}$$

2. Second approach, separating the effect of changes in CIT from changes in sales tax rates,

$$\begin{aligned}
s_{ijt} = & \beta_0 D_{NC} + \beta_1 D_{NC} D_{NE} + \beta_2 D_{NC} D_E \\
& + D_{NC} \tilde{\mathbf{D}}_t \tilde{\Gamma}'_1 \\
& + D_{NC} D_E \tilde{\mathbf{D}}_t \tilde{\Gamma}'_0 + \gamma_j + \varepsilon_{ijt}
\end{aligned} \tag{40}$$

where now $\tilde{\mathbf{D}}_t$ is a vector of year dummies that excludes the year before the fiscal reform of 1983, which is the reference year. Thus, in Equation 39, $\tilde{\Gamma}_{0,t}$ captures the change in tax evasion due to changes in CIT for firms in exempt industries relative to the year before the fiscal reform of 1983, while $\tilde{\Gamma}_{1,t}$ captures the change in tax evasion due to changes in sales tax rates and CIT for firms in taxed industries relative to the year before the fiscal reform of 1983.

In Equation 40, $\tilde{\Gamma}_0$ captures the change in tax evasion due to changes in CIT for all non-Corporations relative to the year before the fiscal reform of 1983, while $\tilde{\Gamma}_1$ captures the change in tax evasion due to changes in sales tax rates for non-Corporations in taxed industries relative to the year before the fiscal reform of 1983.

6.7.1 Comparison of Specifications

- **I prefer first specification with the combined effects of changes in CIT and ST and the difference with respect to the year before the fiscal reform of 1983**

It is more straightforward to interpret the results in terms of changes in tax evasion relative to the year before the fiscal reform of 1983, which is what we are interested in.

- Why are the standard error so much larger in the levels compared to the differences with respect to the year before the fiscal reform of 1983?

First Specification; Joint Effects: Levels and Diff v. 1983

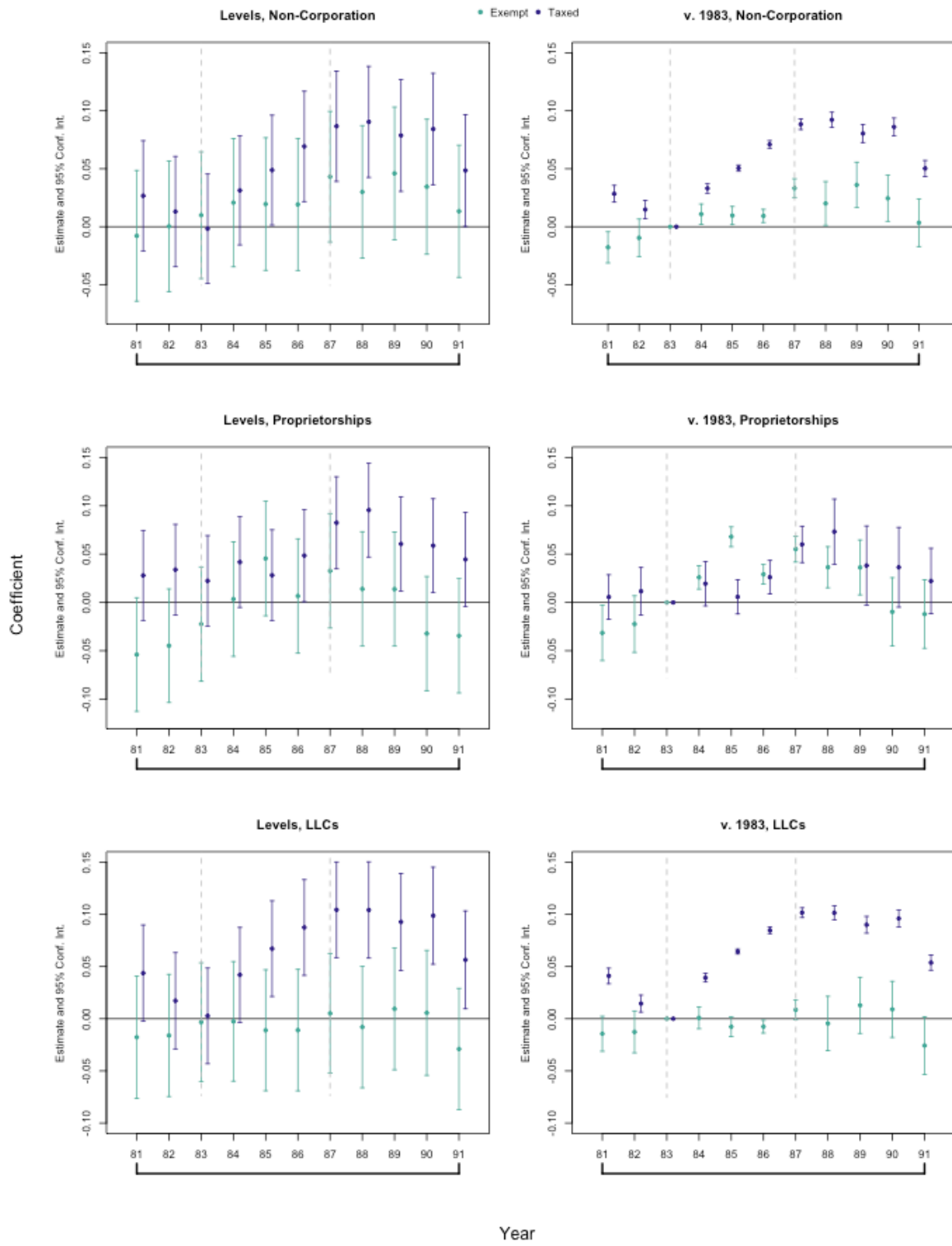


Figure 3

Second Specification; Separated Effects: Levels and Diff v. 1983

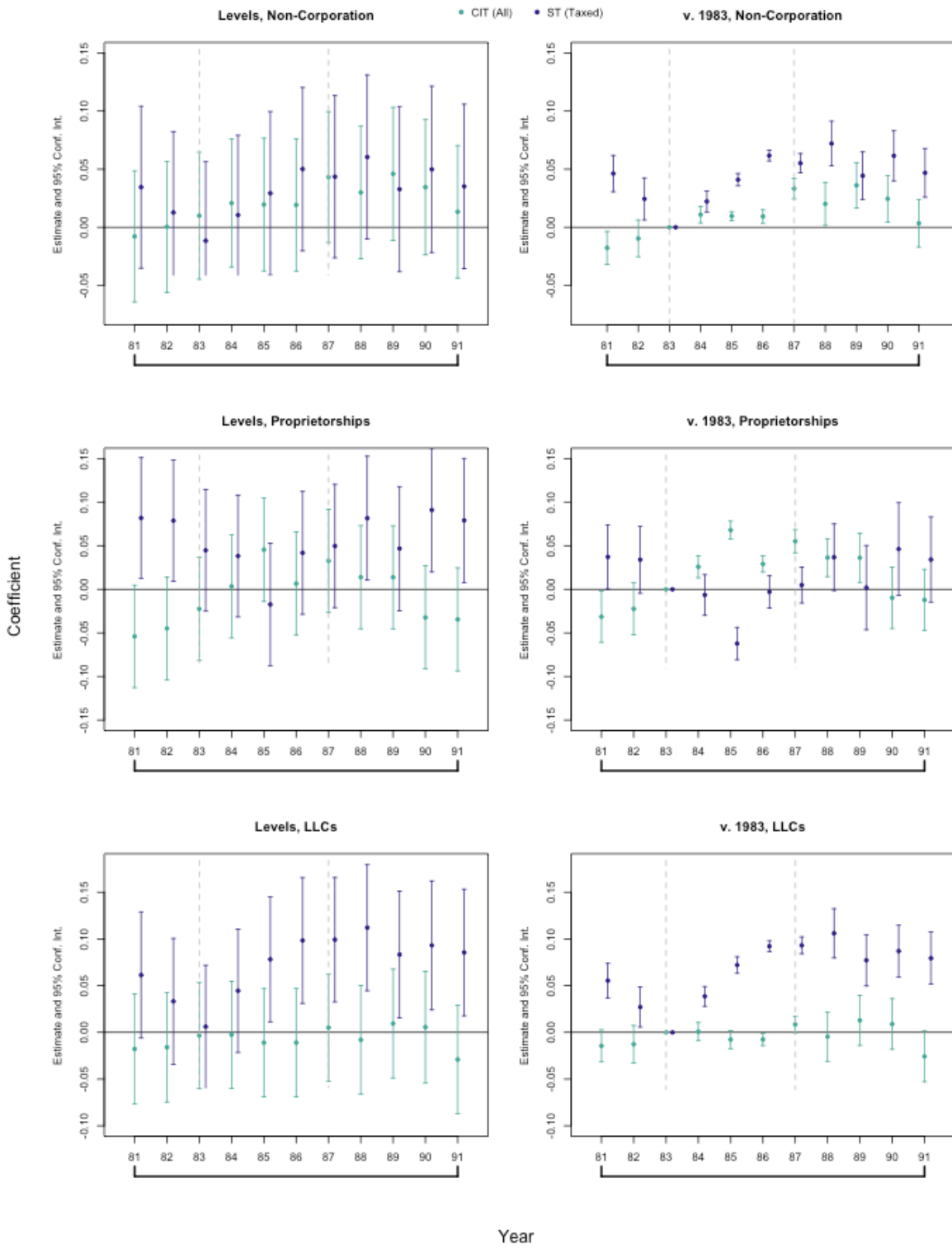


Figure 4

6.7.2 Results

Previous specification,

$$\begin{aligned}
 s_{ijt} = & \beta_0 D_{NC} + \beta_1 D_{NC} D_{NE} \\
 & + \beta_2 D_{NC} D_E \\
 & + D_{NC} D_E \mathbf{D}_t \Gamma'_1 \\
 & + D_{NC} D_{NE} \mathbf{D}_t \Gamma'_2 \\
 & + \gamma_j + \varepsilon_{ijt}
 \end{aligned}$$

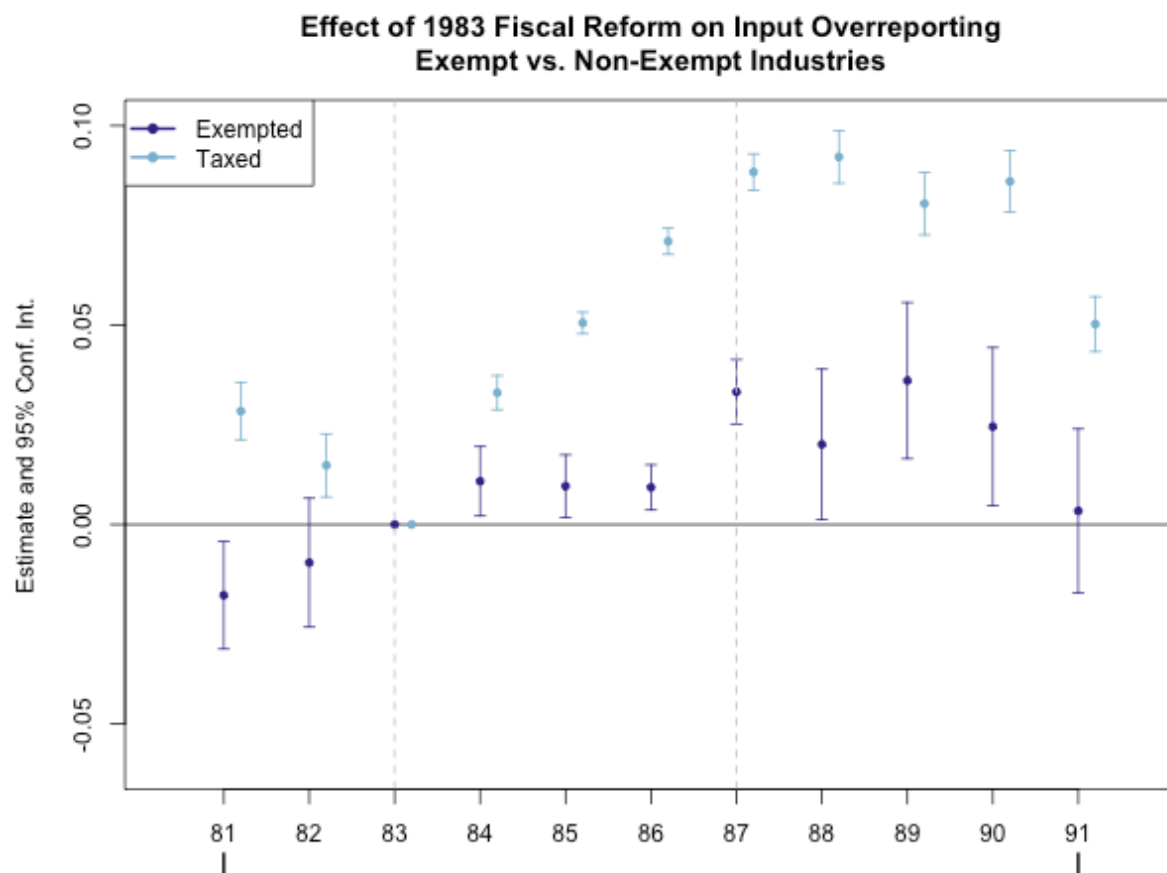


Figure 5

- **After the fiscal reform of 1983, non-Corporations in exempt industries increase tax evasion slightly, while non-Corporations in taxed industries increase tax evasion significantly (Figure 5)**

Since firms in exempt industries are not subject to sales taxes, the effect on tax evasion is due to the changes in individual and corporate income tax rates. The slight increase in tax evasion by non-Corporations in exempted industries reflects the combined effects of the increase in the income tax rate for individuals, which was eight percent on average across the different brackets, and the slight decrease in CIT for LLCs, from 20 to 18 percent.

On the other hand, firms in non-exempt industries face both the changes in sales tax rates and income tax rates. The significant increase in tax evasion by non-Corporations in taxed industries results from the combined effect of the increase in sales tax rates for most industries, and the changes in income tax rates.

Breaking down exempt and non-exempt industries into juridical organizations,

- **LLCs in taxed industries show a significant increase in tax evasion, while LLCs in exempt industries show no significant change (Figure 6)**

LLCs in exempt industries mainly faced a decrease in their CIT. This decrease in would lead to lower incentives to evade taxes. However, since the reduction in CIT is small, it is not unreasonable to find no evidence of significant changes in tax evasion for these industries.

In taxed industries, LLCs faced two opposing effects: the increase in sales tax rates, which would increase incentives to evade taxes, and the decrease in CIT, which would decrease incentives to evade taxes. The evidence suggests that the increase in sales tax rates dominates the effect of the decrease in CIT, leading to a significant increase in tax evasion for LLCs in taxed industries.

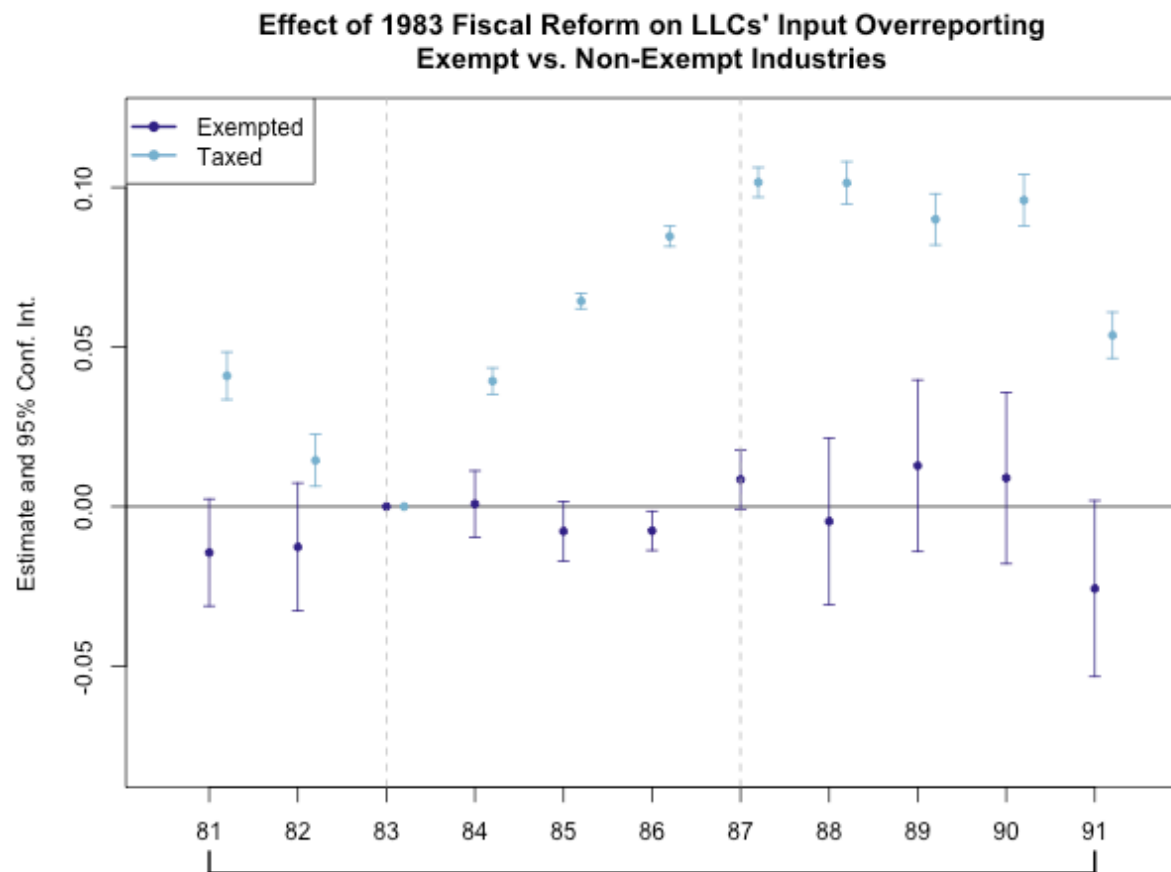


Figure 6

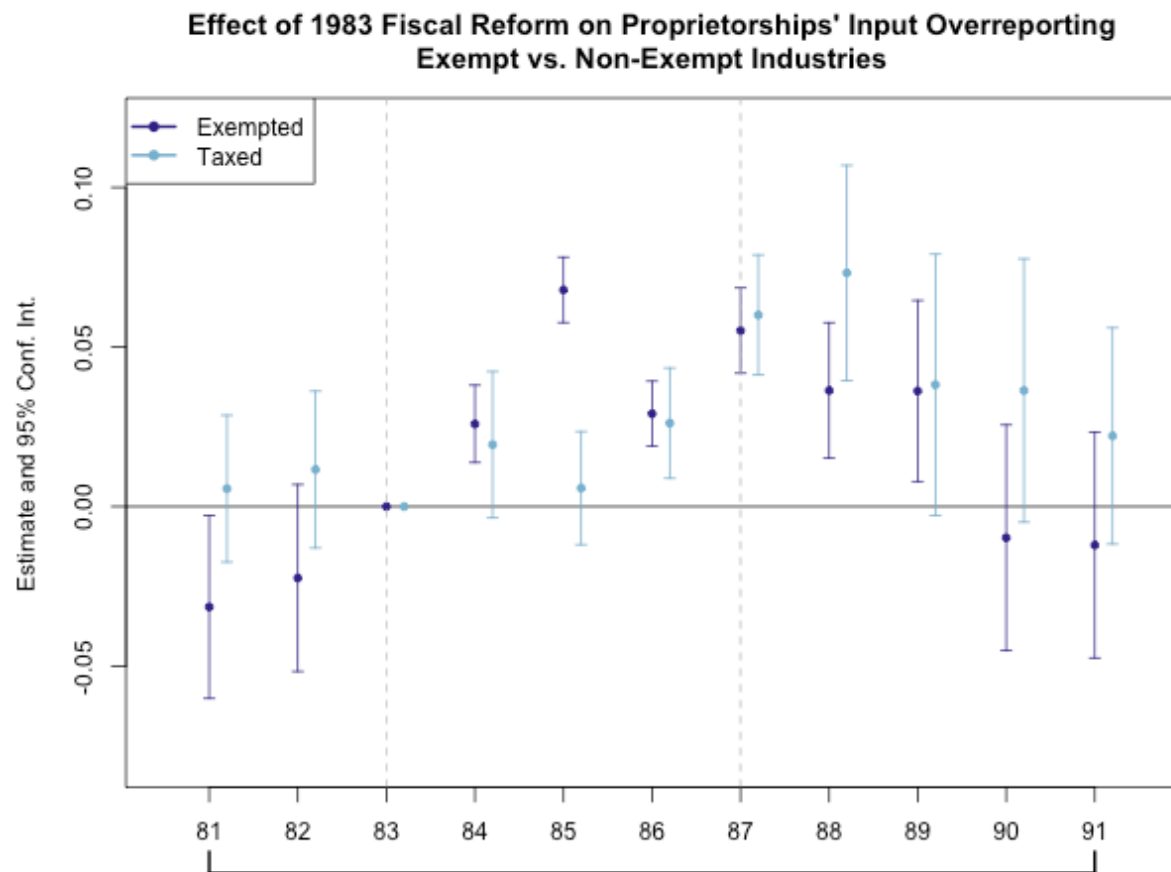


Figure 7

- **Proprietorships in exempt industries show a moderate but significant increase in tax evasion (Figure 7)**

This result and the lack of response from LLCs in exempt industry suggest that Proprietorships were driving the previously discussed increase in tax evasion by non-Corporations in exempt industries. This is even more evident in Figure 8, where we can see that Proprietorships in exempt industries significantly increase tax evasion after 1983, while LLCs in exempt industries show no significant change.

- **Proprietorships in taxed industries mostly behaved no differently than proprietorships in exempt industries (Figure 7)**

Except for 1985, Proprietorships in taxed industries are not significantly different from their counterparts in exempt industries. Their standard errors are generally quite large however, and thus they are either not significantly different from Corporations, except from 1986 to 1988. The larger standard errors might be due to higher heterogeneity in this group of firms.

6.7.2.1 By Exempt vs Taxed Industries

6.8 Tax Evasion and Firm Heterogeneity

As a first attempt to characterize the firms that evade taxes, I try to capture differences in the output elasticities of materials between different types of firms using a simple regression model with interaction terms.

I am interested to see if there are systematic differences between firms with different characteristics. The characteristics of interest are different measures of size, such as revenue, capital, and labour. Other relevant characteristics are the exports and imports of the firm.

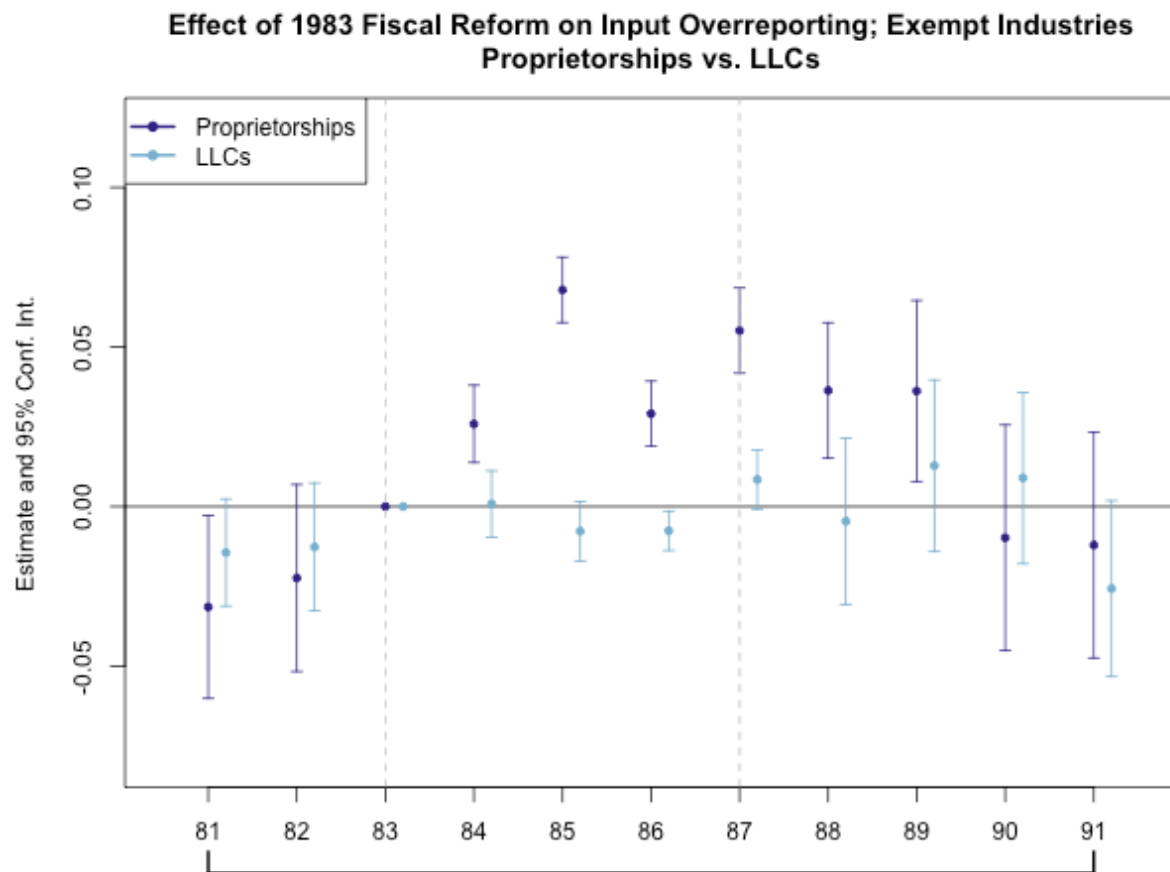


Figure 8

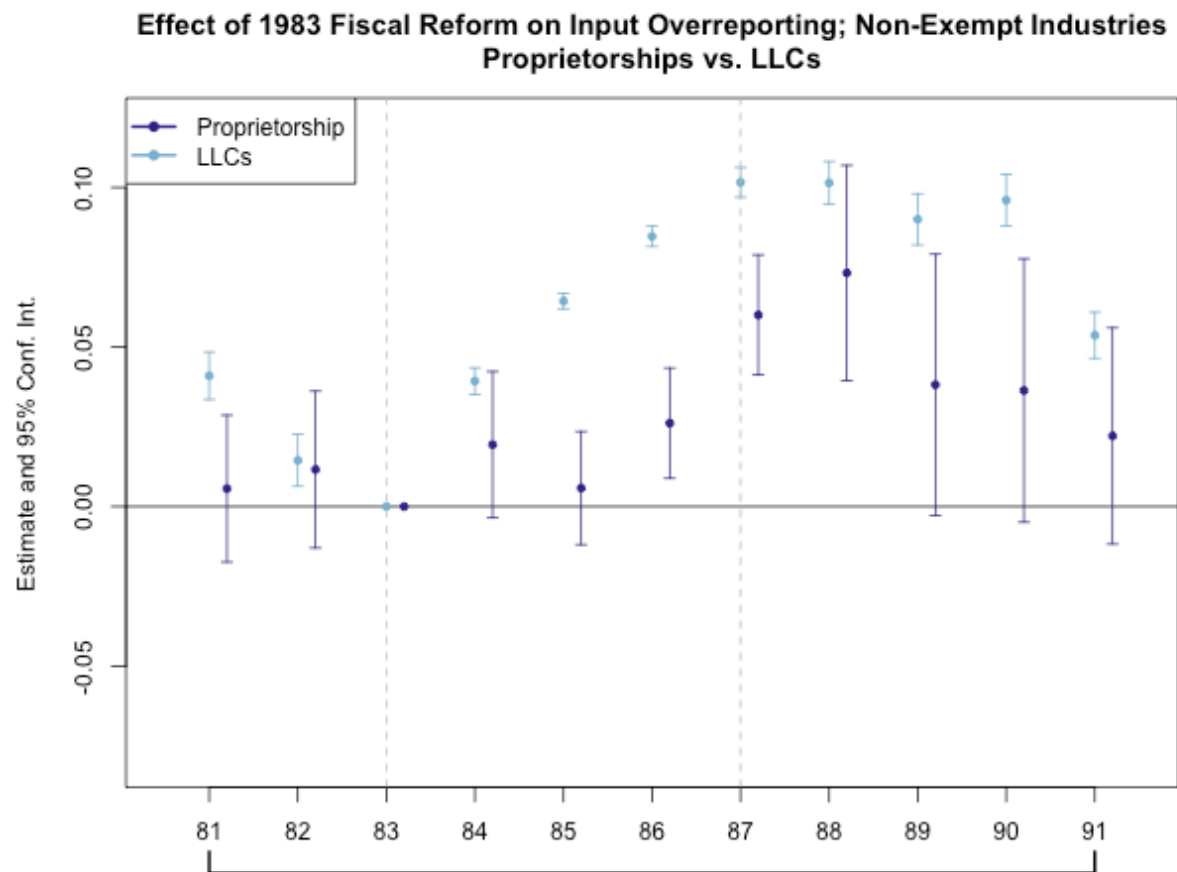


Figure 9

- The literature has documented that, to the contrary of common intuition, larger firms evade more by overreporting their inputs ([Carrillo et al. 2022](#)).

Initially, my prior was that size was inversely related to tax evasion. The intuition was that the larger the firms, the more likely to be caught evading taxes. Higher probability of detection, lower incentives to evade taxes.

This intuition assumes that the probability of detection is increasing with size. However, it is standard in the literature that the probability of detection is independent of any firm characteristic. The probability of detection is a simplification of two probabilities: The probability of being audited by the tax authority, and the probability of being detected cheating conditional on being audited. It is common in the literature to simplify these probabilities, in particular in empirical papers, by assuming that firms are detected every time if they are audited, and that the tax authority audits firms at random. Thus, the probability of detection is independent of firm characteristics, including size. The probability of detection depends only on the level of evasion —the more overreporting— and an exogenous component from the firm side, which is effort from the tax authority. The idea is that you can be audited at random. Then, if you are audited, you will be compared to similar firms. The more you overreport, the more likely you are to be detected.

Another relevant component in the tax evasion decision is the cost of evasion. This cost includes accounting and managing costs associated with tax evasion, including the extra work of accounting employees to keep two books of records, one for the tax authority and one for internal use —the firm would need to know its true profits to make business decisions—, as well as economic costs, such as lost investment opportunities due to investors not liking low profits, or distrusting the firm because of suspected tax evasion. Although previous work has allowed for heterogeneity in the cost of evasion, an explicit relationship

between size and evasion cost has not been explicitly modeled. [Almunia & Lopez-Rodriguez \(2018\)](#) suggest that costs of evasion might be heterogeneous among firms for several reasons, including the number of employees with which the employer needs to collude to evade taxes (this reason would suggest that larger firms might face higher costs of evasion, as more employees need to collude). Other reasons evasion costs might vary are manager's preferences (e.g., risk aversion, ethical considerations), or the time and skills invested in the misreporting activity rather than allocating them to productive activities.

Recently however, new evidence has shown that tax evasion by overreporting is increasing in revenue percentile, but it collapses for the largest firms, in particular, for very large corporations ([Carrillo et al. 2022](#)). The same paper finds that in Ecuador, corporations evaded 1.7% and sole Proprietorships, 11.5%. The authors find that overreporting firms are larger than other firms, with higher revenues, costs, and tax liabilities (makes sense, higher the tax liability, higher the incentives to evade taxes). The authors show that the probability of overreporting increases monotonically in firm revenue. Furthermore, the share of fake deductions out of total deductions due to overreporting increases throughout the revenue distribution, except at the very top. Authors suggest that the sharp drop at the top might be due to corporations having stronger incentives to avoid illegal behaviour, or that they can use more sophisticated and effective measures of tax avoidance, that do not require misreporting expenses.

- Exporters might evade less.

Why? Exports are not subject to sales taxes. Therefore, the higher the share of sales is exported the lower the incentives to overreport their inputs, as their sales tax liability decreases. In both models, the effective sales tax rate τ is lower for exporters.

- Importers might evade more.

Even though imports are subject to sales taxes, firms pay in addition import tariffs. While imports might not be easily overreported due to a larger paper trail, firms might overreport their domestic purchases to compensate for the higher cost of imports. In both models, the effective sales tax rate τ is higher for importers. In particular, import tariffs increase the effective sales tax rate by decreasing the share of purchases that might be deducted from sales tax liability.

6.8.1 Empirical Strategy

Again, for the truth-reporters, corporations,

$$s_{ijt}^C = \gamma_j + \varepsilon_{ijt}$$

For the evaders, non-corporations,

$$s_{ijt}^{NC} = \gamma_j + e_{ijt}(\tau, q, \kappa) + \varepsilon_{ijt}$$

where $e_{ijt}(\tau, q, \kappa)$ is the overreporting of inputs for firm i in industry j at time t , which is a function on the effective sales tax rate τ , the probability of detection q and the cost of evasion κ .

Combining these equations, we can write the following,

$$\begin{aligned}
(1 - D_{NC})s_{ijt} + D_{NC}s_{ijt} &= (1 - D_{NC})(\gamma_j + \varepsilon_{ijt}) \\
&\quad + D_{NC}(\gamma_j + e_{ijt}(q, \kappa) + \varepsilon_{ijt}) \\
s_{ijt} &= \gamma_j + D_{NC}e_{ijt}(\tau, q, \kappa) + \varepsilon_{ijt}
\end{aligned} \tag{41}$$

Since the probability of detection and the cost of evasion are not observed, we could use firms characteristics that are likely to influence them, and thus lead to systematic differences in the overreporting of inputs across firms. For example, if we want to investigate the relationship between size and tax evasion by overreporting inputs, we can use the interaction between a Non-Corporation dummy and different measures of size to capture the heterogeneity in overreporting of inputs across firms due to size.

In theory, I could use any firm characteristic since error terms is not part of the information set of the firm, and thus $\mathbb{E}[\varepsilon_{ijt}|e_{ijt}] = 0$. In practice however I would like to avoid variables that could be mechanically correlated with the explained variable, the input cost share. For example, if I use current revenue as a measure of size, it could be mechanically correlated with the input cost share, as the denominator of the input cost share is revenue.

6.8.2 Firm Size and Tax Evasion

As, previously discussed, size could be correlated with tax evasion. To investigate the effects of size, I use Equation 41 and estimate the following regression model:

$$s_{ijt} = \gamma_j + D_{NC}\mathbf{X}_{it}\Gamma'_0 + \varepsilon_{ijt}$$

where γ_j stands for industry fixed effects, and ε_{ijt} is the error term. $\mathbf{X}_{it}$ is a set of dummy variables for each decile of size measured by labor, capital, revenue last year, and revenue two years ago. I use revenue lagged one and two years to avoid spurious correlation

depending on the construction of the explained variable, which uses current year revenue to compute the input cost share. Γ_0 is a vector of coefficients for the interaction between the Non-Corporation dummy and the deciles of size. The coefficient of interest is Γ_0 , the interaction term between the deciles of each size measure X_{it} and Non-Corporations D_{NC} . Γ_0 is vector of coefficients of interest and can be interpreted as the average overreporting of each size decile for Non-Corporations.

Figure 10 plots the regression results for the deciles of the different characteristics of interest separately.

- **Tax evasion by overreporting is increasing along firm size, but it drops at the last decile (Figure 10).**

What I observe is that tax evasion by overreporting inputs is increasing in firm size, measured by revenue, capital, and labour. In particular, the percentage of inputs costs that firms overreport increases along the deciles of revenue, capital, and labour for Non-Corporations. Furthermore, all these measures of size show a similar pattern: the percentage of overreported inputs increases steadily from the first to the ninth decile, but then it collapses at the tenth decile.

- **This pattern is consistent with the evidence for Ecuador (Figure 11).**

6.9 Marginal Effects

The previous measures are correlated with size. Which one is the most relevant? What is the marginal effect of each size measure holding the others constant?

I investigate these questions by estimating the following regression model:

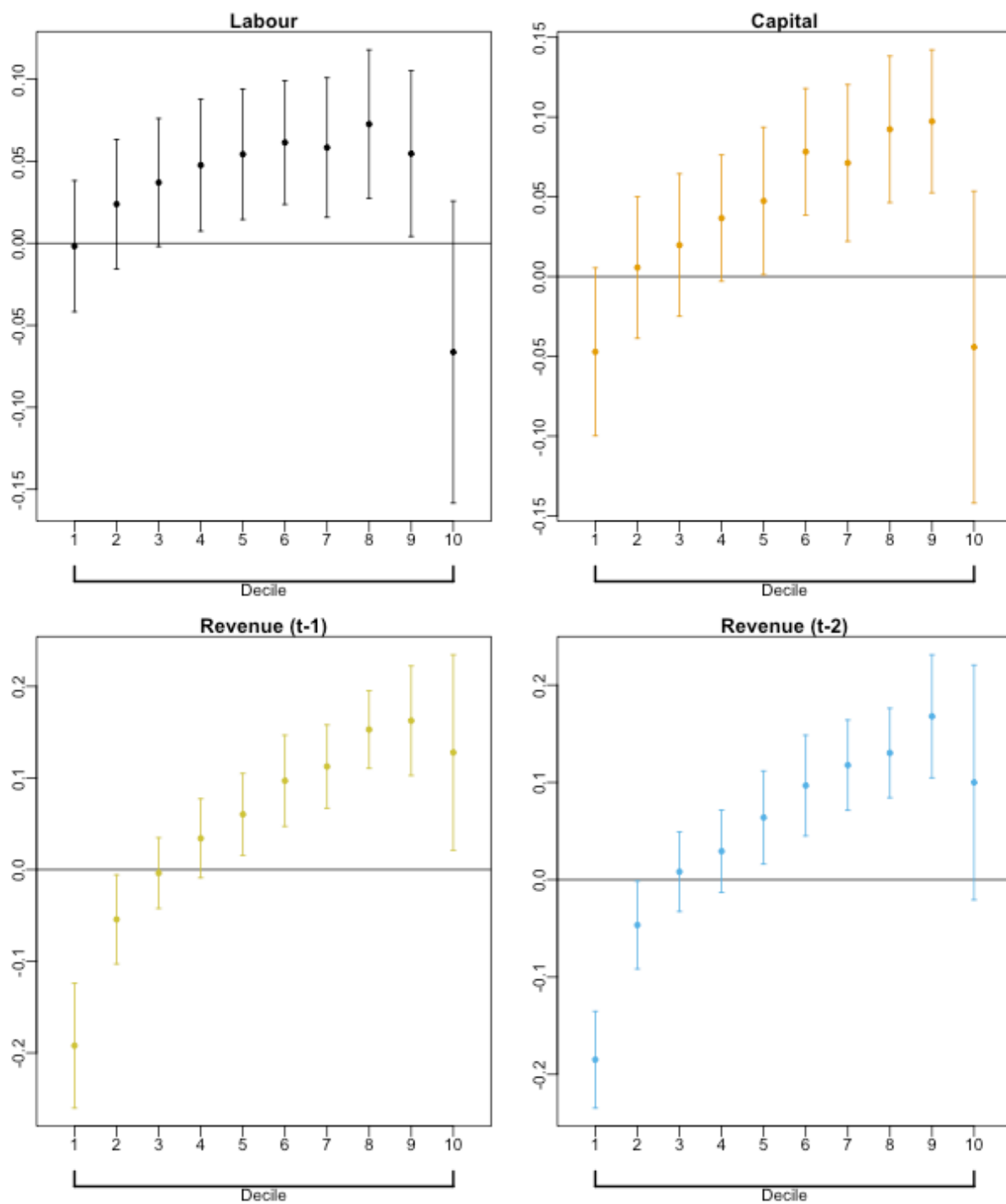


Figure 10

B) Value of Reported Ghost Purchases Over Firm Total Purchases, by Firm Size

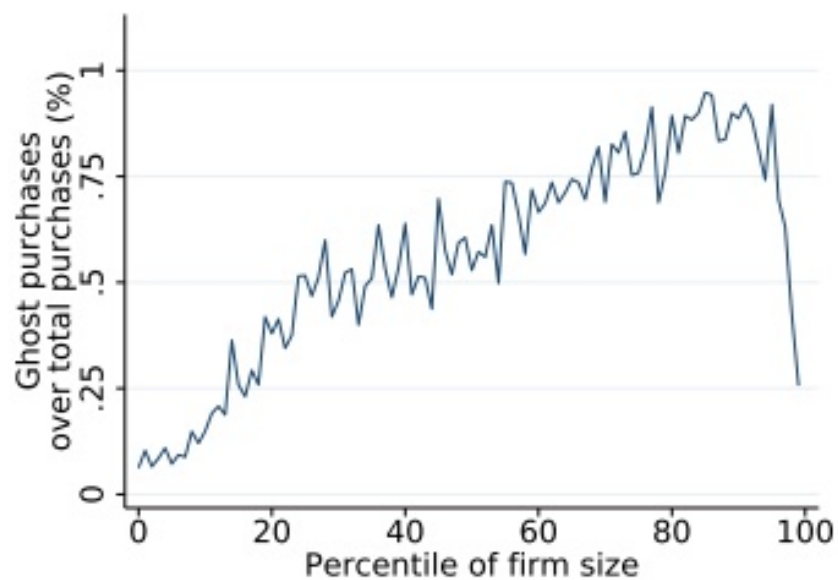


Figure 11: Figure 1B from [Carrillo et al. \(2022\)](#)

$$s_{ijt} = \gamma_j + \beta_0 D_{NC} + D_{NC} \mathbf{X}_{it} \Gamma'_0 \\ + D_{NC} \mathbf{X}_{it}^2 \Gamma'_1 + \varepsilon_{ijt}$$

where the characteristics X_{it} are the percentiles of: labour, capital, revenue (lagged two years), exports, and imports. I include a square term for each characteristic X_{it} to capture non-linearities in the relationship between each characteristic and tax evasion by overreporting inputs.

Figure 12 shows the relevant marginal effects are labour, revenue (lagged two years), and imports. Figure 13 shows the predicted overreporting curves for each characteristic holding the others at their median (50th percentile).

- **Holding all other characteristics constant, increasing the labour percentile decreases tax evasion.**

The reduction in overreporting is more pronounced as we move to higher percentiles of labour. This result suggests that if we take two firms with similar revenue, capital, exports, and imports, the one with more employees evades less by overreporting inputs.

This makes sense for several reasons. **First, the marginal cost of evasion would be higher** for the firm with more employees, as more employees need to collude with the firm to evade taxes illegally. **Second, more employees involved also increases the probability of detection** from within the firm, as the probability that an employee might be willing to anonymously report the illegal activity to the tax authority increases.

- **Moving up in the revenue distribution increases tax evasion by overreporting inputs, holding constant the other characteristics.**

The statistical significance of the quadratic term is likely due to the non-linear pattern

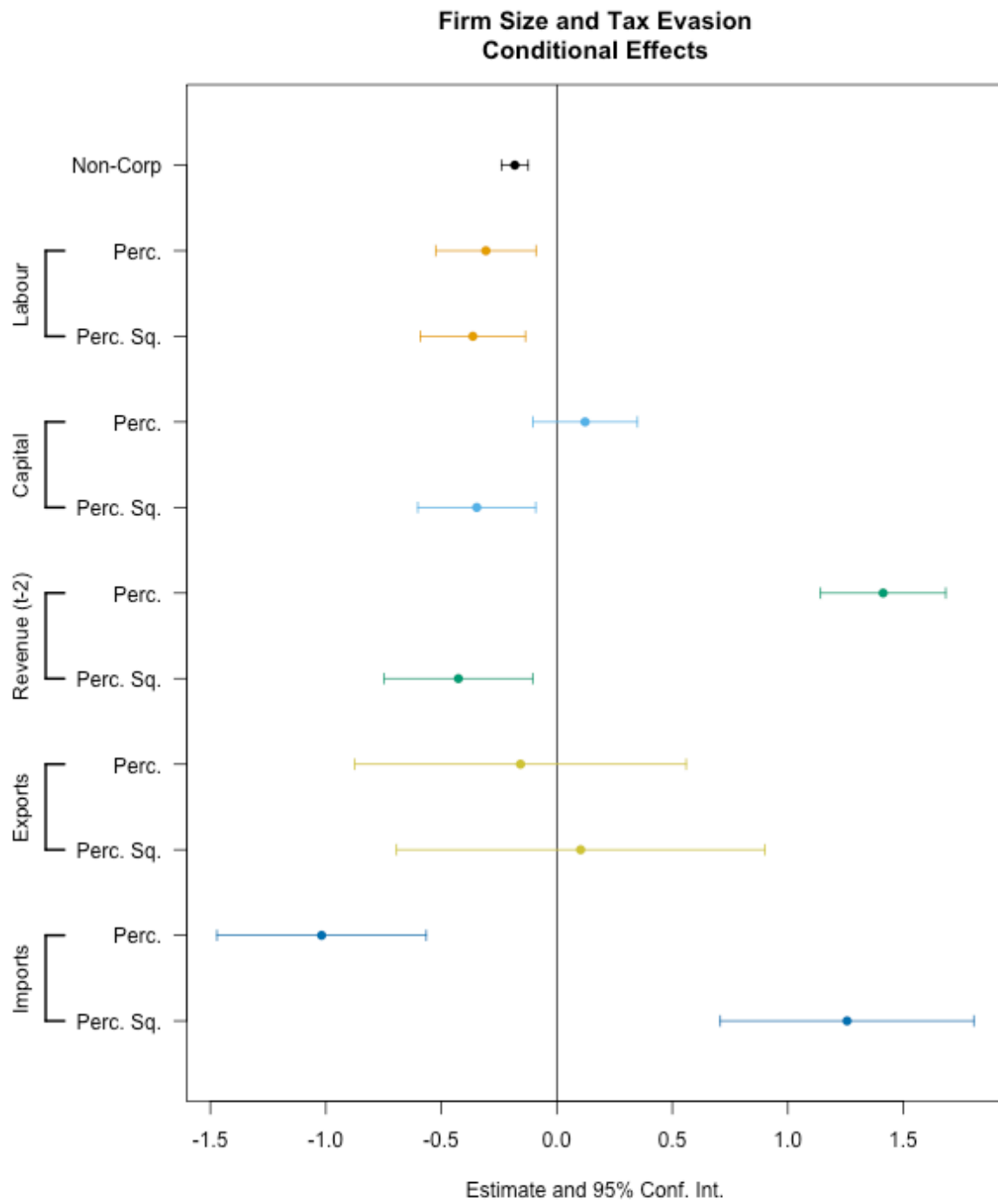


Figure 12

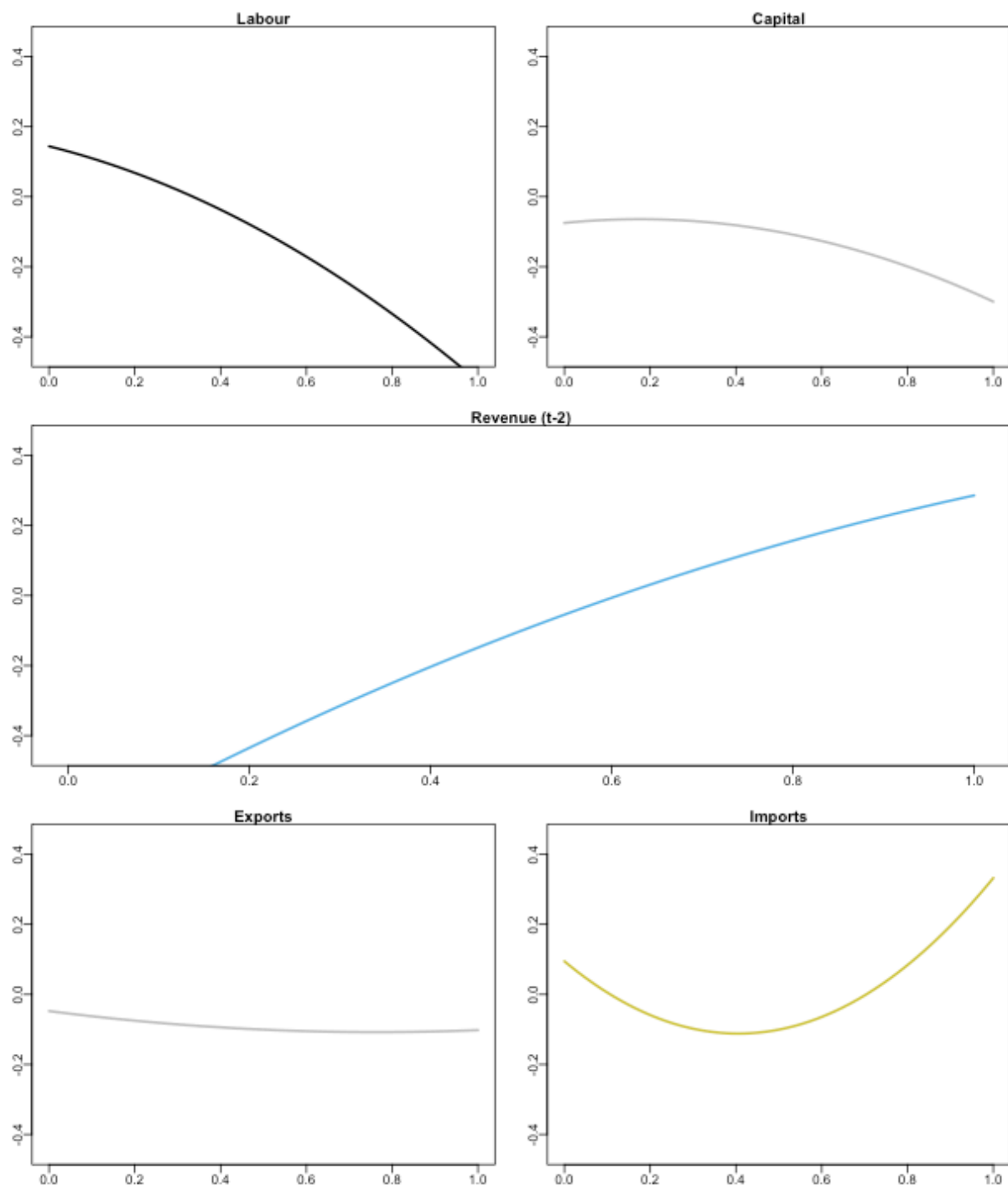


Figure 13

observed in the previous section, where tax evasion increases along the revenue distribution but then collapses at the very top.

For firms with small revenues, the relative cost of overreporting inputs is high, as they have less resources to allocate to tax evasion activities. As firms grow larger in revenue, they can allocate more resources to tax evasion, reducing the relative cost of evasion.

However, for the largest revenue firms, the cost of evasion might increase due to several reasons. The cost of *illegal* tax evasion might abruptly increase relative to more sophisticated *legal* tax avoidance strategies once a firm can implement them (e.g. offshoring profits). The reputational cost of being caught evading taxes might be higher for larger firms, as they are more visible to the public and stakeholders.

- **Finally, imports are negatively correlated with tax evasion by overreporting inputs, holding constant the other characteristics, but the effect is convex.**

By observing Figure 13, we observe that after the 40th percentile, the percentage of overreported inputs increases with imports. This result is consistent with the previous discussion that **importers face higher incentives to overreport their domestic purchases to compensate for the higher costs due to import tariffs**. However, the convexity of the relationship suggests that for firms with low imports, the incentives to overreport domestic purchases are not strong enough to increase tax evasion. **Only for firms with large imports, the incentives become strong enough to increase tax evasion by overreporting inputs.**

Overall, these results suggest that the non-linear pattern observed in the previous section might be explained by;

- a probability of detection that increases slowly along most of the size distribution, but then increases sharply for the largest firms. The probability of detection might in fact be flat for most of the size distribution. It then increases sharply for the largest firms because they are more visible to the tax authority, and because internal detection from within the firm becomes more likely as more employees need to collude to evade taxes.
- a cost of evasion that decreases along most of the size distribution, but then it increases for the largest firms. This is because firms with small revenues have high relative costs of evasion. As firms increase in revenue, they can allocate more resources to tax evasion, reducing the relative cost of evasion. However, for the largest revenue firms, the cost of evasion might increase due to the higher costs of illegal evasion relative to legal avoidance strategies, and higher reputational costs.

6.10 Robustness Check

7 References

(n.d.).

Akerberg, D. A. (2019), ‘Timing assumptions and efficiency: Empirical evidence in a production function context’.

Akerberg, D. A., Caves, K. & Frazer, G. (2015), ‘Identification properties of recent production function estimators’, **83**, 2411–2451.

Akerberg, D., Frazer, G., Luo, Y. & Su, Y. (2021), ‘Under-identification of structural

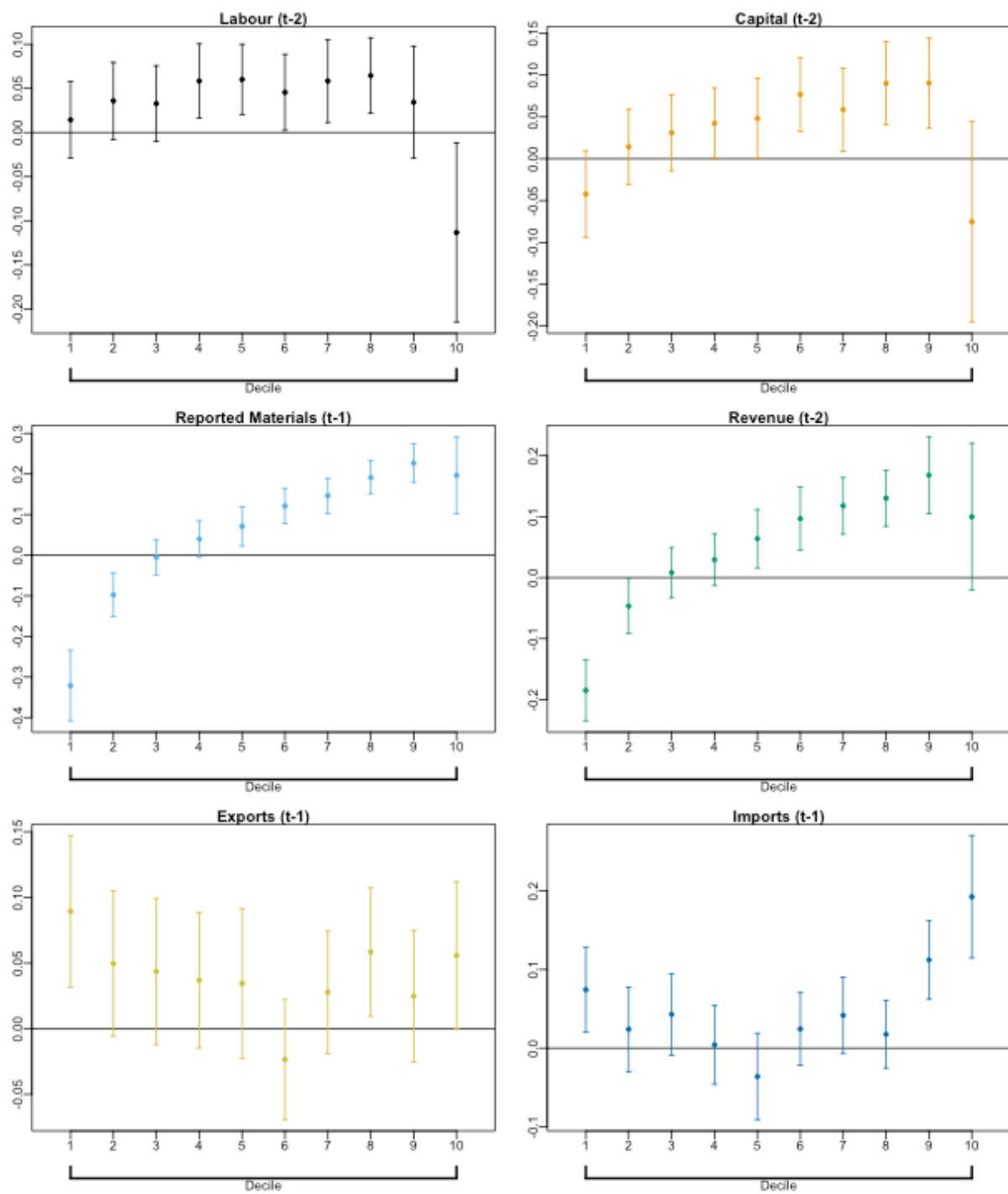


Figure 14

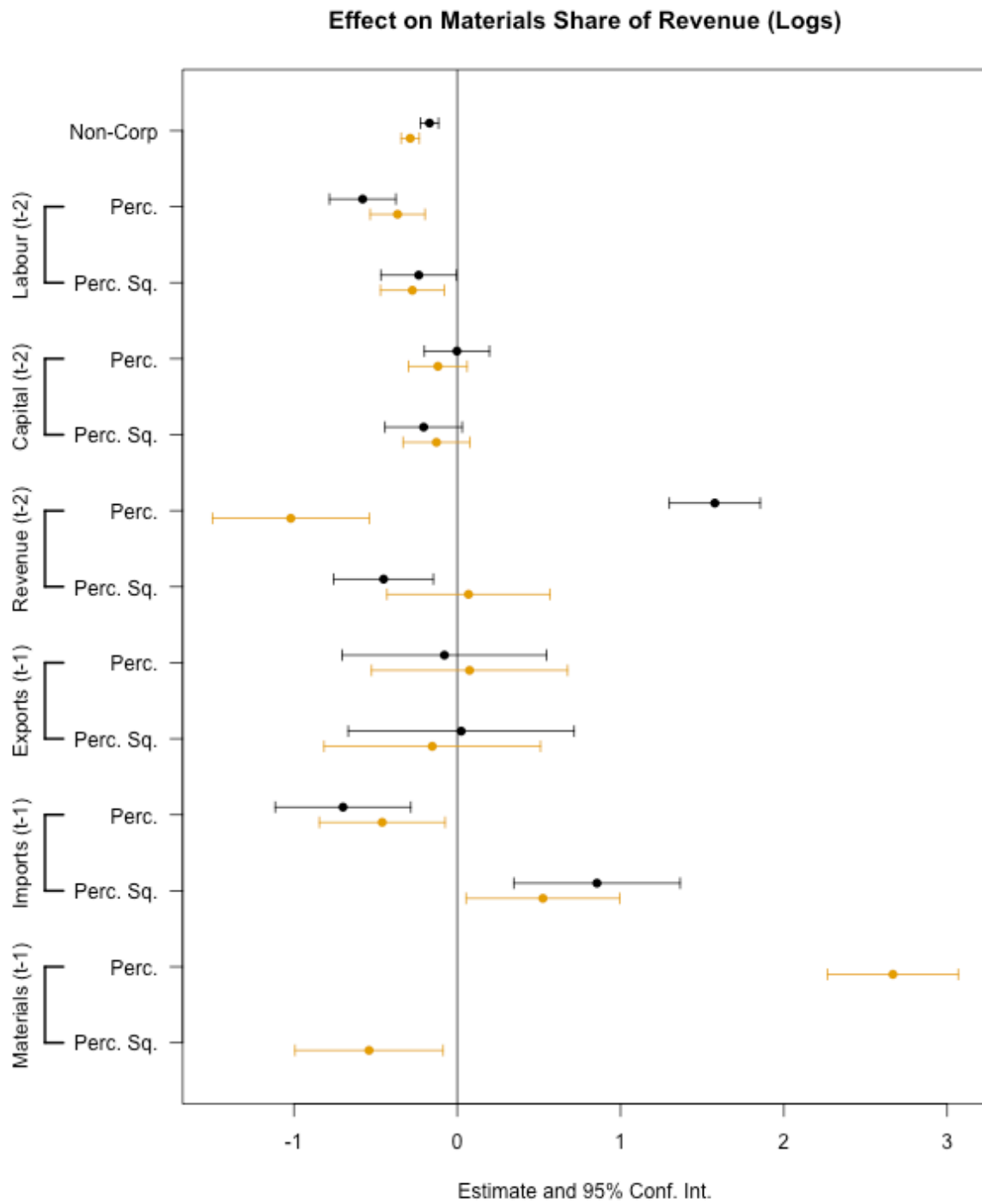


Figure 15

models based on timing and information set assumptions’.

URL: <https://ssrn.com/abstract=3717757>

Allingham, M. G. & Sandmo, A. (1972), ‘Income tax evasion: a theoretical analysis’, *Journal of Public Economics* **1**, 323–338.

Almunia, M. & Lopez-Rodriguez, D. (2018), ‘Under the radar: The effects of monitoring firms on tax compliance †’, *American Economic Journal: Economic Policy* **10**, 1–38.

URL: <https://doi.org/10.1257/pol.20160229>

Carrillo, P., Donaldson, D., Pomeranz, D. & Singhal, M. (2022), ‘Ghosting the tax authority: Fake firms and tax fraud’.

URL: <http://www.nber.org/papers/w30242>

Chen, X. (2007), ‘Large sample sieve estimation of semi-nonparametric models *’.

Dabla-Norris, E., Gradstein, M., Miryugin, F. & Misch, F. (2019), ‘Productivity and tax evasion’.

URL: www.RePEc.org

Eilers, P. H. & Marx, B. D. (1996), ‘Flexible smoothing with b-splines and penalties’, *Statistical Science* **11**, 89–121.

URL: <https://projecteuclid.org/journals/statistical-science/volume-11/issue-2/Flexible-smoothing-with-B-splines-and-penalties/10.1214/ss/1038425655.full>
<https://projecteuclid.org/journals/statistical-science/volume-11/issue-2/Flexible-smoothing-with-B-splines-and-penalties/10.1214/ss/1038425655.short>

Eslava, M., Haltiwanger, J., Kugler, A. & Kugler, M. (2004), ‘The effects of structural reforms on productivity and profitability enhancing reallocation: evidence from colombia’, *Journal of Development Economics* **75**, 333–371. quantities data.

- Gandhi, A., Navarro, S. & Rivers, D. A. (2020), ‘On the identification of gross output production functions’.
- Griliches, Z. & Mairesse, J. (1995), Production functions: The search for identification.
- Hu, Y., Schennach, S. & Shiu, J. L. (2022), ‘Identification of nonparametric monotonic regression models with continuous nonclassical measurement errors’, *Journal of Econometrics* **226**, 269–294.
- Levinsohn, J. & Petrin, A. (2003), ‘Estimating production functions using inputs to control for unobservables’, *Review of Economic Studies* **70**, 317–341.
- McClure, C. E. (1989), Analysis and reform of the colombian tax system, *in* M. Gillis, ed., ‘Tax Reform in Developing Countries’, Duke University Press, Durham and London.
URL: https://pdf.usaid.gov/pdf_docs/pnabp729.pdf
- OECD (2017), Technology tools to tackle tax evasion and tax fraud, Technical report, OECD.
URL: www.oecd.org
- Olley, G. S., Pakes, A. & Steven, G. (1996), ‘The dynamics of productivity in the telecommunications equipment industry’, *Econometrica* **64**, 1263–1297.
- Paulus, A. (2015), ‘Tax evasion and measurement error: An econometric analysis of survey data linked with tax records’.
URL: <http://hdl.handle.net/10419/126489>
- Perry, G. & de Triana, L. O. (1990), ‘The vat in colombia’.
- Rudnick, H. & Velasquez, C. (2021), Learning from power sector reform experiences the

case of colombia, Technical report, The World Bank.

URL: *www.worldbank.org*

Slemrod, J. (2019), ‘Tax compliance and enforcement†’, *Journal of Economic Literature* **57**, 904–954.

Torres, F. S. & Sourdis, C. G. (1994), Reformas tributarias en colombia 1980-1992. aspectos de equidad, eficiencia, y simplificación administrativa, *in* ‘VI Seminario Regional de Política Fiscal’.

Yi, G. Y. (2021), ‘Likelihood methods with measurement error and misclassification’, *Handbook of Measurement Error Models* pp. 99–126.

URL: *https://www-taylorfrancis-com.proxy1.lib.uwo.ca/chapters/edit/10.1201/9781315101279-6/likelihood-methods-measurement-error-misclassification-grace-yi*

Zumaya, M., Guerrero, R., Islas, E., Pineda, O., Gershenson, C., Iñiguez, G. & Pineda, C. (2021), ‘Identifying tax evasion in mexico with tools from network science and machine learning’.

URL: *http://arxiv.org/abs/2104.13353*

8 Industries 381-384 and 369

According to some records, agricultural machinery (382) and transportation equipment (384) were exempted from sales tax before 1984. I investigate in this section if the data can confirm this affirmation. In particular, I look at industry 38 Fabricated Metal Products, Machinery & Equipment:

- 381 Fabricated Metal Products (Structural metal products, Cutlery, Hand tools, etc.)
- 382 Machinery and Equipment (Agricultural machinery, Metalworking machinery,

etc.)

- 383 Electrical Machinery and Apparatus (Electric motors, Generators, etc.)
- 384 Transport Equipment (Motor vehicles, Aircraft, Ships, etc.)

If these industries were exempted, then both industries would be candidates for least likely industries to evade taxes by overreporting input purchases. Furthermore, the increase in the sales tax after 1984 should have increased the incentives to overreport input purchases. All else equal, we would expect to see an increase in the materials share after 1984.

There are some complications when analyzing these industries in the data. First, I have just a vague description of the changes in the exemption. To the best of my understanding, agricultural machinery falls under sic code 382 (3822 in particular) and transportation equipment under sic code 384. Industry 382, however, also includes other types of machinery that might have not been exempted, such as engines and turbines (3821), metal and wood working (3823), special industrial (3824), etc. Likewise, industry 384 a wide range of transportation equipment, like shipbuilding (3841), railroad equipment (3842), motor vehicles (3843), motorcycles (3844), aircraft (3845), and other transportation equipment (3849). Which of these industries were exempted? It is not clear.

Second, the 1983-1984 Colombian tax reforms adjusted the sales rate tax for many other industries besides 381-384 and simultaneously also adjusted the corporate income tax for some juridical organizations. In particular, the reform modified the income tax rate of proprietorship firms, through changes to the individuals' income tax rate, and the limited liability companies (LLCs). Most brackets of the individuals' income tax received an 8 percent increase, while the maximum rate was reduced from 56 to 49 percent. LLCs were granted a small reduction of the CIT rate from 20 to 18 percent.

Table 14: Sales and Purchases Sales Tax Rates for Industries 381-384

SIC	Sales Tax Rate			
	Sales		Purchases	
	Before 1984	After 1984	Before 1984	After 1984
381	8.9	9.7	4.4	5.3
382	5.7	9.7	3.1	4.8
383	7.7	9.9	3.8	5.3
384	5.5	9.5	4.7	5.3

In Table 14, the data shows an increase in the sales and purchases sales tax rate after 1984, but I do not find evidence that these industries were exempted like industries 311 and 312 before 1984.

Given that the increase from non-zero sales tax rate before 1984 and the change in the CIT for Proprietorships and LLCs are not exclusive to industries 381-384, a difference-in-differences approach would capture the effect of both the change in the sales tax and the change in the income tax.

I explore a triple difference approach using industries 311 and 312 as a control group. These industries were exempted from the sales tax before and after 1984, but they faced the same changes in the income tax as industries 381-384. However, I prefer to perform this analysis for all industries in the second paper of the thesis.

8.1 Industry 369

The tax evasion density due to overreporting for industry 369 Non-Metallic Mineral Products (Structural clay products, Cement and plaster, Other non-metallic mineral products) displays an extremely high skewness. This skewness may be the reason behind the small standard deviation of the productivity density estimates.

I find that trimming the data to remove outliers in the materials share distribution helps reduce the skewness of the tax evasion density.

Figure 16 and Table 15 both indicate the presence of outliers in the materials share distribution. There is a somewhat thin tail to the right of the distribution, above 0.75 on the density plot. This value, 0.75 is above the 99.5 percentile, and we can still see some observations above this value up to the maximum at 0.836.

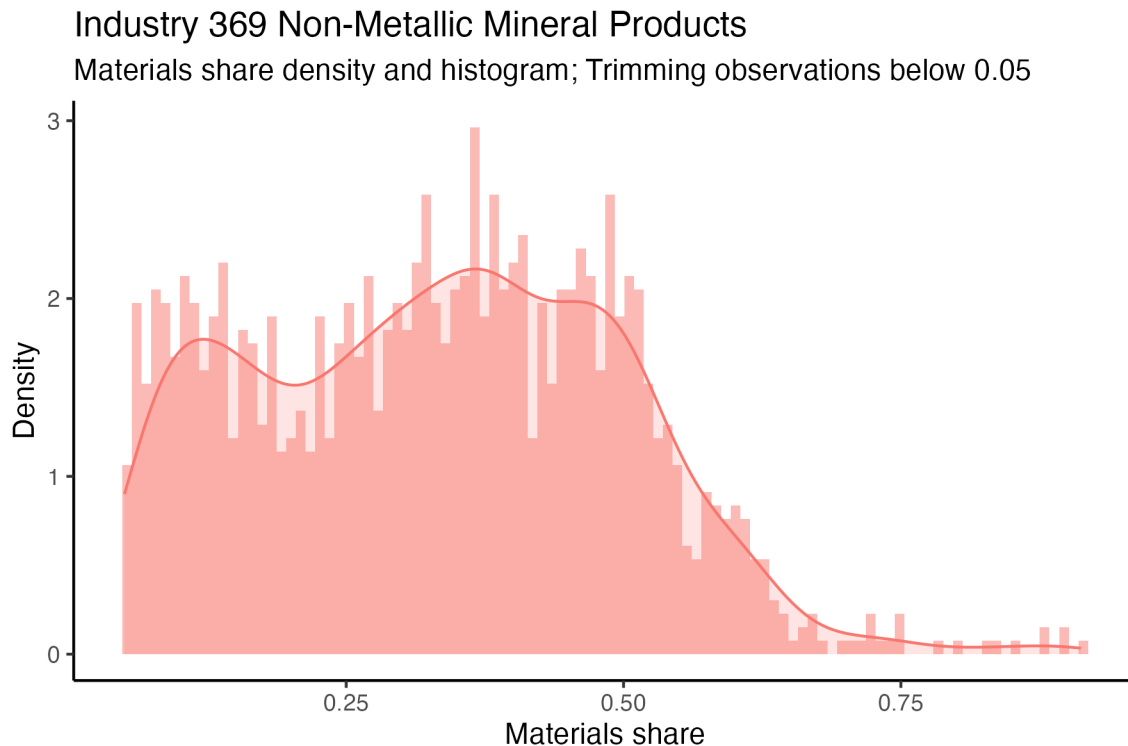


Figure 16

Table 15: Summary statistics for industry 369 Non-Metallic Mineral Products

Obs.	All (N)	Corps. (N)	Min	Q1	Median	Q3	P90	P95	P99	P99.5	Max
1511	274	58	0.05	0.191	0.337	0.457	0.534	0.59	0.725	0.813	0.912

Trimming at 0.75 removes less than 1 percent of the observations, but it removes a significant portion of the right tail. 90th percentile remains at 0.53, 95th percentile decreases from 0.59 to 0.57, and 99th percentile decreases from 0.72 to 0.63.

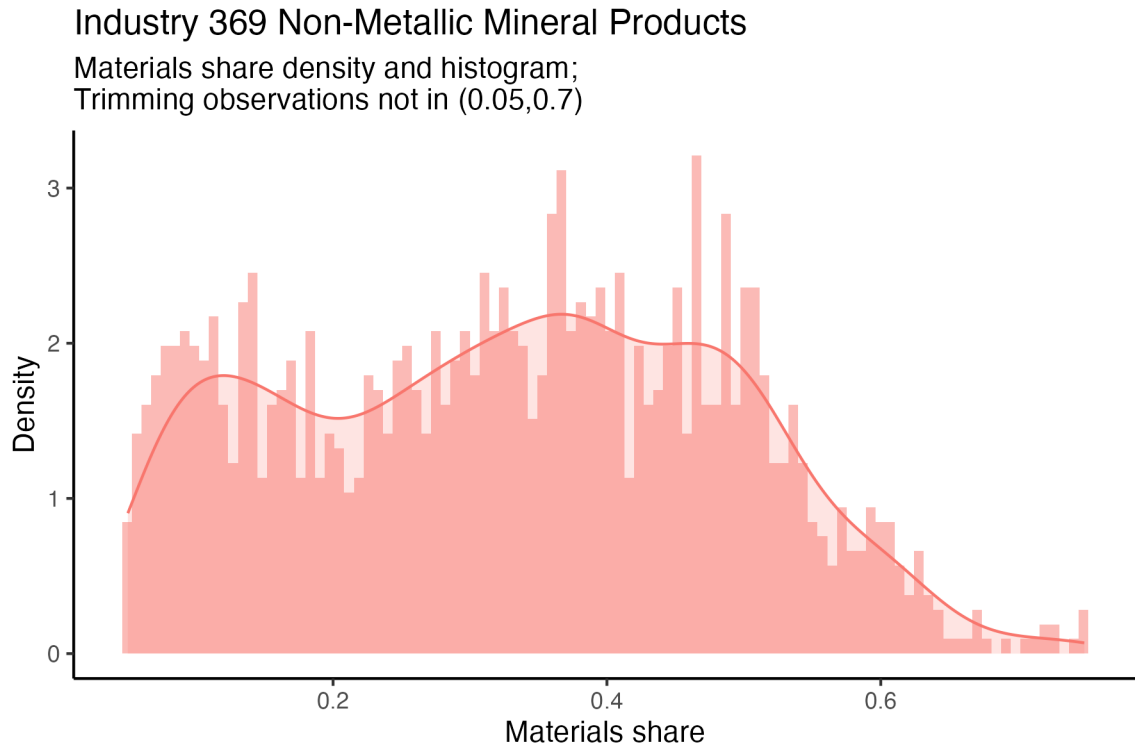


Figure 17

Trimming the data reduces skewness while barely affecting the no tax evasion test.

Table 17 shows that, even though the statistic decreased marginally, we can still reject the null hypothesis of no tax evasion (one side test). The output elasticity of materials of non-evaders and evaders are still statistically different.

Table 16: Summary statistics for industry 369 Non-Metallic Mineral Products. Trimming observations not in (0.05,0.7).

Obs.	All (N)	Corps. (N)	Min	Q1	Median	Q3	P90	P95	P99	P99.5	Max
1501	273	58	0.05	0.191	0.334	0.454	0.53	0.583	0.666	0.716	0.749

Table 17: Summary statistics for the tax evasion test for industry 369 Non-Metallic Mineral Products. Trimming observations not in (0.05,0.7).

SIC	Corps.	Others	Tax Evasion Test
369	0.24	0.29	0.19**
	[0.19, 0.26]	[0.26, 0.31]	[0.04, 0.37]

Table 18 displays that the skewness of the tax evasion density decreased significantly, while the mean remained almost unchanged. The standard deviation is in line with the other industries.

Table 18: Summary statistics for the tax evasion density for industry 369 Non-Metallic Mineral Products. Trimming observations not in (0.05,0.7).

Mean	SD	Q1	Median	Q3	Skewness
0.075	0.012	0.067	0.075	0.083	0.166

9 Getting the Story Straight

Firms face incentives to evade sales taxes by overreporting their input costs. Firms overreport their cost by acquiring fake invoices to claim additional deductions of their sales taxes, maximizing after-tax profits. The higher the sales tax, the greater the incentive to evade. However, the probability of detection and the threat of penalties limit how much firms overreport.

The key incentive for firms is the **effective sales tax rate** — the amount firms ultimately pay or receive from the tax authority. The effective tax rate is calculated as the difference between the total sales tax charged by the firm on its sales and the total sales tax paid by the firm on its inputs, divided by total sales. A positive difference indicates what the firm should pay to the government. If the difference is negative, the firm receives a refund from the tax authority.

$$\begin{aligned}\tau &= \frac{\tau_0 P_t Y_{it} - \tau_1 \rho_t M_{it}}{P_t Y_{it}} \\ &= \tau_0 - \tau_1 \frac{\rho_t M_{it}}{P_t Y_{it}}\end{aligned}$$

In the data, I observe sales taxes paid on sales, $\tau_0 P_t Y_{it}$, and on purchases, $\tau_1 \rho_t M_{it}$. I can directly estimate the effective sales tax rate.

However, if firms are overreporting their inputs, this reported effective sales tax rate will be lower than the true effective sales tax rate,

$$\tau_0 - \tau_1 \frac{\rho_t M_{it}^*}{P_t Y_{it}} < \tau_0 - \tau_1 \frac{\rho_t M_{it}}{P_t Y_{it}}$$

I can use the estimated output elasticity of material inputs to get a corrected estimate of

the effective sales tax rate, since $\beta\mathcal{E} = \mathbb{E} \left[\frac{\rho_t M_{it}}{P_t Y_{it}} \right]$. Thus, I can estimate the true effective sales tax rate as,

$$\tau = \tau_0 - \tau_1 \beta\mathcal{E}$$

In Colombia, the total amount charged by the firm as sales tax depends on the statutory tax rate of each product sold domestically; **exports are exempt** from sales tax. Low statutory sales tax rates and exempt products result in lower sales tax collected by the firm and thus, a lower effective tax rate. Consequently, firms whose products are exempt from sales tax or taxed with **low statutory rates** have low incentives to evade. Likewise, as the share of their exports increases, exporter firms bear decreasing incentives to evade taxes because a larger portion of their sales is exempt from sales tax, reducing the effective sales tax rate.

On the other hand, the total amount paid by the firm as sales tax depends on the statutory tax rate of each of its inputs, including imports. **Imports**, in contrast to exports, are **not exempt**. Hence, the composition of the foreign and domestic inputs does not affect the effective sales tax rate, nor the incentives to evade.

However, the tax system imposed a zero sales tax rate on **unprocessed primary inputs**, such as forestry, mining, fishing, and agriculture. Therefore, firms consuming a high share of primary inputs face a higher effective sales tax rate and thus, greater incentives to evade.

Note that the effective sales tax rate might naturally be **negative** for firms in certain industries. Before the 1983 reform in Colombia, there were four different statutory sales tax rates: 4, 10, 15 and 25 percent. Because the sales tax was applied at the product level, *the total amount paid as sales tax by the firms on its inputs could be larger than the total*

amount charged by the firm as sales tax on its products. This situation would result in a negative effective sales tax rate. Firms in this situation would receive a refund from the tax authority.

Would firms with negative effective tax rates have incentives to overreport inputs to increase their refunds? Unlikely. Although the refunds had to be made within 30 days, *the procedure was complicated, and the tax authority delayed, on average, one year to process the payment.* Suspicion of fraudulent activities might further delay the process. Therefore, firms in this situation did not have incentives to overreport inputs to increase their refunds.

Whether firms in certain industries have incentives to overreport their inputs to evade sales taxes might **not** be **obvious** *ex-ante*. The tax code might be complex, with many exemptions and different statutory rates. Different taxes might generate overlapping or conflicting incentives. Available data might lack details on the products or inputs used by firms, making it difficult to identify the effective sales tax rate.

A first contribution of this paper is to provide practitioners and policymakers a way to identify industries where tax evasion through cost-overreporting might be non-trivial and can be identified from data. The simple model employed in this paper to investigate the mechanisms of tax evasion provides conditions that can be tested with data. As a result, I propose a statistical test where the null hypothesis states that firms in an industry of interest do not evade sales taxes by overreporting; the alternative states that firms do commit tax evasion Section [6.1](#).

Failing to reject the null hypothesis does not necessarily imply that firms in this industry are not overreporting. If a researcher suspects this practice on a particular industry, he would have to investigate if the assumptions of the model are not unreasonable for said

industry. The main assumption comes from the production function method, even though they might differ in their productivity, evaders and non-evaders share a common technology. With this in mind, in any dataset using SIC codes, it should be considered that the SIC codes group firms with similar products, but some categories might group products that require different technologies. Furthermore, most firms are multiproduct, and might be classified under a certain SIC code because it produces more of a certain product marginally.

9.1 Showing the Evidence

9.1.1 Overall tendencies

What makes sense:

- Downward sloping relationship between the average materials overreporting (test coefficient) and the sales tax rate of purchases (τ_1), reported ($\tau_1 \exp(s_{it})$) and corrected ($\tau_1 \beta$) sales taxes paid on purchases as a share of sales Figure 18, Figure 19.

As the share of purchases sales taxes increases, the effective tax rate decreases, reducing the incentives to evade. Thus, firms overreport less of their inputs. Alternatively, as the sales tax rate on purchases increases, the share of the firm's total cost that can be deducted from sales tax liability increases, reducing the effective tax rate, and thus, the incentives to evade. Therefore, firms overreport less of their inputs.

- When the effective sales tax rate is corrected using the output elasticity of materials inputs instead of the reported share of material, there is slight but positive relationship between the test coefficient and the effective sales tax rate. If left uncorrected, there seems to be no relationship between the test coefficient and the effective sales tax rate ?@fig-fig-test-effective.

Table 19: Intermediate elasticities for Corporations (Non-Evaders) and non-Corporations (Evaders) for different industries. Estimates were obtained using GNR (2020) method assuming a CD functional form. Intermediates are defined as raw materials. ε is defined as **measurement error**. The Tax Evasion Test displays the coefficient of the statistical test, where the null hypothesis states that there is no tax evasion through cost overreporting. Among the industry characteristics, Sales is the industry's market share of the total sales in Colombia between 1981 and 1991. Sales Tax Rate is the average sales tax reported rate. Exports is the average share of exports of the industry total sales, and Exporters is the share of firms exporting 10 percent or more of their total sales. Bias-corrected bootstrap confidence intervals at 95% significance level are displayed in brackets. *, **, *** denote significance at the 10%, 5%, and 1% levels (one-sided).

SIC	Elasticities		Test					
	Corps.	Others	Tax Ev.	N	N (Corps.)	Sales (Mkt %)	Sales (τ_0)	P
311	0.6	0.61	0.02*	888	147	23.6	0.2	
	[0.57, 0.62]	[0.59, 0.62]	[-0.01, 0.08]					
313	0.29	0.35	0.18***	130	74	9.7	8.4	
	[0.26, 0.3]	[0.32, 0.39]	[0.05, 0.32]					
321	0.39	0.44	0.13***	470	71	7.4	8.0	
	[0.36, 0.42]	[0.42, 0.45]	[0.09, 0.24]					
351	0.34	0.37	0.09	116	68	7.5	7.0	
	[0.28, 0.37]	[0.29, 0.43]	[-0.13, 0.33]					
352	0.41	0.44	0.06**	300	100	8.5	6.8	
	[0.38, 0.43]	[0.42, 0.47]	[-0.01, 0.16]					

Table 20: Intermediate elasticities for Corporations (Non-Evaders) and non-Corporations (Evaders) for different industries. Estimates were obtained using GNR (2020) method assuming a CD functional form. Intermediates are defined as raw materials. ε is defined as **measurement error**. The Tax Evasion Test displays the coefficient of the statistical test, where the null hypothesis states that there is no tax evasion through cost overreporting. Among the industry characteristics, Sales is the industry's market share of the total sales in Colombia between 1981 and 1991. Sales Tax Rate is the average sales tax reported rate. Exports is the average share of exports of the industry total sales, and Exporters is the share of firms exporting 10 percent or more of their total sales. Bias-corrected bootstrap confidence intervals at 95% significance level are displayed in brackets. *, **, *** denote significance at the 10%, 5%, and 1% levels (one-sided).

SIC	Elasticities		Test				
	Corps.	Others	Tax Ev.	N	N (Corps.)	Sales (Mkt %)	Sales (τ_0)
31 Food & Beverages							
311	0.6	0.61	0.02*	888	147	23.6	0.2
	[0.57, 0.62]	[0.59, 0.62]	[-0.01, 0.08]				
312	0.58	0.57	-0.01	188	41	5.3	0.2
	[0.49, 0.64]	[0.51, 0.63]	[-0.15, 0.14]				
313	0.29	0.35	0.18***	130	74	9.7	8.4
	[0.26, 0.3]	[0.32, 0.39]	[0.05, 0.32]				
32 Textile, Apparel & Leather							
321	0.39	0.44	0.13***	470	71	7.4	8.0
	[0.36, 0.42]	[0.42, 0.45]	[0.09, 0.24]				
322	0.35	0.42	0.2***	737	22	2.5	8.1
	[0.3, 0.4]	[0.4, 0.43]	[0.07, 0.32] 99				
323	0.56	0.47	-0.19	99	13	1.2	8.1
	[0.5, 0.58]	[0.45, 0.5]	[-0.26, -0.02]				

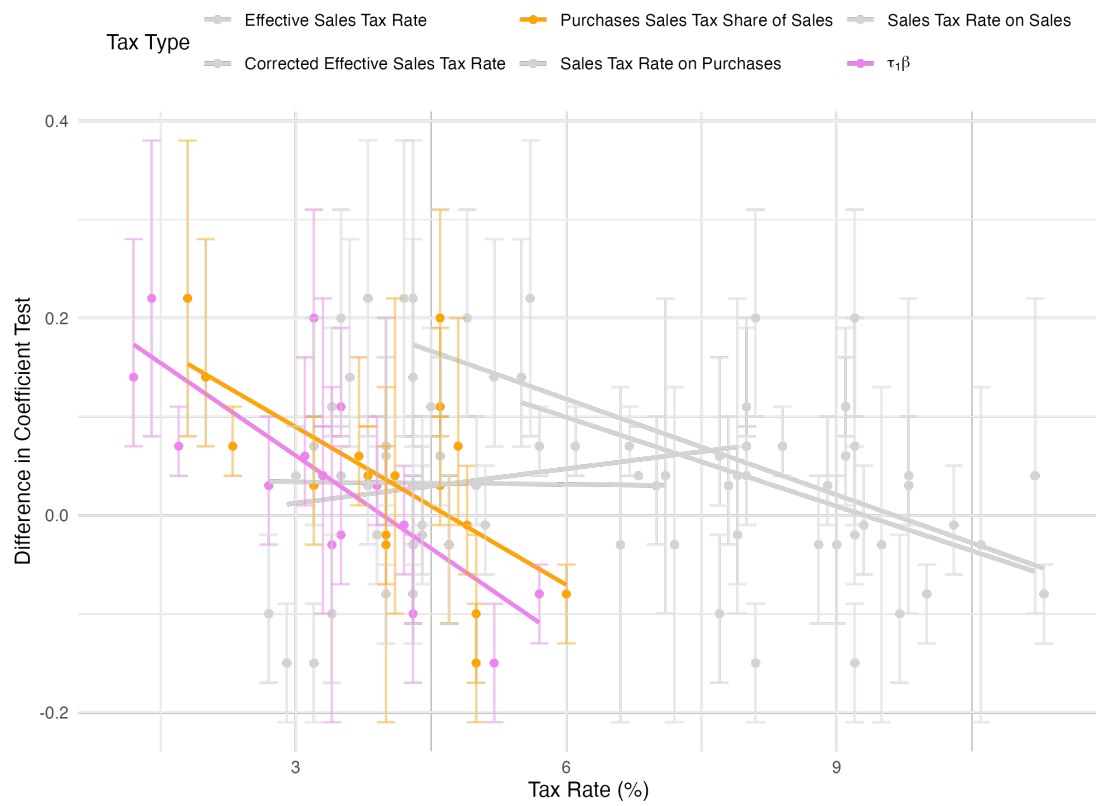


Figure 18: Test coefficient vs Tax Rates — Purchases Sales Taxes Share of Sales

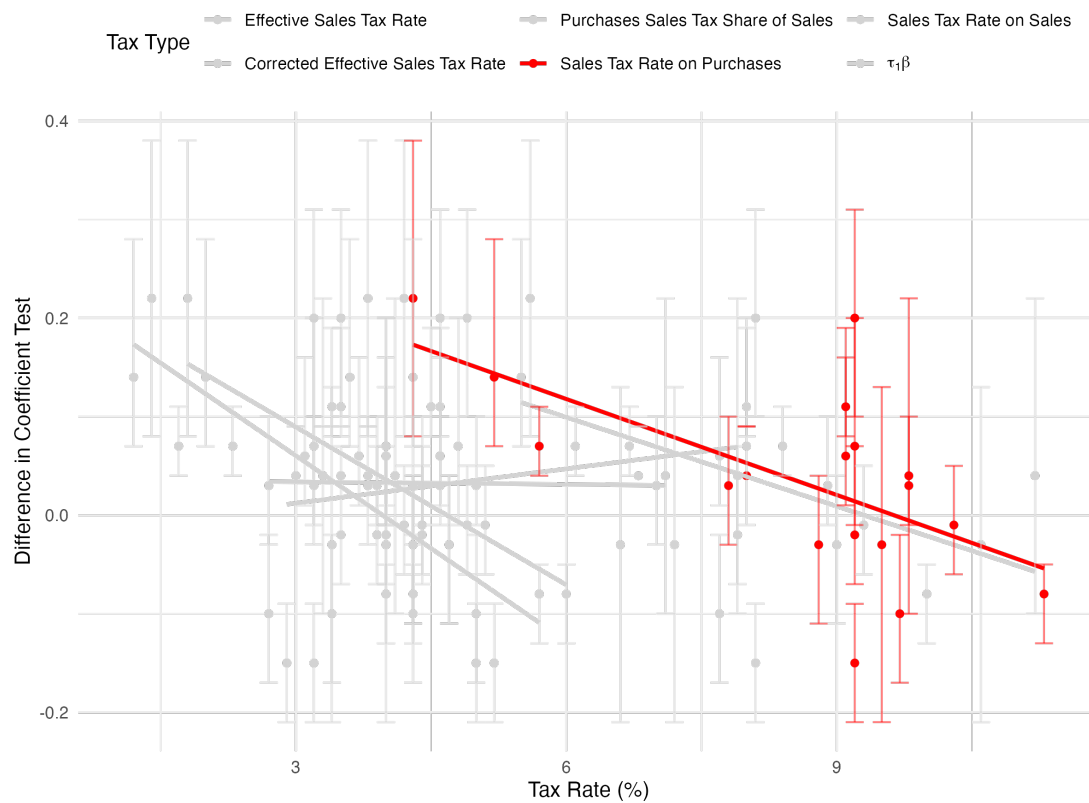


Figure 19: Test coefficient vs Tax Rates — Sales Tax Rate of Purchases

As the effective sales tax rate increases, the incentives to evade increase, leading firms to overreport more of their inputs. The flat relationship between the uncorrected effective sales tax rate and the overreporting might suggest a rule of thumb used by firms to decide how much to overreport based on the sales tax rate on sales. In other words, given the sales tax rate on sales, firms might have a target effective sales tax rate in mind, and overreport enough inputs to reach said target effective sales tax rate. Figure 20 reveals that most uncorrected effective sales tax rates are clustered between 3 and 5 %, while the corrected effective sales tax rates are more dispersed. These results should be taken with caution, as the relationship is weak, and there are few observations for high effective sales tax rates, that are not statistically significant.

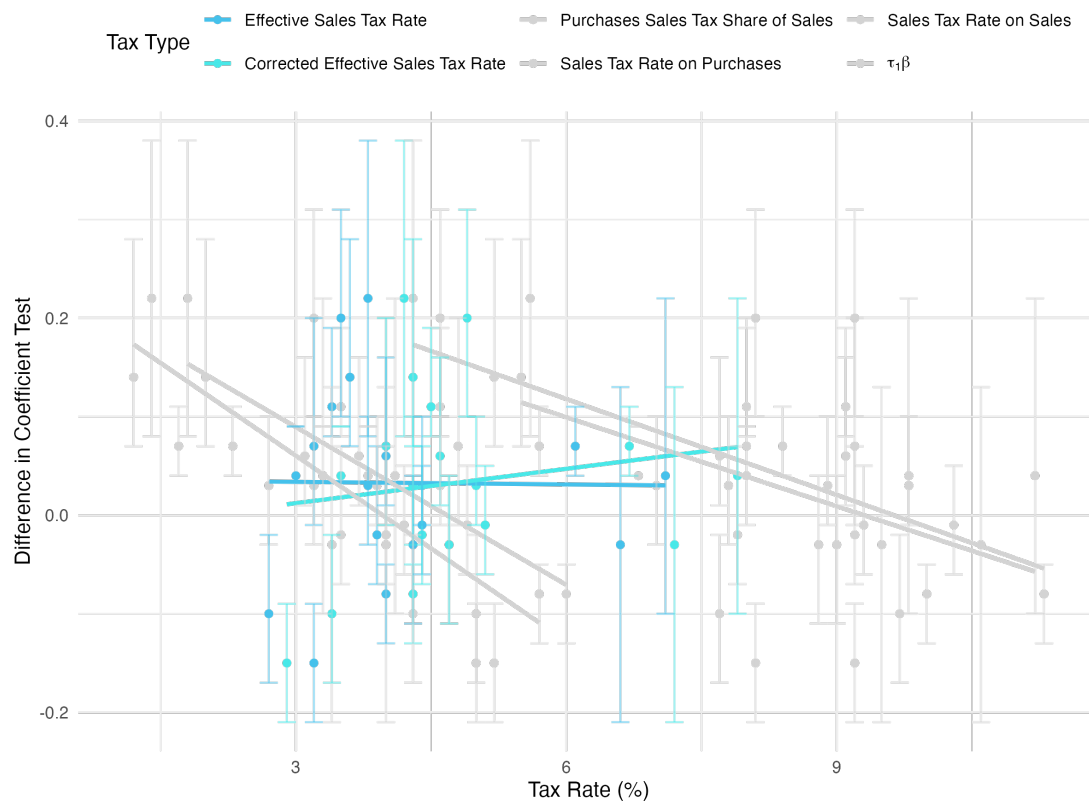


Figure 20: Test coefficient vs Tax Rates — Effective Sales Tax Rate

What does not make sense:

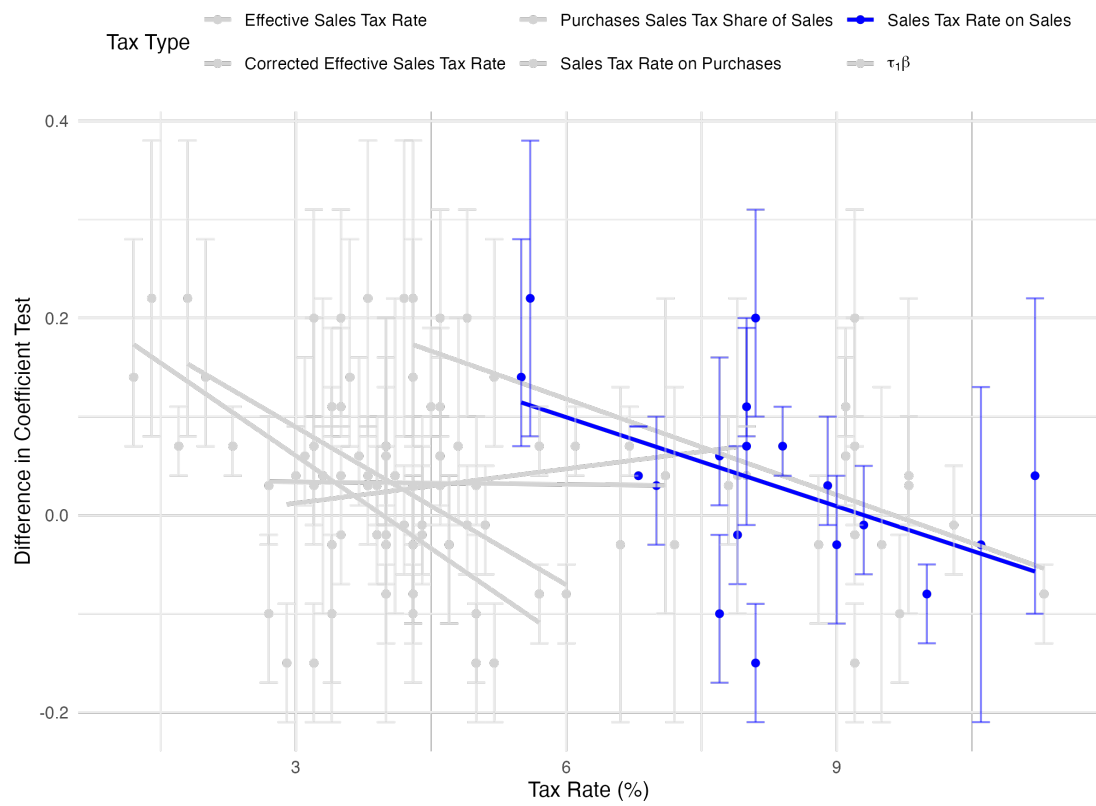


Figure 21: Test coefficient vs Tax Rates — Sales Tax Rate on Sales

- Sales tax rate of sales displays a negative relationship with the test coefficient.

This is counterintuitive, as higher sales tax rates lead to higher effective sales tax rates, increasing the incentives to evade, and thus, leading firms to overreport more of their inputs.

However

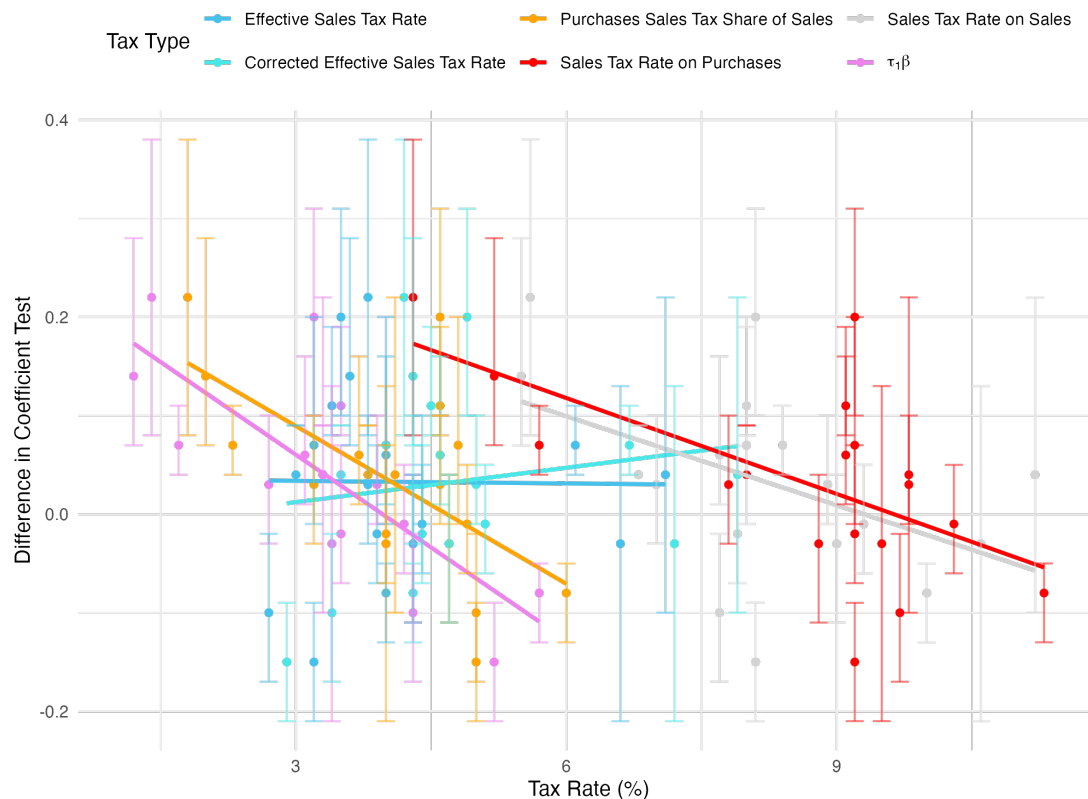


Figure 22: Test coefficient vs Tax Rates

9.1.2 Industry-level results

31 Food & Beverages:

- 311 Food Products (Meat, Dairy, Sugar, Bakery, etc.)
- 312 Food Products (Animal feeds, Others)
- 313 Beverages (Soft drinks, Alcoholic beverages, etc.)

In the Food and Beverages industry classification, industry 313 beverages displays sus-

picious activity of overreporting. This finding is consistent with the model because, in contrast to 311 and 312, industry 313 is not exempt from the sales tax. In fact, the effective sales tax rate for industries 311 and 312 is negative, implying that these firms were receiving refunds from the government. In comparison, industries in 313 industry would have higher incentives to overreport because of their positive effective sales tax.

32 Textile, Apparel & Leather:

- 321 Textiles
- 322 Wearing apparel
- 323 Leather and Fur Products
- 324 Footwear

Table 20 also shows evidence for potential tax evasion by overreporting in industries 321, 322, and 324. Again, from the theory perspective, these firms face positive effective tax rates, tax rates generate incentives to evade. This is consistent with the model. Corporations in industries 321, 322, and 324 display intermediate elasticities statistically smaller than the rest of the firms and the tax evasion test display a significantly positive coefficient.

What about industry 323? The effective sales tax rate is positive but does not display evidence of evasion according to the test, moreover the coefficient of the test is negative and statistically significant as indicated by the confidence intervals.

Most likely, it is not a good idea to compare corporations with the rest of the firms in this industry. First, looking at the sample size, we note that the number of corporations in this industry is small. While this is also true for industry 324, looking more closely, we can also note that industry 323 groups classifications 3231 (Leather Tanning and Finishing), 3232 (Fur dressing and Dyeing), and 3233 (Leather products, except footwear and wearing

apparel). In contrast, industry 324, only includes 3240 (Footwear, except plastic). The two most important groups of 323 are 3231 (30% of all the firms) and 3233 (60%). However, the composition of these subindustries is different between corporations (60/40) and non-corporations (22/73).

Although technologies between small and large firms were similar at the core, the processes were significantly different which raises the question on the validity of the comparison between corporations and the other firms. During this period, small firms were typically artisanal producers. These smaller establishments relied primarily on traditional local tanning methods based on vegetable extracts such as quebracho and chestnut. The process is long, typically requiring several weeks, however the products are highly valued because their durability and the distinctive patina it acquires as it ages. Their products were destined for the local market.

In contrast, large firms used a chrome-based mineral tanning, mechanized drums and automated finishing machinery. Chromium tanning is faster and allows for mass scale consistent quality. The product is softer and water resistant. These characteristics make chrome-based tanning leather suitable for exporting to the US and European markets. However, chromium disposal requires advanced waste water treatment necessary to remove heavy metals and prevent chromium discharge into the environment. Furthermore, these firms need to comply with strict environmental regulations including regular monitoring, hazard assessment, and reporting.

The data shows that corporations are more oriented to the foreign market than the rest of the firms, with highest average share of exports (28 vs. 13%), highest share of exporters (80 vs 35%).

33 Wood & Wood Products; 34 Paper & Printing:

- 331 Wood Products except furniture (Sawmills, Containers, etc.)
- 332 Wood Furniture (Wooden furniture and fixtures)
- 341 Paper and Paper Products
- 342 Printing, publishing and allied industries

In these categories, we find that industries 331 and 342 display suspicious activity of over-reporting. However, results should be taken with caution in particular for industry 331 because the sample of corporations is relatively small.

Again, failing to reject the null hypothesis of no tax evasion could only mean that cost overreporting cannot be identified with the data available using corporations as reference group.

35 Chemicals, Petroleum, Rubber & Plastic; 36 Non-Metallic Mineral Products:

- 351 Industrial Chemicals (Basic chemicals, Fertilizers, Synthetic resins and fibers)
- 352 Other Chemicals (Paints and lacquers, Drugs, Soap and cosmetics, etc.)
- 356 Other Plastic Products (not classified elsewhere)
- 369 Non-Metallic Mineral Products (Structural clay products, Cement and plaster, Other non-metallic mineral products)

I find significant evidence for industry 369 Non-Metallic Mineral Products, and some evidence for industry 352. Results should be taken with caution as both industries include different types of products, processes might vary significantly from one another. On the other hand, the sample of corporations is not small and there are observations across all subindustries, suggesting that the comparison between corporations and the rest of the firms might not be unreasonable.

Regarding industry 356, the perspicaz reader might notice that the coefficient of the tax evasion test is negative and statistically significant. However, in this category is not clear the composition of the subcategories included, as it covers other plastic products not classified elsewhere. Unlike other categories, this one only includes one subcategory, 3560 (Plastic products, not elsewhere classified). As in other industries, corporations have higher average share of exports and their share of exporters is higher than the non-incorporated firms. Does this matter? Likely. How? It is not clear.

38 Fabricated Metal Products, Machinery & Equipment; 39 Other Manufacturing:

- 381 Fabricated Metal Products (Structural metal products, Cutlery, Hand tools, etc.)
- 382 Machinery and Equipment (Agricultural machinery, Metalworking machinery, etc.)
- 383 Electrical Machinery and Apparatus (Electric motors, Generators, etc.)
- 384 Transport Equipment (Motor vehicles, Aircraft, Ships, etc.)
- 390 Other Manufacturing (Jewelry, Musical instruments, etc.)

I cannot reject the null hypothesis of no tax evasion by overreporting for these industries. It is interesting to note that agricultural machinery, transportation equipment, and equipment were exempt from sales tax since 1974. In 1984, exemptions for agricultural machinery and transportation equipment were removed.

If these results are valid, this could imply that there is a dynamic component to the tax evasion decision. Meaning that the decision of tax evasion in the current period depends critically on the past period. For example, if a firm did not evade in the previous year, it might be less likely to evade in the current year. The fact that there is no evidence of tax evasion might suggest that this effect is persistent over time.

How does this dynamic decision could enter the economic model? Firms in the agricultural machinery and transportation equipment sectors started facing incentives to evade after 1984. However, they have been providing truthful information to the tax authority for over a decade. They know that the probability of detection for them is high, given that the authority can easily compare their records with previous years and detect anomalies.

9.1.3 Results Summary

I find that four out of the top five (nine of the to top twenty) industries display activities consistent with tax evasion through cost overreporting in Colombia between 1981 and 1991. Firms might overreport up to 22 percent of their costs. Consisten with the model, firms in industries with exempt products and negative effective sales tax rates display minor or no evidence of tax evasion. Furthermore, the evidence could suggest that the tax evasion decision is dynamic, as there is a lack of evidence of tax evasion in industries that lost exemption after a decade.

Of course, the results should be taken with the usual precautions. The main assumption of the model is that evaders and non-evaders share a common technology. This assumption might be not unreasonable in most cases. However, commonly available data might group firms with similar products, but the technological processes might differ.

On the other hand, failing to reject the null hypothesis of no tax evasion does not necessarily mean that firms in this industry are not overreporting; it might mean that overreporting cannot be identified with the data available at this industry level using corporations as the reference group. A researcher suspecting tax evasion in a particular industry should exhaustively investigate if the assumptions of the model are not unreasonable for said industry. In particular, she should inform herself about the industry technological processes

[illegible]

10 Non-random Auditing

Firms might believe however that their probability of being detected evading taxes is increasing in their size. In other words, the larger the firm, the more likely it gets audited by the tax authority. This scenario corresponds to a strategic audit, as opposed to a random

Table 21: Comparison of different methods to estimate the bias corrected confidence intervals and the p-values. The null hypothesis states that firms do not commit tax evasion through cost-overreporting. The alternative is that they do. The first column is a difference of the means test (includes error of the estimator, but assumes independence of the samples). The different methods shown in the rest of the columns are combinations of two different procedures when sampling the bootstrap —sampling from corporations and non-corporations, or sampling full data— and two different methods to estimate the coefficient of the test — using all the firms or only the non-incorporated firms. When not fixing the number of corporations, Bayesian sampling was used to ensure there would be at least one corporation needed to estimate the first-stage parameter. The numbers in parenthesis are the standard errors of the mean difference test. The numbers in brackets represent the bias-corrected bootstrap confidence intervals at 95% significance level. *, **, *** denote significance at the 10%, 5%, and 1% levels (one-sided).

SIC	Mean Diff.	Bootstrap			
		Two Group Sampling		Bayesian	
		All	Others	All	Others
311	0.02**	0.02*	0.02*	0.02	0.02
	(0.011)	[-0.01, 0.06]	[-0.01, 0.08]	[-0.03, 0.07]	[-0.04, 0.08]
312	-0.01	-0.01	-0.01	-0.01	-0.01
	(0.034)	[-0.09, 0.11]	[-0.15, 0.14]	[-0.16, 0.12]	[-0.2, 0.15]
313	0.18***	0.07***	0.18***	0.07**	0.18***
	(0.026)	[0.03, 0.12]	[0.05, 0.32]	[0.01, 0.12]	[0.03, 0.34]
321	0.13***	0.11***	0.13***	0.11***	0.13***
	(0.018)	[0.06, 0.19]	[0.09, 0.24]	[0.02, 0.18]	[0.02, 0.25]
322	0.2***	0.2***	0.2*** 111	0.2**	0.2**
	(0.039)	[0.06, 0.32]	[0.07, 0.32]	[0.03, 0.35]	[-0.03, 0.36]
323	0.19	0.15	0.19	0.15	0.19

Table 22: Comparison of industry characteristics between corporations and non-corporations.

sic_3	n_Corp	n_Non-Corp	avg_age_Corp	avg_age_Non-Corp	sales_sales_tax_rate_Corp
311	147	783	27	20	0.5
312	41	161	26	21	0.2
313	74	61	31	31	4.5
321	71	422	26	17	8.2
322	22	724	24	14	7.9
323	13	90	35	17	6.5
324	9	186	35	14	8.1
331	13	160	21	18	6.6
332	12	187	25	15	9.5
341	41	119	26	20	8.7
342	40	324	25	22	5.1
351	68	55	24	17	7.3
352	100	216	30	24	7.1
356	53	241	20	15	9.6
369	58	223	29	19	4.9
381	85	569	24	17	9.0
382	46	330	26	20	8.1
383	57	168	23	18	9.3
384	42	209	26	17	8.7
390	18	163	24	21	9.6

Table 23: Rejection frequencies of the $\mathcal{V}_{it}$ for Corporations and Non-Corporations. The rejection frequencies of the one tail test are the average number of observations that fell above the 95% region of the $f_{\varepsilon}(\varepsilon)$ distribution recovered in the first stage. The rejection frequencies of the two tail test are the average number of observations that fell above the 97.5% or below the 2.5% region.

SIC	One Tail		Two Tail	
	Corp	Non-Corp	Corp	Non-Corp
311	5.1	9.3	5.1	8.7
312	5.1	3.2	5.1	6.8
322	4.4	26.3	3.5	8.9
381	4.9	4.6	4.9	6.3
321	4.9	21.4	5.1	12.7
382	5.0	5.2	5.0	5.4
384	5.0	0.6	4.7	4.3
383	4.9	2.6	5.1	2.9
313	5.0	17.9	5.0	12.9
341	5.0	4.4	5.0	8.1
324	5.1	10.4	5.1	11.2
342	5.1	6.3	5.1	5.9
369	5.1	12.5	4.8	12.8
332	5.3	4.3	5.3	8.5
390	4.4	1.1	4.4	1.0
351	4.8	11.2	4.8	10.5
352	4.9	12.1	5.0	9.9
356	5.2	3.9	5.2	8.6
331	5.8	26.1	4.8	22.7
323	4.7	1.5	3.8	12.6

audit.

What matters for firms is what they believe about their audit probability. We can model the probability of detection as $q(e_{it}, M_{it})$. So, now the probability of detection depends not only on the firm's overreporting of inputs e_{it} , but also on the firm's level of true intermediate inputs, M_{it} .

Other measures of size may be preferred, such as revenues, because that is a popular way for tax authorities to classify firms for taxing purposes. I use intermediate inputs because they are directly related to the firm's size in terms of output, revenues, or market share and to simplify the identification of the model.

Then, the problem of the firms becomes,

$$\begin{aligned}
\max_{K_{it}, L_{it}, M_{it}, e_{it}} & (1 - q(e_{it}, M_{it})) \left[P_t \mathbb{E}_\varepsilon[Y_{it}] - \rho_t M_{it} - r_t K_{it} - w_t L_{it} \right. \\
& \left. - \tau (P_t \mathbb{E}_\varepsilon[Y_{it}] - \rho_t (M_{it} + e_{it})) - \kappa_i(e_{it}) \right] \\
& + q(e_{it}, M_{it}) \left[(1 - \tau)(P_t \mathbb{E}_\varepsilon[Y_{it}] - \rho_t M_{it}) - r_t K_{it} - w_t L_{it} \right. \\
& \left. - \kappa_i(e_{it}) - \phi \rho_t \tau e_{it} \right] \\
\text{s.t. } & Y_{it} = K_{it}^{\alpha_K} L_{it}^{\alpha_L} M_{it}^\beta \exp(\omega_{it} + \varepsilon_{it})
\end{aligned} \tag{42}$$

The first order conditions render the following equations,

$$\beta K_{it}^{\alpha_K} L_{it}^{\alpha_L} M_{it}^{\beta-1} \exp(\omega_{it}) \mathcal{E} = \frac{\rho_t}{P_t} \left(1 + \frac{\tau}{(1 - \tau)} (1 + \phi) \frac{\partial q(e_{it}, M_{it})}{\partial M_{it}} e_{it} \right) \tag{43}$$

$$\rho_t \tau \left(1 - (1 + \phi) \left(q(e_{it}, M_{it}) + \frac{\partial q(e_{it}, M_{it})}{\partial e_{it}} e_{it} \right) \right) = \kappa'_i(e_{it}) \tag{44}$$

We can still manipulate Equation 43 to obtain an equation in terms of the output elasticity of intermediate inputs as in Gandhi et al. (2020),

$$\beta \mathcal{E} \exp\{-\varepsilon_{it}\} = \frac{\rho_t M_{it}}{P_t Y_{it}} \left(1 + \frac{\tau}{(1-\tau)} (1+\phi) \frac{\partial q(e_{it}, M_{it})}{\partial M_{it}} e_{it} \right) \quad (45)$$

As in the base case, Equation 44 will not hold for Corporations, for any positive $e_{it} > 0$. Consequently, Corporations will choose not to evade taxes, $e_{it} = 0$. This implies that for these firms, Equation 44 becomes,

$$\rho_t \tau \leq \kappa'_i(0) \quad (46)$$

Likewise, for these firms Equation 45 becomes,

$$\beta \mathcal{E} \exp\{-\varepsilon_{it}\} = \frac{\rho_t M_{it}}{P_t Y_{it}} \quad (47)$$

Now, I have four equations. I can try to parametrize $q(e_{it}, M_{it})$, $\kappa_i(e_{it})$, normalize $\phi = 0$, and estimate the parameters of this function along with the other parameters of the model.

First, the probability of detection needs to have the following properties, $q(0, M_{it}) = 0$, $q(\infty, M_{it}) = 1$, $\frac{\partial q(e_{it}, M_{it})}{\partial e_{it}} > 0$, and $\frac{\partial q(e_{it}, M_{it})}{\partial M_{it}} > 0$.

One possible functional form that satisfies these properties is, $q(e_{it}, M_{it}) = 1 - \exp\{-\lambda_0 e_{it} - \mathbb{1}[e_{it} > 0] \lambda_1 M_{it}\}$, with $\lambda_0, \lambda_1 > 0$. Then, for the evaders the derivatives are given by, $\frac{\partial q(e_{it}, M_{it})}{\partial e_{it}} = \lambda_0 \exp\{-\lambda_0 e_{it} - \lambda_1 M_{it}\}$, and $\frac{\partial q(e_{it}, M_{it})}{\partial M_{it}} = \lambda_1 \exp\{-\lambda_0 e_{it} - \lambda_1 M_{it}\}$.

Likewise, the cost of evasion function needs to satisfy the following properties, $\kappa_i(0) = 0$, $\kappa'_i(e_{it}) > 0$. In addition, I need to allow for Corporations to have larger costs of evasion

than non-Corporations. One possible functional form that satisfies these properties is,

$$\kappa_i(e_{it}) = (\gamma_0 + \gamma_1 \mathbb{1}[i \in \text{Corp}])e_{it}$$

Then, the derivative is given by, $\kappa'_i(e_{it}) = \gamma_0 + \gamma_1 \mathbb{1}[i \in \text{Corp}]$.

With these functional forms, I can rewrite the first order conditions as moments to estimate the parameters $\lambda_0, \lambda_1, \gamma_0, \gamma_1$ along with the other parameters of the model, $\alpha_K, \alpha_L, \beta$, and the densities of the error term $f_\varepsilon(\varepsilon)$ and of overreporting $f_e(e)$.

The four equations are given by,

$$\beta \mathcal{E} \exp\{-\varepsilon_{it}\} = \frac{\rho_t M_{it}}{P_t Y_{it}} \quad (48)$$

$$\rho_t \tau \leq \gamma_0 + \gamma_1 \quad (49)$$

for the non-evaders, and

$$\beta \mathcal{E} \exp\{-\varepsilon_{it}\} = \frac{\rho_t M_{it}}{P_t Y_{it}} \left(1 + \frac{\tau}{(1 - \tau)} \lambda_1 \exp\{-\lambda_0 e_{it} - \lambda_1 M_{it}\} e_{it} \right) \quad (50)$$

$$\rho_t \tau (1 - (1 - \exp\{-\lambda_0 e_{it} - \lambda_1 M_{it}\} + \lambda_0 \exp\{-\lambda_0 e_{it} - \lambda_1 M_{it}\} e_{it})) = \gamma_0 \quad (51)$$

for the evaders, where $e_{it} > 0$.

10.1 Implementation

The challenge again is that as in the base model, e_{it} and ε_{it} are unobserved. Similarly, what we observe in the data is $M_{it}^* = M_{it} + e_{it}$, the reported intermediate inputs.

I can start with the non-evaders. For these firms, reported intermediate inputs are the true inputs. By taking logs, I can use Equation 48 to identify β and $f_\varepsilon(\varepsilon)$ as in the baseline model.

$$\ln\left(\frac{\rho_t M_{it}}{P_t Y_{it}}\right) = \ln(\beta) + \ln(\mathcal{E}) - \varepsilon_{it} \quad (52)$$

Next, I can use Equation 49 to identify γ_0 and γ_1 .

Here I have several options depending how τ is defined. In the model, τ is the sales tax rate, and raw materials can be deducted at the same rate. However, in Colombia firms might face two different sales tax rates, one of sales, τ_0 , and a different for purchases, τ_1 . In the data, I observe paid sales taxes of sales, $\tau_0 P_t Y_{it}$, and purchases, $\tau_1 \rho_t M_{it}$.

What matters for the firm's decision to evade is the effective tax rate, which is given by, $\tau_{it} = \frac{\tau_0 P_t Y_{it} - \tau_1 \rho_t M_{it}}{P_t Y_{it}} = \tau_0 - \tau_1 \frac{\rho_t M_{it}}{P_t Y_{it}}$. The intermediates cost share of revenue is $\frac{\rho_t M_{it}}{P_t Y_{it}} = \beta \mathcal{E}$, so the effective tax rate can be rewritten as,

$$\tau_{it} = \tau_0 - \tau_1 \beta \mathcal{E} \quad (53)$$

where the subscript it indicates that there would be an estimate for each firm-year.

Then, I can rewrite Equation 49 as,

$$\tau_{it} \rho_t \leq \tilde{\gamma} \quad (54)$$

where $\tilde{\gamma} = \gamma_0 + \gamma_1$.

For the evaders, what I observe in the data is $M_{it}^* = M_{it} + e_{it}$. I can rewrite Equation 50 and Equation 51 substituting $e_{it} = M_{it}^* - M_{it}$, using Equation 53, and taking logs as,

$$\ln \left(\frac{\rho_t M_{it}}{P_t Y_{it}} \right) = \ln(\beta \mathcal{E}) - \ln \left(1 + \frac{\tau_{it}}{(1 - \tau_{it})} \lambda_1 \exp\{-\lambda_0(M_{it}^* - M_{it}) - \lambda_1 M_{it}\} (M_{it}^* - M_{it}) \right) - \varepsilon_{it} \quad (55)$$

$$\ln(\tau_{it} \rho_t) = \ln(\gamma_0) + \lambda_0(M_{it}^* - M_{it}) + \lambda_1(M_{it}) - \ln(1 - \lambda_0(M_{it}^* - M_{it})) \quad (56)$$

How to estimate? Decovluting Equation 55 by Non-Parametric MLE, using β and the distribution of ε_{it} from the non-evaders, subject to Equation 56 and Equation 54.

$$\begin{aligned} & \max_{\lambda_0, \lambda_1, \gamma_0, \gamma_1, f_{M|M^*}(M|M^*)} \sum_{i,t: e_{it} > 0} \ln \left(\int \hat{f}_\varepsilon \left(\ln \left(\frac{\rho_t M_{it}}{P_t Y_{it}} \right) + \ln \left(1 + \frac{\tau_{it}}{(1 - \tau_{it})} \lambda_1 \exp\{-\lambda_0(M_{it}^* - M_{it}) - \lambda_1 M_{it}\} \right) \right) f_\varepsilon \right) \\ & \text{s.t. } \mathbb{E}_{M_{it}^*} \left[\ln(\tau_{it} \rho_t) - \ln(\gamma_0) - \int \left(\lambda_0(M_{it}^* - M_{it}) - \lambda_1(M_{it}) + \ln(1 - \lambda_0(M_{it}^* - M_{it})) \right) f_\varepsilon \right] \\ & \mathbb{E}[\tau_{it} \rho_t - \tilde{\gamma}] \leq 0 \end{aligned}$$

11 Tax Evasion and Resource Missallocation

A popular way to decompose total productivity introduced by Olley et al. (1996) is the following :

$$\varphi_t = \bar{\varphi}_t + \sum_{i=1}^I (s_{it} - \bar{s}_t) (\varphi_{it} - \bar{\varphi}_t)$$

where $\varphi_{it} = \exp\{\omega_{it} + \varepsilon_{it}\}$ is the firm-level productivity, s_{it} is firm' i share of output at time t . $\bar{\varphi}_t$ and $\bar{s}_t$ represent the unweighted mean productivity and unweighted mean share.

Intuitively, the more output is allocated to high productivity firms the higher the total productivity will be. Alternatively, the smaller the shares allocated to low productivity firms the greater the total productivity.

Ideally, we would like resources to be allocated to high productivity firms. Hence, the link between the decomposition of total aggregate productivity and missallocation.

In the tax evasion context, the productivity of misreporting firms would look lower than it really is. How would this affect our measures of total productivity? If misreporting firms have small shares, their corrected productivity measure might be closer to the mean reducing their contribution to total productivity, leading to lower measured levels of total productivity.

In other words, tax evasion through cost overreporting might lead to biased measures of total aggregate productivity.

However, if we think about the marginal firm that is struggling to stay in the market, the additional income gained through misreporting might help her survive longer. Therefore, tax evasion increases missallocation in the sense that a share of the output is allocated to firms that otherwise would not be in the market.

There are alternative ways to decompose total productivity that account for the entry and exit of firms (Foster et al. 2008). In addition, alternative measures to aggregate productivity have been used elsewhere (e.g. aggregate demand shocks in [Eslava et al. \(2004\)](#)).

For example, we can decompose aggregate productivity like this

$$\Delta\varphi_t = \sum_{i \in C} \bar{s}_i \Delta\varphi_{it} + \sum_{i \in C} (\bar{\varphi}_i - \bar{\varphi}) \Delta s_{it} + \sum_{i \in N} s_{it} (\varphi_{it} - \bar{\varphi}) - \sum_{i \in X} s_{it-1} (\varphi_{it-1} - \bar{\varphi})$$

where Δ is the change from $t - 1$ to t , and the sets C, N , and X represent the continuing, entering, and exiting firms.

In the case of corporate tax evasion a relevant measure could be aggregate profitability. Low productivity firms are surviving not because they are technologically efficient but because their profitability is marginally higher when they evade taxes. Decomposing an aggregate profitability measure by market shares and firm dynamics would indicate how exiting firms are contributing to the aggregate measure. Intuitively, again higher levels of aggregate profitability correspond to larger shares of the market being allocated to high profitability firms.

In the absence of tax evasion, firms overreporting costs would have lower after-tax profits. The aggregate profitability level would be higher if these firms have small market shares and remain in the sample. On the other hand, the aggregate profitability level would decrease if the small-share and low-profitability firms were to exit the market.

Therefore, in the case of tax evasion, the key aspect of aggregate profitability is the counterfactual decision of low-profitability firms to exit the market in addition to the corrected profits if firms were not to overreport their costs.

Consequently, the economic model should account for firm entry and exit along with tax evasion.

11.1 Implementation

Given I can only obtain the distribution of productivity non-parametrically, one way to recover the covariance term between productivity and size is the following:

$$Cov(\varphi_{it}, s_{it}) = E[\varphi_{it}s_{it}] - E[\varphi_{it}]E[s_{it}]$$

Clearly, I can estimate the expectation of φ_{it} and s_{it} using the estimated non-parametric distribution. While, I can directly compute the expectation of size from data.

For the joint expectation, All I need is the conditional distribution of φ_{it} given s_{it} . This can be estimated non-parametrically dividing size in different groups, and then estimating the productivity within each group.

To see this, note that

$$\begin{aligned} E[\varphi_{it}s_{it}] &= \int \int \varphi_{it}s_{it}f(\varphi_{it}, s_{it})d\varphi_{it}ds_{it} \\ &= \int \left(\int \varphi_{it}s_{it}f(\varphi_{it}|s_{it})d\varphi_{it} \right) f(s_{it})ds_{it} \\ &= E_{s_{it}} \left[\left(\int \varphi_{it}s_{it}f(\varphi_{it}|s_{it})d\varphi_{it} \right) \right] \end{aligned}$$

12 Lit Review Notes

For personal reference, to be removed from the final version.

12.1 [Dabla-Norris et al. \(2019\)](#)

Productivity improvements by firms can causally lead to lower tax evasion, i.e., underreporting sales for tax purposes. In other words, an increase in firm productivity leads to a higher share of sales reported for tax purposes. The authors use the self-reported share of declared income as a proxy for tax evasion.

Limitations: Using the World Bank Enterprise (47 emerging and developing countries) survey question to get the percentage of sales that are reported for tax purposes, which is

used to measure tax evasion. In addition, productivity is measured as the log of sales per worker. No access to quantities data, they use LP but with sales and workers.

Undisputed (?) facts: 1) Tax evasion is higher in poor countries; 2) Tax evading firms tend to be less productive (?).

Productivity gaps between firms that comply with taxes and regulations and those which do not are significant, (between 25 and 50 %) (Amin et al. 2019; Fajnzylber, 2011; Busso et al., 2012). These large productivity gaps can translate into low-economy wide productivity and growth: 1) if, by not complying with existing taxes and regulations, firms enjoy a potentially large implicit subsidy that allows them to stay in business despite low productivity, or 2) to expand their market share at the expense of more productive firms (Farrell, 2004; Bobbio, 2016).

Almeida and Carneiro (2009): Tax evasion may also increase firms' efficiency (productivity?). Avoiding onerous taxes might confer firms more flexibility in their employment and production decisions.

Kleven et. al (2016): Larger firms are less likely to avoid taxes. The firm's use of business records increases as it hires more employees. They show that the government may be able to enforce higher tax compliance even in the presence of a low threat of audit due to the increased ease of whistleblowing as the firm grows.

Tax evasion is hidden: it is difficult to obtain precise measures (Kundt et al., 2017; Slemrod and Weber, 2012).

Loayza and Rigolini (2006) and La Porta and Shleifer (2014): informality is larger in lower GDPpC countries

Tax evasion and informality are associated with lower income levels and productivity

(Dabla-Norris and Feltenstein, 2003; Loayza, 1996; Sarte, 2000)

12.2 Paulus (2015)

The author models discrepancies between tax records and income survey data as individual tax evasion and measurement error jointly. Using data for Estonia, the author finds that people in the bottom and the top part of earnings distribution evade much more and about 12% of wages and salaries in total are underreported. The identifying assumption, i.e., location condition, is that people working in the public sector cannot evade taxes. In other words, measurement error is unrelated to the sector where the individual works while the same does not hold for tax compliance.

The main constraint for empirical research on tax evasion is the lack of suitable data. This is worse for developing countries. A measure of undeclared income for individuals is needed. This kind of data is usually unreliable and very difficult/or expensive to obtain.

Baldini et al. (2009) use linked income data with tax records at the individual level to estimate tax evasion. However, the author assumes people report their true income in the survey to obtain a measure of non-reporting. Furthermore, in the measurement error literature, several studies assume the opposite, e.g. Krueger (1991) and Bollinger(1998); survey of lit Chen et al. (2011) and Bound et al. (2001). Linked data is referred to as Validation data in the measurement error literature. Later studies have started to consider matching errors (Kapteyn and Ypma, 2007; Meijer et al., 2012) and errors in register data (Abowd and Stinson, 2013).

Pissarides and Weber (1989) pioneered a methodology to study underreported income. They assume that the underreported income for one population group is inferred from a comparison with a reference group assumed to have negligible non-compliance but to be

similar in other respects. These types of studies assume implicitly that the underreporting of income in a survey corresponds to the underreporting of income to the tax authority.

12.3 Slemrod (2019)

Tax evasion policies have been a greater priority in developed countries since the 2008 Great Recession (IMF, 2015). For example, the UK announced efforts to combat tax avoidance, tax evasion, and ‘imbalances’ in the tax system that would bring in £5 billion in additional revenue each year.

Tax evasion matters because it affects both the resource cost of raising taxes and the distribution of the tax burden, i.e. core interests of public economics. Policy question? Does curbing evasion improve the equity and efficiency of the public finances, given the cost of doing so? If so, what’s the best way to go about it?

Tax evasion creates horizontal inequity because people evade taxes by different proportionate amounts because of differences in personal characteristics —risk aversion, tax system, and honesty— and because of different opportunities and potential rewards for evasion. Likewise, efforts to combat tax evasion might create vertical inequity. For example, going after lower-income households such as the IRS fighting fraud related to the earned income tax credit (EITC) than going after unreported foreign accounts that are more likely to be held by high-income households.

Tax evasion creates efficiency costs, e.g., the individual’s cost of implementing and hiding non-compliance, the tax authority’s cost of combating non-compliance, etc. Tax evasion also creates socially inefficient incentives to engage in activities that facilitate tax evasion. For example, tax-noncompliance opportunities attract people to self-employment who would otherwise be employees.

Tax evasion creates misallocations. For example, it can cause companies that otherwise would not find it attractive to do so to set up operations in a tax haven to facilitate or camouflage evasion.

Curtailing tax evasion cost shapes optimal policy—it is not optimal to completely eradicate tax evasion because of the cost of doing so. The presence of evasion also alters the optimal setting of tax rates—because it affects the marginal efficiency cost of so doing—and the choice of the tax base—because different tax bases are more or less susceptible to evasion.

Slemrod and Weber (2012) review macroeconomic measures of the informal economy.

The standard tax evasion framework is the model of tax deterrence by Allingham and Sandmo (1972), who adapted Becker’s (1968) model of criminal behavior to tax evasion. A relatively recent simplified framework is where the maximand is $y(1 - t) + te - c(e, \alpha)$, y is income, e is the understated tax liability, t is a linear tax rate, and c is the private cost of evasion which includes the utility cost of bearing risk and the expected value of punishment. The private cost may depend on the amount of attempted evasion and α , the vector of enforcement instruments, such as the extent of auditing.

The model predicts that a risk-averse agent will evade taxes if the expected return is positive. Additionally, an increase in either the probability of detection or the penalty if detected will reduce evasion. Importantly, the effect of the tax rate is less clear, for even its sign depends on the penalty function (Yitzhaki, 1974).

What matters about p is the perception of taxpayers. Much work is focused on what alters the taxpayer’s perception of the enforcement environment.

This framework naturally applies to individuals, and small, single-owned firms. Large firms might be risk-neutral rather than risk-averse.

The challenge of measuring tax evasion is the incentive of evaders to conceal their behavior. Many taxpayers are unwilling to respond accurately maybe because of the threat of punishment or social shame. For this reason, even the most credible empirical studies of evasion don't actually have a reliable measure of evasion.

Tax gap studies by auditing (real differences between countries' tax gaps or differences in methodologies?):

- U.S. Overall gross tax-gap average over 2008-10 was \$458 billion USD, 18.3% of estimated tax liability (IRS, 2016). IRS National Research Program (NRP).

- UK: overall tax gap from 2013-2014 6.4% of true liability: 6.4% for the corporation income tax 5.0% individual income tax, and 11.1% value-added tax (VAT). Small- and medium-sized enterprises account for over half of the overall tax gap (HMRC, 2015). Her Majesty's Revenue and Customs (HMRC, 2015) Toro et al. (2013).

- Denmark's overall tax gap is only 2.2 percent of net income (Kleven et al., 2011).

- OECD(2015) other countries.

Sharp variations in the compliance rate by the extent of information reporting to the tax authority provide compelling evidence that the probability of successful evasion play a key role in the deterrence of tax evasion. For example, in the US, the estimated non-compliance rate when there is little to no third-party-reported information (e.g., self-employment income) is 63%, compared to 19% and 7% for 'some' and 'substantial' information reporting, respectively. Small businesses represent a large portion of the tax gap in individual income with approximately 47% of underreporting. In contrast, only 1 % of non-compliance is estimated for employee wages and salaries —substantial information reporting. Kleven et al. (2011) find a similar disparity, but lower, non-compliance rates in Denmark, where 14.9

% for self-employment income but less than 1% for third-party reported items. In other words, when the information reporting renders the likelihood of evading taxes successfully near zero, evasion is eradicated; when the likelihood is higher, evasion proliferates.

Administrative data: Canada, UK, Brazil, Chile, China, Costa Rica, Ecuador, India, Pakistan, Rwanda, Tunisia, and Uganda. [US access](#)

RCT's review: Hallsworth (2014)

Bérgolo et al. (2018) RCT with firms using letters. Value-added-tax compliance, 20,000 Uruguayan small and medium-sized firms. The authors find that adding to the baseline letter a paragraph with statistics about the probability of being audited and the penalty rates increased tax compliance by about 6.3 percent (about one-fourth of its current level), and adding a paragraph that informs firms that evading taxes increased the probability of being audited increased tax compliance by about 7.4 percent.

Ortega and Scartascini (2018) Colombia, individuals. Method of communication. The authors find sizable differences across delivery methods. Personal visits by a tax inspector are more effective than the impersonal methods; they are, alas, also much more expensive. In a follow-up study, Ortega and Scartascini (2015) find that the effect of phone calls falls between those of the impersonal methods and the personal visits.

Almunia and Lopez-Rodriguez (2018) study the behavior of Spanish firms in response to a notch in enforcement intensity that arises because the Spanish Large Taxpayer's Unit (LTU) monitors firms with revenues above €6 million for all major taxes. Sure enough, there is significant bunching of firms just below the threshold. The bunching is more pronounced for firms that produce intermediate goods, which makes sense because their transactions create more of a paper trail than firms that sell to final consumers, and thus

the discontinuous increase in enforcement intensity affects these firms more than retailers. Examining data from Ecuador, Carrillo, Emran, and Rivadeneira (2012) find evidence of bunching in reported tax liability just above the withholding threshold, suggesting that some firms manipulate their self-reported tax liability and possibly real economic choices to minimize expected tax payments subject to the discontinuity in the audit probability. Third-party data on sales and intermediate input costs reported by large firms who act as withholding agents indicate that bunching is indeed associated with tax evasion: self-reported sales are smaller than third-party reports for more than 10 percent of firms. “withholding”, large firms withhold and remit to the tax authority a fixed share of their purchases from small firms (and individuals), who can then apply the withheld amount as a credit against their self-reported tax liability. While withholding does not change the firms’ true tax liability, there is typically a discontinuity in the audit probability at the withholding rate; firms seeking tax refunds (because self-reported tax liability is lower than the withheld amount) are audited at a higher rate than firms making additional tax remittances.

In Chile, firms can only claim tax credits for inputs bought from tax-compliant suppliers, the invoice–credit VAT system has a built-in (albeit imperfect) self-enforcement mechanism for firm-to-firm transactions. Pomeranz (2015) tests this hypothesis by mailing increased-audit-threat letters to over 100,000 randomly selected Chilean firms, using a sample of over 300,000 firms receiving no letter as the control group. Consistent with theoretical predictions on the self-enforcement mechanism, the increase in VAT receipts (and therefore the inferred level of evasion) induced by the letters is concentrated at the level of sales from firms to final consumers, for which there is no paper trail.

In Ecuador, Carrillo, Pomeranz, and Singhal (2017) examine the effect of a change in

the tax authority's use of third-party information on reported corporate tax revenues. In the authors' study, the Ecuadorian tax authority informed some firms of the discrepancy between the two reports and offered them the opportunity to file an amended return. The authors compare the reporting behavior of firms before and after the notification and find that 24 percent of firms underreported revenue in the years when the government did not use the third-party-verified information. In the three rounds of the experiment, 11–19 percent of notified firms filed an amended return.

In amended returns, firms correctly reported their revenues but also increased their reported costs almost one-for-one with the increase in reported revenues (96 cents for each dollar!), nearly eliminating any impact on apparent tax liability. This study also reveals that reported costs were lower than third-party information on costs, which at first blush seems to be at odds with a model of firms that seek to maximize after-tax profits. The authors propose that this behavior is consistent with firms who may believe that the probability of an audit is a function of firm size and profits. To appear small, and lower the odds of being audited, some firms may underreport both revenues and costs.

Kumler, Verhoogen, and Frías (2013) study the effects on evasion of a 1997 pension reform in Mexico that tied younger workers' retirement benefits more closely to the wage reported by employers. This reform provided a new incentive for this group of workers to ensure that their employers accurately reported their wages, which in turn might lower payroll tax evasion by firms. To examine the impact of this initiative, the authors combine two sources of data on wages: administrative data from the Mexican Social Security Institute (IMSS) and household survey data from the Encuesta Nacional de Empleo Urbano (ENEU). The authors find that, consistent with a decline in evasion, the gap in median or mean wage within a cell between the ENEU data and the IMSS data fell for younger age groups after

the pension reform but, as predicted, for older workers not affected by this reform there was no decrease in the gap between the two income reports.

In Pakistan, corporations either pay a tax on profits or turnover, depending on which liability is greater. This effectively implies that at a profit rate lower than the ratio of the turnover tax rate to the profit tax rate, firms cannot deduct costs. Because in Pakistan a large portion of evasion is through misreporting of costs, this tax regime trades off loss in production efficiency (as firms move from a neutral profit tax to a distorting output tax regime) for a gain in revenue collection. Best et al. (2015) use administrative data on the universe of corporations in Pakistan to estimate the elasticity of taxable income using the bunching of firms below the threshold profit rate. They find clear evidence of such bunching, whose location shifts along with changes in tax rates that move the threshold. Using the analysis-of-bunching methodology, they estimate that turnover taxes reduce evasion by between 60–70 percent.

Relevance of tax evasion Widespreadness of tax evasion Seminal model of tax evasion Real effects of tax evasion firm-to-firm sales less likely to evade taxes small firms more likely to evade taxes evidence of misreporting of costs (Pakistan) Chile and Colombia tax evasion evidence

12.4 Carrillo et al. (2022)

The authors study tax evasion through the use of ghost firms — fake firms that issue fraudulent receipts so that their clients can claim false deductions. Using transaction-level data from Ecuador, the research documents that ghost transactions are widespread; prevalent among large firms —(!) measured as revenue level [related to size]— and firms with high-income owners —owners with experience in tax evasion? Sophisticated or less

risk-averse owners?—; and exhibit suspicious patterns: bunching at the end of the fiscal year [tax evasion decision is realized post realization of productivity and true input optimization], at round numbers, and just below financial system thresholds [no real economic activity behind].

Globally widespread but extremely limited researched problem. “Ghost firms” —also known as “invoice mills”, or “missing traders”— issue fraudulent receipts that allow the receivers to claim additional tax deductions on value-added (VAT) and corporate income taxes (OECD, 2017; Keen and Smith, 2006) Appendix B.

Data: firm-to-firm transaction-level data with 800 ‘ghost firms’ and firm owners of Ecuadorian firms data.

Widespread and quantitatively important, not limited to small, semiformal firms, benefits those at the top of the income distribution. In 2015, 4.7% of potential clients took deductions from a ghost firm. Annual ghost deductions were 14.1% of the value of their purchase deductions [huge number!]. Larger firms are more likely to engage in ghost transactions and feature larger shares of ghost transactions in total input costs [a function of size]. Ghost deductions are disproportionately used by firms owned by high-income individuals.

Process: Given a list of identified ghost firms (2016) by the Ecuadorian Internal Revenue Service (SRI), the authors identified ghosts’ clients —firms with at least one purchase from ghosts firms— with administrative data -which includes annual corporate and individual income tax filings in Ecuador (2010-2017) and monthly purchase annexes (2010-2015). The sample is restricted to firms that are required to file purchase annexes: incorporated firms (88%) and large sole proprietorships (12%) (i.e., annual sales > \$100,000, annual costs > \$80,000, or capital above \$60,000)

1. Tax deductions based on fake receipts from ghost firms are widespread and large. 10.4% of unique firms filed deductions based on receipts from at least one identified ghost firm. On average, 3.6% of all purchases of ghost clients are from ghosts, amounting to 10.4% of the value of their purchase deductions. These shares are the highest, 4.6% and 14.1%, for 2015, the year before the ghost list was collected. Earlier ghosts might have disappeared by the time the list was established. Total tax evasion was \$2.1 billion in value, 1.7% of tax evaded for corporations and 11.5% for sole proprietorships.
2. Evasion through ghost firms is more prevalent among relatively large firms. Ghost clients have higher revenues, costs, and tax liabilities. The probability of engaging in cost overreporting through ghost firms increases monotonically in firm revenue—maybe only because of more transactions. Share of ghost deductions also increases throughout much of the size distribution, except at the very top—very large corporations might have stronger incentives to avoid illegal behavior or they can use more sophisticated avenues of tax avoidance that do not require evasion using fake receipts (Bustos et al., 2022)
3. Ghost deductions are most prevalent in firms owned by high-income individuals. The probability of having ownership of a ghost client and the amount of ghost purchases attributed to individuals increases with individuals' income. The amount of additional tax as a share of owners' income is over 56 times higher in the top 5% than in the bottom 80% of the income distribution, and almost 170 times higher in the top 1%. Ghost-firm-enabled tax evasion tends to reduce disproportionately the effective taxation of firms owned by rich individuals.
4. The number and value of ghost transactions are clustered towards the end of the tax

year —the same as the calendar year in Ecuador. In December, there are twice as many ghost transactions as during the six previous months of the year. Non-ghost transactions are only 6% higher in December. This is consistent with firms assessing their annual revenues at the end of the year and then utilizing ghost transactions to achieve a target reported profit level or rate for tax purposes. Similar to US firms spending more on capital investment towards the end of the fiscal year to reduce tax obligations (Xu and Zwick, 2022).

5. Ghost transactions bunch at round numbers more prevalently than non-ghost transactions. This is consistent with ghost transactions representing false activity (Klimek et al, 2018; Nigrim, 2018).
6. Ghost transactions exhibit bunching below the financial system payment threshold. In Ecuador, transactions above \$5,000 (gross-of-VAT) are required by law to be made through the formal financial sector (e.g. electronic transfers, checks, or credit cards). Ghost transactions bunch just below \$5,000. In contrast the distribution of non-ghost transactions is smooth through the \$5,000 threshold.

Lit on tax evasion via ghost firms is extremely sparse. Waseem (2020) Pakistan. Policy: VAT rates reduced to zero. Ghost firms decreased. Most ghost deductions are claimed by exporters. Ghost deductions are a large share of over-claimed refunds for exporters. Mittal et al. (2018) develop a Machine Learning algorithm to detect ghost firms.

Most recent work on tax evasion focuses on firms' revenue misreporting (Slemrod, 2019; Pomeranz and Vila-Belda, 2019), but few have studied firms' cost misreporting. Recent firms' revenue misreporting studies have found that enforcement policies targeting revenue underreporting led to increasing reported costs —to offset tax liabilities—, but there is no offsetting behavior —reduction in reported revenue— in the study by Carrillo et al. (2022).

This is consistent with third-party information reporting —firm-to-firm transaction level data is used in this study— decreasing the incentive to evade by increasing the probability of getting caught.

Other studies finding bunching at the thresholds are: - VAT reporting thresholds: Onji, 2009; Asatryan and Peichl, 2017; Luksic and Mittal, 2019; Liu et al., 2021. - Different tax regimes and regulations: Hasegawa et al., 2013; Best et al., 2015; Almunia and Lopez-Rodriguez, 2018; Lopez-Luzuriaga, 2019; Chen et al., 2021; Clifford and Mavrokonstantis, 2021; Overview: Kleven, 2016.

Intervention: Tax authority challenged deductions from firms with ghost transactions (ghost clients) during 2010-2015 by notifying them via email.

[What goes into the cost overreporting function? Tax authority chooses effort (b(type)-c(type))e (?)]Tax authorities target firms with higher potential tax recovery. In this study, notified firms were larger than typical incorporated ghost clients —authors only had access to the data on incorporated firms that were notified. Notified firms had 2.4 times higher median tax liability, the median amount of ghost deductions of these firms was \$181,000 (mean \$338,000) and the median share of ghost deductions out of their total purchases was 26% (mean 38%). [conditional on being large, the overreporting level does affect your probability of being audited.]

[Evidence of overreporting in intermediates] Adjusting firms reduced most strongly non-labor costs: domestic purchases, other production costs, and imports. Some offsetting behavior was observed; Firms increased labor, inventory, and financial costs. However, overall total costs still decreased on average by 72 cents for every dollar contested in the notification.

Many firms did not respond to the notifications. A portion might be explained by failure in communication (e.g., letters not reaching the firm or the right person within the firm). Firms might choose not to amend knowing the tax authority has limited capacity for follow-up enforcement. Adjusting firms tend to be somewhat smaller than all the notified firms. The intervention resulted in large reductions in firms' reported costs. The reduction in costs increased tax liabilities by \$20.6 million, 81% increase on the \$25.4 million originally filed by adjusting firms. On average, reported costs were reduced by \$229,000

Firm-to-firm transactions —third-party reporting: firm's sales to other firms can be cross-checked— are less like to underreport revenue. Firms did not adjust revenues down to offset tax liability resulting from the decrease in costs —as in other studies where the enforcement strategies pushing firms to truthfully report their revenues led firms to make large offsetting adjustments by increasing reported costs (Asatryan and Peichl, 2017; Carrillo et al., 2017; Slemrod et al., 2017; Almunia and Lopez-Rodriguez, 2018; Mascagni et al., 2018; Naritomi, 2019; Li and Wang, 2020)

No evidence that the intervention was followed by firms going out of business or de-formalizing; consistent with ghost clients being large and established.

International prevalence of false invoicing (ghost firms) - Mexico: between 2014 and 2018, the estimated annual tax revenue lost due to ghost firms was 5 billion MXN (SAT), 0.03% of Mexico's GDP; 8,500 identified ghost firms in 2018 (Senado de la Republica, 2019). - Chile: historically, false invoices represent between 15% and 25% of total VAT evasion. In 2004, fake-invoice tax evasion was 114 billion CLP, 0.2% of Chile's GDP (Gonzalez and Velasquez, 2013; Jorrat, 2001; CIAT, 2008). - Colombia: fake-invoice tax evasion was around 2,300 billion COP, 0.2% of Colombia's GDP (Portafolio, 2019). - Poland: In 2016, fake-invoice

tax evasion amounted to 103,850 million PLN, 5.6% of Poland's GDP (Poland's Minister of Finance, 2018). - Asia: South Korea (Krever, 2014) and China (Hashimzada et al., 2010; Barboza, 2013). - Africa: Kenya (Kenya Revenue Authority; Mak'Osewe, 2019); Rwanda (Mascagni et al., 2019; 2021); African Tax Administration Forum (Brain and Nwankwo, 2018).

12.5 Mclure (1989)

12.5.1 Colombian Tax System

The tax system of Colombia has been under constant study and revision since the 1960s.

During the early 1960s, a Fiscal Survey of Colombia (1965) led by Professor Milton Taylor was prepared as part of the ongoing program of the Organization of American States and the Inter-American Development Bank. This study led to important administrative innovations, but few concrete substantive reforms.

In 1970, Professor Richard Bird prepared a report, influential in terms of policy, based on his experience as a resident advisor to the government of Colombia on tax and fiscal policy during the period of 1964-1966.

In 1968, Professor Richard Musgrave was commissioned by President Carlos Lleras Restrepo to appraise the tax system and make recommendations for reform with the help of Colombian and foreign experts. The resulting document, Fiscal Reform for Colombia: Final Report and Staff Papers of the Colombian Commission on Tax Reform, significantly shaped the 1974 and subsequent tax reforms. This report has also become a classic in the field of tax reform in developing countries.

In late 1974 Colombia undertook a major fiscal reform drawing heavily on the recommen-

dations of the Musgrave Commission.

Other influential documents during this period were La Reforma Tributaria 1974 (Gillis and McLure, 1974) and a World Bank report (McLure, 1982). Finally, there exists a report on the taxation of income from business and capital (McLure, Mutti, Thuronyi, and Zodrow, 1989?)

12.5.2 Income tax

At the time of the Taylor Mission, Colombia's system of *income and complementary taxes* consisted of a basic income tax, a net wealth tax, and an excess profit tax.

In the early 1960s, the base of the Colombian income tax was quite broad. It included all "enrichment"; it included corporate dividends and shares in the earnings of limited liability companies and partnerships.

During this period, high-income individuals could evade the individual income tax on dividends and interest on bearer securities, since a withholding tax of only 12 percent was levied on such income.

The income of individuals was taxed under a graduated schedule consisting of 56 rates, ranging from 0.50 to 51 percent. There was a complex system of personal deductions for expenditures on medical, educational, and other professional services; these deductions depended on the income and number of children of the taxpayer.

Tax incentives were provided for, among others, industrial enterprises. In addition, income tax exemptions, limitations, or reductions were granted to oil companies, certain mining companies, large-scale agricultural improvements, cattle-raisers, Colombian airlines, certain basic industries, manufacturers using products from *Paz del Rio* steel plant, and exporters

other than coffee, petroleum, bananas, hides and precious metals.

Juridical persons were subject to graduated rates. Corporate rates (on *Sociedades anónimas*), which applied to all foreign entities, were 12, 24, and 36 percent. The rates applied to limited liability companies (*Sociedades de Responsabilidad Limitada*), the business forms used for many of the most important Colombian business ventures, were only 4, 8, and 12 percent, and the income of partnerships was taxed at rates of 3 and 6 percent. Whereas the income of proprietorships was taxed only in the hands of the owner under the individual tax, the income of juridical persons was taxed directly to the firm. The entire net income of limited liability companies (like that of partnerships) was taxed to the partners, whether or not distributed; by comparison, only dividends were taxed to owners of corporate shares.

The tax system of Colombia suffered from many administrative problems in 1960. A serious problem involved the interplay of an overly short statute of limitations (two years), a system of penalties and interest that encouraged false and delinquent returns, and an overburdened tax administration. In extreme cases, taxpayers could avoid taxation completely by deliberately reporting false information, and waiting for the statute of limitations to preclude correction by the fiscal authorities.

12.5.3 Net wealth tax

Since 1953, Colombia has imposed a tax on net wealth (*impuesto complementario de patrimonio*), apparently to increase the tax burden related to income from capital and to induce a more productive use of land. The net wealth tax applies to individuals but not to corporations, partnerships, and limited liability companies. Taxable wealth was that held in Colombia, net of indebtedness. There were numerous exceptions, which paralleled, for the most part, those under the income tax. For example, there were exemptions for livestock

kept for breeding purposes, certain agricultural investments, assets of Colombian airlines, investments in motor vehicle assembly or manufacturing plants, investments in securities of the *Paz del Rio* steel plant, and investments in certain basic industries and certain iron and steel fabricators.

12.5.4 Excess-Profits Tax

Colombia imposed an excess-profits tax beginning in 1935; this tax was repealed in 1974.

12.5.5 The Taylor Mission Proposals

The Taylor Mission produced an impressive analysis of the distortions and inequities that flow from the disparate treatment of individuals, corporations, limited liability companies, and partnerships. It noted particularly the case for integration of the corporate and individual income taxes. However, it concluded that such a policy, integrating corporate and individual income taxes, was not warranted at the time. McLure rationalizes this contradiction with political realism because the double taxation had been recently introduced in 1953. The Mission's recommendation focused on the interplay between the corporate income tax, two of the supplementary taxes on income, and the excess-profits tax. The Mission favored a tax increase in limited liability companies, which it acknowledges were used as "tax avoidance device".

The Mission devoted particular attention to the agricultural sector because of the potential revenue being lost and the potential role of tax policy to stimulate efficient use of rural resources. The Mission proposed a presumptive calculation of income from agriculture equal to 10 percent of the assessed value of agricultural property.

The Taylor Mission also proposed administrative changes such as withholding for wages,

salaries, interests, and dividends, the requirement of advance tax payments based on estimated liabilities, self-assessment, extension of the statute of limitations, and rigorous application of penalties.

12.5.6 Musgrave Commission's Proposals

For the most part, the Taylor and Musgrave recommendations for income tax reform were broadly consistent.

The Musgrave Commission repeated the call for heavier penalties for tax evasion. The Commission also suggested strategies to improve taxation of the *hard-to-tax* group such as small traders, independent professionals, and agriculture.

In contrast to the Taylor Mission, the integration of the taxes on business and on individuals was rejected by the Musgrave Commission. Instead, the Commission proposed that the income of corporations and limited-liability companies be subject to a unified system of taxation. The proposed system would consist of taxation at the entity level and at the individual shareholder level only on income that is distributed. Partnerships would be taxed at the entity level at a substantial lower rate, and subject to certain conditions limited-liability companies would be given the option to being taxed like partnerships; the partners' share in partnership profits would also be subject to individual taxation, whether distributed or not.

12.5.7 Sales tax and VAT

Colombia introduced a national sales tax in 1965. This was a single-stage manufacturers' tax levied at rates of 3 to 10 percent on finished domestic goods and imports. Buyers' certification that goods would undergo further processing prevented the transaction from

being taxed to avoid double taxation. This system led inevitably to double taxation and evasion. In 1966, this sales tax was converted into a crude credit-type value added tax (VAT), but one that extended only through the manufacturing level and allowed no credit for tax paid on capital goods.

12.5.8 1974 reforms

The 1974 reforms were greatly influenced by the Musgrave recommendations, which were considered and refined by local experts during the intervening half-decade. These reforms rationalized many aspects of the income and net wealth taxes, and eliminated the tax on excess profits, added a calculation of presumptive income based on net wealth, and further improved the system of internal indirect taxes.

Income. The 1974 reforms eliminated many exemptions and other forms of preferential treatment for non-labor income, including the exclusion of interest on public debt, exemptions for automobile producers and private electrical companies, and deductions for reserves for investment. Exemptions remained intact for the airlines, publishing, and reforestation sectors, and for various activities in selected regions (primarily “frontier” and other less developed ones).

The 1974 reforms replaced graduated rates with flat-rate taxes for business income. But the taxation of corporations and limited liability companies was not unified. Instead, the income of corporations was taxed at 40 percent and that of limited liability companies and partnerships at a rate of 20 percent. As before, shareholders of corporations paid tax only on dividends, whereas owners of limited liability companies and partnerships were required to pay tax currently on the share of the profits of the company imputed to them.

The 1974 reforms extended the concept of presumptive income from only to the agricultural

sector (1973) to the economy as a whole. Income was presumed to be no less than 8 percent of net wealth, defined generally in the same way as for the net wealth tax.

Internal indirect tax. The 1974 reforms eliminated the possibility of buying otherwise taxed items on an exempt basis simply by certifying to the seller that they were to be processed further; under the new law, the tax credit (invoice) system was extended to all such transactions. This created administrative headaches because of the many claims for refunds that had to be processed.

The law expanded the base of the VAT to reduce the regressivity of the tax at the upper end of the income scale. The VAT was applied to many services, including parking lots, insurance, international air fares, photographic developing and photocopying, telegraphic and telephone services, and fees of social clubs. Likewise, the degree of progressivity of the rate schedule applied to various goods, ranging from “wage” goods (6%) to luxuries (35%) was increased. The 1974 reforms also provided more widespread exemptions, for example, for almost all food and for selected agricultural machinery.

As under previous law, no credit was allowed on capital goods, in part to compensate for underpricing of capital resulting from various non-tax policies, including an undervalued exchange rate. On the other hand, the law contained anomalous provision that exempted imports of capital goods destined for “basic” industries. The combination of these policies placed domestic producers of capital goods at a competitive disadvantage relative to foreign producers of imported capital goods.

12.5.9 1974-1986 Period

During this period of continued tinkering with the tax system in Colombia, some changes represented improvements, but the more important ones represented retrogression. These

counter-reforms were in large a response to the pleadings of politically powerful economic groups.

For example, law 20 of 1979 effectively exempted cattle raising from income tax and arbitrarily reduced the value of breeding stock and dairy cattle for net wealth and presumptive income tax purposes by 50%.

Law 9 of 1983 introduced a tax credit of 10% of dividends received was provided as a means of reducing double taxation of dividends.

In addition, Law 9 of 1983 instituted a measure of presumptive income equal to 2 percent of gross receipts to supplement the measure based on net wealth. This reform was aimed specifically at the commerce and service sectors; the former were thought to evade the wealth-based presumptive tax by systematically understating inventories. In addition, it extended the presumptive income tax to limited liability companies.

Tax returns were not required if 80 percent of income was from labor subject to withholding, if the taxpayer was not an owner of shares in a limited liability company (ownership of corporate shares was allowed), and if the taxpayer was not liable for sales tax, among other requirements.

In 1983, the value-added tax (VAT) was extended to the retail level, with a *simplified system* being made available to small retailers in the effort to ease compliance costs and administrative burden. Services like hotels, computing services, maintenance, and rental of goods and fixtures were brought into the scope of the tax. Moreover, there was some unification of rates to alleviate the difficulties of dealing with differential rates at the retail level which led to an increase in the rate of (non-creditable) tax applied to domestically produced capital goods. Imported capital goods continued to be exempt from tax. In 1984,

exemptions for agricultural machinery, transportation equipment, and certain other goods were eliminated.

12.5.10 The 1986 reforms

The 1986 reform reduced tax rates dramatically. The top tax rate applied to individual income was reduced from 49 to 30%; the same rate (30%) applied to the income of corporations and limited liability companies.

Expense deductions of independent professionals were limited to 50% of income in an effort to curtail a major source of abuse.

The 1986 reforms unified the taxation of corporations and limited liability companies by taxing both at a rate of 30%. It integrated the taxation of companies and individuals by exempting corporate dividends and participation in profits of limited liability companies from tax at the individual shareholder/owner level. Consistent with this treatment, losses, exempt income, and tax credits of companies cannot be used to offset income of their owners.

12.6 Fiscal Survey of Colombia

Juridical persons in Colombia (p. 27-30).

- Corporation (*sociedad anónima*), the typical association of capital and is the counterpart of the United States corporation. The Colombian corporation is a separate juridical entity, having its own name, domicile, and property. Corporations are subject to the Superintendent of Corporations and are closely supervised, being required, for example, to have an auditor and to maintain a legal reserve. Foreign corporations, partnerships, and other companies or entities are taxed as corporations. Also taxable

as corporations are limited stock partnerships (*sociedades en comandita por acciones*), which are limited partnerships with the partnership interest represented by shares of stock.

- General partnerships (*sociedades colectivas*), which include mining partnerships (*sociedades ordinarias de minas*), *de facto* or constructive partnerships (*sociedades de hecho*), organized associations, associations, and foundations.
- The third type of entity includes the ordinary limited partnership (*sociedad en comandita simple*), i.e., one other than a limited stock partnership; and the limited liability company (*sociedad de responsabilidad limitada*). The limited liability company is quite important (1960s) in Colombia. It combines certain features of a corporation and a partnership. It is organized in the same manner as a general partnership but with the additional requirement that the articles of association must indicate that the personal liability of the partners is limited to the capital contributed. The limited liability company offers some advantages as compared to a corporation, chiefly, the lower cost of organization and taxation (until the 1980s reforms) and comparative freedom of operation. Under Colombian law, the limited liability company is an association of persons, not of capital, and it is not subject to the superintendent of Corporations unless more than one-third is owned by a corporation.

Profits from individual proprietorships [were] taxed only once under the income tax system, but the dividends of corporations [were] taxed in full, resulting in a double income tax burden on distributed corporate profits. On the other hand, for both partnerships and limited liability companies, all income after payment of the income tax at the business level is construed to be distributed and is taxed again at the individual level.

12.7 Perry & de Triana (1990)

Development of a Value Added Tax in Colombia

12.8 Torres & Sourdis (1994)

12.8.1 Reforma 1983

El objetivo de la reforma de 1983 fue reducir el déficit fiscal y evitar financiación con recursos de emisión. Asimismo, la reforma de 1983 buscaba reactivar la economía aumentando la inversión mediante incentivos tributarios. Con el objetivo de la reactivación económica, se amplió y reestructuro la base tributaria y se simplificó el cobro de impuestos. Se redujo el número de personas obligadas a declarar renta y se extendió la renta presuntiva al comercio y los intermediarios financieros. Se eliminó la doble tributación para las sociedades anónimas (Corporations) y se redujeron las tasas para las sociedades limitadas y las personas naturales.

El IVA se extendió al comercio y detal. Se unificaron y simplificaron las tarifas.

12.8.2 Reforma 1986

Objetivo: devolver la equidad y la neutralidad del impuesto de renta, fortalecer la administración tributaria y adecuar la estructura tributaria a la movilidad internacional de capitales.

Con este objetivo, se igualó las alternativas de inversión de las empresas (impuestos sobre rendimientos de capital), se unificaron las tarifas tributarias para todo tipo de sociedades tanto nacionales como extranjeras; se eximió del pago de la retención en la fuente al 90% de los asalariados del país.

En el campo administrativo, la contribución más importante fue la delegación en la red bancaria de la recepción y recaudo del impuesto de renta (income tax).

12.8.3 Reforma 1990

Objetivo: Incrementar el ahorro de la economía, desarrollar el mercado de capitales y fomentar la repatriación de los capitales fugados. El objetivo también incluía disminuir la dependencia fiscal de la actividad económica externa y adecuar la estructura tributaria a la apertura comercial.

Se desgravó la utilidad obtenida en la enajenación de acciones a través de bolsa, se exoneró del impuesto de renta a los fondos de inversión, fondos comunes y de valores y se redujeron las tarifas de remesas y renta para la inversión extranjera. Se dieron amnistías fiscales y se redujo el impuesto de saneamiento para fomentar la repatriación de capitales.

Se amplió la base del IVA y se elevó la tarifa general de 10% al 12%.

12.8.4 Impacto en recaudos

Grafico 1. Se observa una caída en los ingresos tributarios como porcentaje del PIB desde 1980 hasta 1983, el punto más bajo. A partir de 1984, los ingresos incrementan hasta 1987, donde se observa un declive hasta 1990. Por último, se observa un incremento para 1991.

12.8.5 Evasión

Grafico 3. Impuesto a la renta (Income tax). Se muestra una reducción sostenida a lo largo del tiempo, de 1980 a 1991. Se muestra también un decremento de 1980 a 1983, luego un incremento desde 1983 hasta 1988, luego decremento del 88 al 90 y finalmente un gran incremento del 90 al 91.

Grafico 4. IVA, evasión y recaudo. Se aprecia un decremento del 42% de la tasa recaudo efectivo/recaudo potencial de 1984 hasta 1990 y luego un incremento en 1991.

12.9 Almunia & Lopez-Rodriguez (2018)

Effects of size-dependent tax enforcement on firm's tax compliance. The authors exploit quasi-experimental variation generated by a Large Taxpayers Unit (LTU) in Spain, which monitors firms with more than 6 million euros in reported revenue. Firms strategically bunch below the eligibility threshold in order to avoid stricter tax enforcement. The response is stronger in sectors where transactions leave more paper trail, suggesting that the monitoring effort and the traceability of information reported by firms are complements.

Following the advice of multilateral organizations such as the IMF and the OECD, many tax administrations have established special units to monitor and improve the tax compliance of the largest taxpayers. The benefits of establishing Large Taxpayers Units (LTU) include higher tax revenues and efficiency.

In Spain, the LTU focuses exclusively on firms with total operating revenue above 6 million euros, a threshold established in 1995 and it has remained constant in nominal terms ever since. The LTU has more auditors per taxpayer than the rest of the tax authority, and these auditors are on average more experienced and better trained to deal with the most complex taxpayers. Thus, while firms above and below the eligibility threshold face the same tax schedule, monitoring effort changes discretely at this arbitrary cutoff.

In the model, firms make production and tax-reporting decisions given a tax rate on reported profits. Underreporting revenue lowers tax liabilities, but it also implies some resource costs (e.g., keeping two sets of accounting books or foregoing business opportunities). The detection probability increases endogenously with the amount evaded by firms,

depending on the monitoring effort of the tax authority (resources devoted to analyzing tax returns and other paper trails generated by business activities) and the enforcement technology (technology to process the information generated by business transactions). The model assumes monitoring effort jumps up discretely at a certain threshold of reported revenue, while enforcement technology remains constant. This creates an incentive for firms to bunch below the LTU cutoff to avoid stricter tax enforcement. Firms are heterogeneous in their cost of evasion and in the effectiveness of monitoring efforts, which depend on the traceability of firms' transactions. Intermediate input sales (firm-to-firm) leave more paper trail and thus are easier to trace than firm-to-consumer sales. In line with this, the model predicts a stronger bunching response by upstream (firm-to-firm) firms compared to downstream (firm-to-consumer) firms.

The authors use the Commercial Registry (Registro Mercantil) from the Banco de España. The dataset contains firm-level information on annual net operating revenue, input expenditures, number of employees, payroll taxes, total value added, and the tax base and liability in the corporate income tax, covering more than 80 percent of registered businesses in Spain with operating revenue between 3 and 9 million euros between 1995 and 2007. The LTU threshold remained constant at 6 million euros.

Empirical finding 1: The authors find that a significant number of firms report revenues just below the LTU threshold. Their estimates indicate that firms reduce their reported revenue by 2 percent of their total revenue on average. The marginal buncher's response is bounded between 6.4 and 8.7 percent of the total reported revenue.

EF 2: The average bunching response is significantly higher in upstream sectors than in downstream sectors. The authors argue that the effectiveness of an additional monitoring effort is higher the easier it is to trace firms' transactions, which suggests that information

trails and monitoring efforts are complements.

EF 3: The authors suggest that they identify patterns that are not compatible with firms lowering real output (real response) suggesting firms are merely underreporting revenue (evasion response). They find that at the LTU threshold, the average ratio of material expenditures over total revenue shifts down sharply, while the average ratio of the wage bill (net of payroll taxes) as a share of total revenue shifts upward. The argument is that if bunchers reduced their true output, both ratios would jump up at the threshold because they are more productive than other firms below the threshold (?). Firms might also overreport material inputs and underreport labor costs to evade VAT and payroll tax (PRT). This doesn't make sense. This is precisely what would be expected if firms were reporting true output but overreporting inputs and underreporting labor.

The authors also document a positive persistent gap in the average reported taxable profit and significant jumps in the VAT and PRT between the two enforcement regimes.

[Almunia & Lopez-Rodriguez \(2018\)](#) conclude that Spanish firms misreport both revenue and inputs. In their argument, the authors do not discard the case in which firms report their true output and only misreport inputs. However, upstream firms are likely to fall in this case, only input but not revenue misreporting. While buying firms have incentives to overreport input expenses, selling firms have incentives to underreport sales. Hence, due to opposite incentives, upstream firms might find it hard to underreport their revenue. If the expected benefits of staying below the LTU threshold and evading taxes outweigh the costs of forgone production and evasion, bunching upstream firms are likely to reduce their true output and misreport their inputs.

This is not so clear in the introduction, where the authors just state that the cross-sectional

patterns they observe in the data are not compatible with a real response (firms lowering their true output to avoid being monitored by the LTU). Maybe the reason for not clarifying they find evidence of both revenue and input misreporting in the introduction is because they focus on revenue misreporting and have nothing to say about input misreporting.

The authors do not consider the case where firms only misreport inputs and reduce their true output to stay below the LTU threshold. To underreport revenue, firms need to coordinate with customers. Customers must agree to not require an invoice, to avoid leaving a paper trail. This is likely to happen with downstream firms, firms that sell to final customers. However, with upstream firms, firms selling to other firms, this is unlikely the case. The reason is that, unlike final customers, other firms have incentives to overreport their input consumption to reduce their VAT and CIT.

It might be likely bunching upstream firms reduce their true output if the expected benefits of staying below the LTU threshold outweigh the cost of forgone production and evasion costs. The authors indeed find lower bunching for upstream firms than for downstream firms. Moreover, in the online appendix, the authors show that the ratios of intermediate expenses over revenue and labor expenses over revenue of wholesale (upstream) and retail (downstream) firms follow the general evasion patterns. Intermediates upward sloping then dropping at the cutoff; labor downward sloping then jumping at the cutoff.

I argue that they cannot distinguish between the scenario where firms only misreport inputs and the one where firms can misreport both revenue and inputs. Another point they missed is that the input expense over revenue ratio is equal to the output elasticity of inputs. In turn, this elasticity depends on the shape of the production function and is productivity-free if the productivity is Hicks-neutral. This is key to their argument, because, if it might be true that bunchers are more productive than firms below the LTU threshold, their input

shares are not necessarily lower. If the production function is Cobb-Douglas, for example, the shares are constant below and above the threshold.

Under multiple taxes, firms have incentives to underreport revenue and overreport intermediate expenses to evade Value-Added Tax (VAT) and Corporate Income Tax (CIT), and to underreport labor expenditure to evade the Pay-Roll Tax (PRT). To study the mechanism of the bunching response under multiple taxes, the authors compare with the data their model's 'predictions' on the input expense share of revenue under three different scenarios. The idea is that in the absence of the LTU, firms around the threshold should have similar average ratios of input expenditure as a fraction of revenue. Since the probability of detection is higher under the LTU, firms above the threshold have a smaller incentive to evade taxes. Once the LTU is introduced, some firms, the bunchers, above the threshold have incentives to bunch below to avoid stricter monitoring by the LTU and have larger expected profits. The heterogeneity arises because firms differ in monitoring effectivity by the authority and evasion costs.

Under the first scenario, there is no misreporting. The bunchers reduce their input consumption to lower their output and stay below the LTU threshold. The authors argue that because the bunchers are more productive than the firms below the threshold, their input expenses share of revenue ratio is lower. Therefore the average ratio of input expenses over revenue should be lower the closer to the cutoff. Then, at the threshold, there would be a discrete upward jump for both ratios. In brief, the predictions of this scenario are (i) a downward trend of both ratios as we get closer to the revenue threshold along the revenue axis, and (ii) a discrete upward jump at the cutoff. Gap: share equals output elasticity of inputs, do not depend on productivity, but on PF shape.

In the second scenario, the authors allow for revenue misreporting but no input misreport-

ing. According to the authors, the bunchers' input expenses share of revenue ratios are mechanically greater than the firms below the cutoff. As a consequence, both ratios go up and then drop at the cutoff. Gap: share should be lower for bunchers than firms below the cutoff. Implicitly, authors are assuming that bunchers underreport revenue enough to make their input shares larger than firms below the cutoff, but is this guaranteed?

Finally, the authors allow for misreporting revenues and inputs. In this scenario, One goes up, the other goes down. Gap:

They conclude that it is likely that firms underreport revenue and misreport inputs (over-reporting inputs, underreporting labor). This is not clear in the introduction as they have nothing to say about input misreporting. Moreover, they don't claim causality or significance. They just say that given their model's testable implications and what they observe in the data simply suggests the third scenario is the more likely to be happening.

They take the input over revenues and labor over revenue ratios and look at what would happen just before the LTU threshold and after the threshold. In their argument, higher productivity firms would use a lower amount of inputs, hence, in the absence of misreporting, these shares should be smaller for firms with higher revenues (higher productivity). However, they miss the point that actually these ratios are productivity free and depend on the output elasticity of inputs.

However, this ratio is equal to the output elasticity of inputs and it is free of productivity. Indeed, using the authors' notation, we have that for the intermediates,

$$\begin{aligned}\psi f_x(x, z) &\geq w \\ \frac{f_x(x, z)x}{f(x, z)} &\geq \frac{wx}{y}\end{aligned}$$

In this equation, equality holds only when firms produce the optimum quantity. When they don't, then the intermediate's input expense over the revenue ratio is lower than the elasticity. For the sake of clarity, suppose the production function is Cobb-Douglas, $f(x, z)\psi = x^\alpha z^{1-\alpha}\psi$. In this case, we have

$$\alpha \geq \frac{wx}{y}$$

Because they are producing less output than the optimal, bunchers' input cost over revenue is lower than the input elasticity of output.

12.10 Rudnick & Velasquez (2021)

In the early 1990s, Colombia's publicly-owned power sector was financially distressed and poorly managed, becoming a risk for the country's macroeconomic stability. The sector reached a tipping point during extensive demand rationing episodes in 1992. Amidst the paradigm transformation promoted by the 1991 Constitution, Colombia launched an ambitious power sector reform package in 1994, introducing private sector participation, independent regulation of distribution and transmission, and competition in power generation and retailing.

Colombia's power sector was heavily distressed in the early 1990's. Financially unsustainable utilities placed an enormous burden on the country's fiscal gap. Efficiency was very low overall, particularly in areas with high electricity theft and fraud. Furthermore, two major blackout periods, first in 1983 and then in 1992-1993, pushed power sector reform to the top of the country's agenda in 1993 (Larsen, Dyner, Bedoya, & Franco, 2004).

To tackle these challenges, a comprehensive reform package was launched with enactment

Table 24: Fortran implementation of the Non-Parametric GNR(2020) method to estimate production functions. Comparison with Stata code and replication data from GNR(2020), and the working data of this paper and my own Fortran implementation code.

NP GNR									
	Replication Data			Working Data					
	Stata			Stata			Fortran		
	m	k	l	m	k	l	m	k	l
311	0.67	0.12	0.22	0.67	0.12	0.22	0.67	0.12	0.22
321	0.54	0.16	0.32	0.54	0.16	0.32	0.54	0.16	0.32
322	0.52	0.05	0.42	0.52	0.05	0.42	0.52	0.05	0.42
331	0.51	0.04	0.44	0.51	0.04	0.44	0.51	0.04	0.44
381	0.53	0.10	0.43	0.53	0.10	0.43	0.53	0.10	0.42

of Laws 142 and 143 of 1994. These laws established the framework for vertical unbundling and private sector participation, a new regulatory framework and rate-system governed by principles such as efficiency and financial sufficiency, and introduction of wholesale competition, among other measures.

Table 25: Fortran implementation of the Non-Parametric GNR(2020) method to estimate production functions. Comparison of the error standard deviation and $\mathcal{E}$ between the Stata code and replication data from GNR(2020), and the working data of this paper and my own Fortran implementation code.

NP GNR						
	Replication Data		Working Data			
	Stata		Stata		Fortran	
	$\mathcal{E}$	err sd	$\mathcal{E}$	err sd	$\mathcal{E}$	err sd
311	1.05	0.24	1.05	0.24	1.05	0.24
321	1.03	0.23	1.03	0.23	1.03	0.23
322	1.02	0.22	1.02	0.22	1.02	0.22
331	1.04	0.25	1.04	0.25	1.04	0.25
381	1.02	0.20	1.02	0.20	1.02	0.20

TABLE 2
AVERAGE INPUT ELASTICITIES OF OUTPUT
(Structural vs. Uncorrected OLS Estimates)

	Industry (ISIC Code)											
	Food Products (311)		Textiles (321)		Apparel (322)		Wood Products (331)		Fabricated Metals (381)		All	
	GNR	OLS	GNR	OLS	GNR	OLS	GNR	OLS	GNR	OLS	GNR	OLS
Colombia:												
Labor	.22 (.02)	.15 (.01)	.32 (.03)	.21 (.02)	.42 (.02)	.32 (.01)	.44 (.05)	.32 (.03)	.43 (.02)	.29 (.02)	.35 (.01)	.26 (.01)
Capital	.12 (.01)	.04 (.01)	.16 (.02)	.06 (.01)	.05 (.01)	.01 (.01)	.04 (.02)	.03 (.01)	.10 (.01)	.03 (.01)	.14 (.01)	.06 (.00)
Intermediates	.67 (.01)	.82 (.01)	.54 (.01)	.76 (.01)	.52 (.01)	.68 (.01)	.51 (.01)	.65 (.02)	.53 (.01)	.73 (.01)	.54 (.00)	.72 (.00)
Sum	1.01 (.01)	1.01 (.01)	1.01 (.02)	1.03 (.01)	.99 (.01)	1.01 (.01)	.99 (.04)	1.00 (.02)	1.06 (.01)	1.05 (.01)	1.04 (.00)	1.04 (.00)
Mean (capital)/mean (labor)	.55 (.08)	.27 (.07)	.49 (.09)	.27 (.06)	.12 (.04)	.04 (.02)	.08 (.05)	.08 (.05)	.23 (.04)	.11 (.04)	.40 (.03)	.23 (.01)
Chile:												
Labor	.28 (.01)	.17 (.01)	.45 (.03)	.26 (.02)	.45 (.02)	.29 (.02)	.40 (.02)	.20 (.01)	.52 (.03)	.32 (.02)	.38 (.01)	.20 (.01)
Capital	.11 (.01)	.05 (.00)	.11 (.01)	.06 (.01)	.06 (.01)	.03 (.01)	.07 (.01)	.02 (.01)	.13 (.01)	.07 (.01)	.16 (.00)	.09 (.00)
Intermediates	.67 (.00)	.83 (.01)	.54 (.01)	.75 (.01)	.56 (.01)	.74 (.01)	.59 (.01)	.81 (.01)	.50 (.01)	.71 (.01)	.55 (.00)	.77 (.00)
Sum	1.05 (.01)	1.05 (.00)	1.10 (.02)	1.06 (.01)	1.08 (.02)	1.06 (.01)	1.06 (.01)	1.04 (.01)	1.15 (.02)	1.10 (.01)	1.09 (.01)	1.06 (.00)
Mean (capital)/mean (labor)	.39 (.03)	.28 (.03)	.24 (.04)	.22 (.04)	.14 (.03)	.12 (.03)	.18 (.03)	.12 (.05)	.25 (.03)	.21 (.04)	.43 (.02)	.42 (.02)

NOTE.—Standard errors are estimated using the bootstrap with 200 replications and are reported in parentheses below the point estimates. For each industry, the numbers in the first column are based on a gross output specification using a complete polynomial series of degree two for each of the two nonparametric functions (D and C) of our approach (GNR). The numbers in the second column are also based on a gross output specification and are estimated using a complete polynomial series of degree two with OLS. Since the input elasticities are heterogeneous across firms, we report the average input elasticities within each given industry. The “Sum” row reports the sum of the average labor, capital, and intermediate-input elasticities, and the “Mean (capital)/mean (labor)” row reports the ratio of the average capital elasticity to the average labor elasticity.

Figure 24: [Gandhi et al. \(2020\)](#) Table 2 (GNR2020)

Table 26: Fortran implementation of the Cobb-Douglas GNR(2020) method to estimate production functions. Comparison with Stata code and replication data from GNR(2020), and the working data of this paper and my own Fortran implementation code.

CD GNR									
	Replication Data			Working Data					
	Stata			Stata			Fortran		
	m	k	l	m	k	l	m	k	l
311	0.62	0.20	0.20	0.62	0.20	0.20	0.62	0.20	0.20
321	0.52	0.21	0.27	0.52	0.21	0.27	0.52	0.21	0.27
322	0.44	0.19	0.29	0.44	0.19	0.29	0.44	0.19	0.29
331	0.46	0.10	0.44	0.46	0.10	0.44	0.46	0.10	0.44
381	0.51	0.18	0.36	0.51	0.18	0.36	0.51	0.18	0.36

Table 27: Fortran implementation of the Cobb-Douglas GNR(2020) method to estimate production functions. Comparison of the error standard deviation and $\mathcal{E}$ between the Stata code and replication data from GNR(2020), and the working data of this paper and my own Fortran implementation code.

CD GNR						
	Replication Data		Working Data			
	Stata		Stata		Fortran	
	$\mathcal{E}$	err sd	$\mathcal{E}$	err sd	$\mathcal{E}$	err sd
311	1.11	0.33	1.11	0.33	1.11	0.33
321	1.06	0.30	1.06	0.30	1.06	0.30
322	1.16	0.43	1.16	0.43	1.16	0.43
331	1.10	0.39	1.10	0.39	1.10	0.39
381	1.05	0.29	1.05	0.29	1.05	0.29

Table 28: First State of Cobb-Douglas production function intermediate elasticities estimates using GNR Stata estimator and R estimator with trimmed data at 0.05. **All firms.**

Industry	Inter.	Working Data					
		Stata			R		
		m	$\mathcal{E}$	err sd	m	$\mathcal{E}$	err sd
311	Intermediates	0.65	1.07	0.31	0.65	1.07	0.31
	Materials	0.56	1.08	0.35	0.56	1.08	0.35
	Deductibles	0.56	1.12	0.39	0.56	1.12	0.39
321	Intermediates	0.52	1.06	0.30	0.52	1.06	0.3
	Materials	0.39	1.12	0.41	0.39	1.12	0.41
	Deductibles	0.43	1.09	0.37	0.43	1.09	0.37
322	Intermediates	0.45	1.13	0.41	0.45	1.13	0.41
	Materials	0.37	1.15	0.45	0.37	1.15	0.45
	Deductibles	0.37	1.15	0.45	0.37	1.15	0.45
331	Intermediates	0.47	1.09	0.38	0.47	1.09	0.38
	Materials	0.35	1.14	0.48	0.35	1.14	0.48
	Deductibles	0.39	1.12	0.43	0.39	1.12	0.43
381	Intermediates	0.51	1.05	0.29	0.51	1.05	0.29
	Materials	0.37	1.09	0.39	0.37	1.09	0.39
	Deductibles	0.41	1.07	0.35	0.41	1.07	0.35

13 Fortran GNR

13.1 Non-Parametric Production Function

13.2 Cobb-Douglas Production Function

13.3 CD GNR + trimming

13.4 CD GNR + Trimming + Corporations

13.5 CD GNR + Trimming + Measurement Error

13.6 Industry Characteristics

14 CD GNR Intermediates

Observations with value of zero for some intermediates were driving down the estimates of the output elasticity of intermediates. This is a common problem when using data from surveys. To avoid this problem, I trimmed the observations with a share of intermediates below 0.05.

These observations increase the variance of the error term, which in turn increases the value of $\mathcal{E}$. The higher the value of $\mathcal{E}$, the lower the value of the elasticity of intermediates.

Figure 25 show that at when trimming observations with a share of intermediates below 0.05, the estimates of $\mathcal{E}$ and the output elasticity of intermediates start to stabilize. That is the changes in the values of these variables are small. In addition, Figure 26 shows that at this threshold, the change in percentage of observations dropped is also small.

A more detailed picture of this is displayed in `?@tbl-cd-gnr-inter-trim`. The table com-

Table 29: First State of Cobb-Douglas production function intermediate elasticities estimates using GNR Stata estimator and R estimator with trimmed data at 0.05. **Only Corporations.**

Industry	Inter.	Working Data					
		Stata.			R		
		m	$\mathcal{E}$	err sd	m	$\mathcal{E}$	err sd
311	Intermediates	0.67	1.06	0.28	0.67	1.06	0.28
	Materials	0.55	1.08	0.35	0.55	1.08	0.35
	Deductibles	0.57	1.09	0.35	0.57	1.09	0.35
321	Intermediates	0.51	1.03	0.24	0.51	1.03	0.24
	Materials	0.35	1.09	0.37	0.35	1.09	0.37
	Deductibles	0.40	1.07	0.33	0.4	1.07	0.33
322	Intermediates	0.25	1.59	0.82	0.25	1.59	0.82
	Materials	0.31	1.11	0.41	0.31	1.11	0.41
	Deductibles	0.34	1.09	0.38	0.34	1.09	0.38
331	Intermediates	0.50	1.03	0.24	0.5	1.03	0.24
	Materials	0.29	1.09	0.42	0.29	1.09	0.42
	Deductibles	0.37	1.05	0.30	0.37	1.05	0.3
381	Intermediates	0.54	1.04	0.27	0.54	1.04	0.27
	Materials	0.37	1.09	0.40	0.37	1.09	0.4
	Deductibles	0.41	1.08	0.37	0.41	1.08	0.37

Table 30: First State of Cobb-Douglas production function intermediate elasticities estimates using GNR Stata estimator and R estimator with trimmed data at 0.05. **Corporations vs. All.** ε is the **measurement error**.

Industry	Inter.	Working Data (R)					
		Corps.			All		
		m	$\mathcal{E}$	err sd	m	$\mathcal{E}$	err sd
311	Intermediates	0.71	1	0.28	0.69	1	0.31
	Materials	0.6	1	0.35	0.61	1	0.35
	Deductibles	0.62	1	0.35	0.63	1	0.39
321	Intermediates	0.53	1	0.24	0.55	1	0.3
	Materials	0.39	1	0.37	0.43	1	0.41
	Deductibles	0.43	1	0.33	0.47	1	0.37
322	Intermediates	0.4	1	0.82	0.51	1	0.41
	Materials	0.35	1	0.41	0.42	1	0.45
	Deductibles	0.37	1	0.38	0.43	1	0.45
331	Intermediates	0.51	1	0.24	0.51	1	0.38
	Materials	0.32	1	0.42	0.4	1	0.48
	Deductibles	0.39	1	0.3	0.44	1	0.43
381	Intermediates	0.56	1	0.27	0.53	1	0.29
	Materials	0.41	1	0.4	0.41	1	0.39
	Deductibles	0.44	1	0.37	0.44	1	0.35

Table 31: First State of Cobb-Douglas production function intermediate elasticities estimates using GNR Stata estimator and R estimator with trimmed data at 0.05. **Corporations vs. All.** ε is the **measurement error**. Top evading industries.

Industry	Inter.	Working Data (R)					
		Corps.			All		
		m	$\mathcal{E}$	err sd	m	$\mathcal{E}$	err sd
322	Intermediates	0.4	1	0.82	0.51	1	0.41
	Materials	0.35	1	0.41	0.42	1	0.45
	Deductibles	0.37	1	0.38	0.43	1	0.45
369	Intermediates	0.41	1	0.41	0.37	1	0.54
	Materials	0.24	1	0.61	0.28	1	0.62
	Deductibles	0.35	1	0.41	0.34	1	0.46
331	Intermediates	0.51	1	0.24	0.51	1	0.38
	Materials	0.32	1	0.42	0.4	1	0.48
	Deductibles	0.39	1	0.3	0.44	1	0.43
321	Intermediates	0.53	1	0.24	0.55	1	0.3
	Materials	0.39	1	0.37	0.43	1	0.41
	Deductibles	0.43	1	0.33	0.47	1	0.37
351	Intermediates	0.55	1	0.3	0.55	1	0.32
	Materials	0.34	1	0.63	0.35	1	0.62
	Deductibles	0.44	1	0.4	0.44	1	0.41

Table 32: Intermediate elasticities for different groups of firms using GNR method assuming a CD functional form. Data trimmed at 0.05 of the intermediates share cost of gross output. ε is defined as **measurement error**.

Inds.	Intermediate	m			
		Corps.	Exp.	Imp.	All
311	Materials	0.6	0.61	0.35	0.61
312	Materials	0.58	0.57	0.49	0.58
313	Materials	0.29	0.35	0.36	0.31
321	Materials	0.39	0.44	0.37	0.43
322	Materials	0.35	0.42	0.41	0.42
323	Materials	0.56	0.47	0.22	0.48
324	Materials	0.43	0.46	0.22	0.46
331	Materials	0.32	0.41	0.42	0.4
332	Materials	0.36	0.35	0.42	0.35

Table 33: One side test of tax evasion through inputs overreporting. Different non-evaders groups used as reference.

Inds.	Corporations		Exporters		Materials
	Materials	Deductibles	Materials	Deductibles	
311	0.1111 (0.0045)***	0.0294 (0.005)***	-0.1489 (0.0124)	-0.0416 (0.0056)	-0.2007 (0.009)
312	-0.0098 (0.0158)	0.0251 (0.0115)**	0.0516 (0.0257)**	0.0648 (0.0127)***	-0.1104 (0.009)
313	0.0659 (0.0123)***	0.0377 (0.0111)***	0.2794 (0.0123)***	0.3439 (0.0111)***	0.1218 (0.011)
321	0.1262 (0.0078)***	0.0885 (0.007)***	0.0604 (0.0111)***	0.115 (0.0079)***	0.016 (0.011)
322	1.27 (0.0068)***	0.6404 (0.0068)***	0.1215 (0.0145)***	0.1103 (0.009)***	-0.1187 (0.009)
323	-0.1534 (0.0133)	-0.1635 (0.0137)	-0.181 (0.0179)	-0.1646 (0.0152)	-0.1544 (0.009)
324	0.0664 (0.0086)***	0.0578 (0.0084)***	-0.1202 (0.0188)	-0.098 (0.0099)	-0.0588 (0.009)
331	0.2177 (0.0154)***	0.1205 (0.0139)***	0.086 (0.0198)***	0.1038 (0.0148)***	-0.0852 (0.009)
332	-0.028 (0.0136)	-0.0284 (0.0127)	-0.1286 (0.0142)	-0.1522 (0.0133)	-0.2887 (0.009)

compares the estimates of the output elasticity of intermediates and other statistics of the first stage of GNR(2020) assuming a Cobb-Douglas Production Function for different definitions of intermediates and different thresholds for trimming observations.

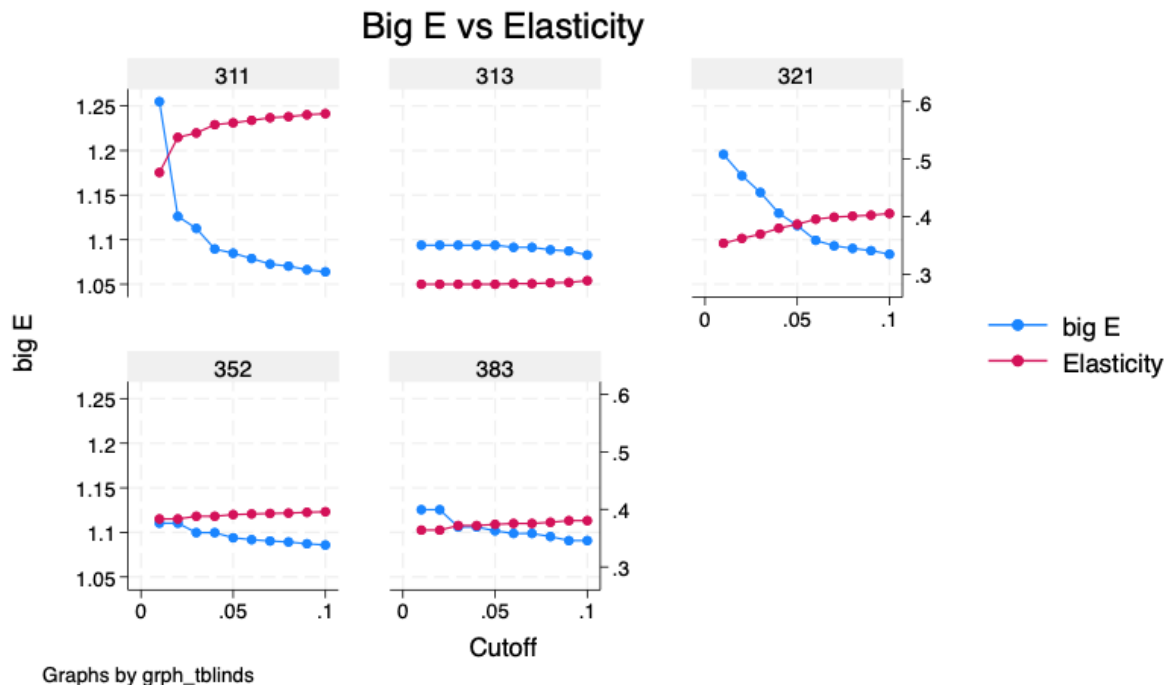


Figure 25: Estimates of $\mathcal{E}$ and the output elasticity of intermediates (raw materials) trimming observations with a share of intermediates below different thresholds going from 0.01 to 0.1 . The elasticities of intermediates were estimated assuming a Cobb-Douglas production function using the first stage of GNR(2020).

15 Density

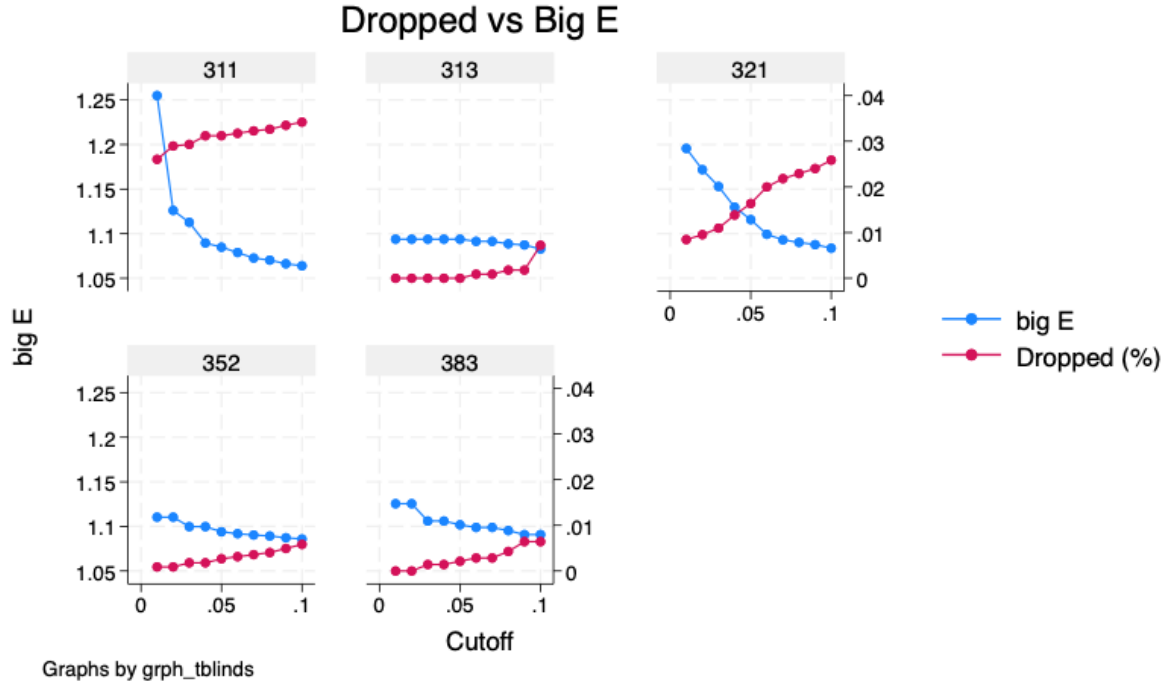


Figure 26: Estimates of $\mathcal{E}$ and percentage of observations dropped by trimming observations with a share of intermediates (raw materials) below different thresholds going from 0.01 to 0.1 . The elasticities of intermediates were estimated assuming a Cobb-Douglas production function using the first stage of GNR(2020).

Table 34: Estimates of Cobb-Douglas Production Function parameters using GNR(2020) and statistics of the first stage for different definitions of intermediates. *m s e* stands for raw materials, services, and energy, *mats* for raw materials, and *ded* for deductible intermediates (raw materials, electricity, fuels, and repair and maintenance services). Observations were trimmed below a share of intermediates of 0.05.

Ind.	Inter.	m	l	k	bigE	si_mean	err_mean	err_sd
311	m_s_e	0.6488	0.1827	0.1832	1.0652	0.7219	0	0.3091
	mats	0.5630	0.2084	0.2443	1.0848	0.6416	0	0.3470
	ded	0.5567	0.2194	0.2337	1.1232	0.6631	0	0.3948
321	m_s_e	0.5183	0.2658	0.2088	1.0557	0.5689	0	0.2996
	mats	0.3869	0.3412	0.2604	1.1158	0.4633	0	0.4136
	ded	0.4273	0.3237	0.2470	1.0934	0.4949	0	0.3720
352	m_s_e	0.5749	0.2133	0.2415	1.0437	0.6201	0	0.2718
	mats	0.3909	0.2858	0.3527	1.0940	0.4574	0	0.3922
	ded	0.4114	0.2797	0.3360	1.0877	0.4756	0	0.3746
313	m_s_e	0.4324	0.1658	0.3086	1.0721	0.4893	0	0.3479
	mats	0.2826	0.2943	0.3395	1.0938	0.3347	0	0.4104
	ded	0.3216	0.2912	0.3203	1.0767	0.3699	0	0.3726
383	m_s_e	0.5261	0.3666	0.1892	1.0445	0.5694	0	0.2786
	mats	0.3739	0.4889	0.2523	1.1018	0.4427	0	0.4069
	ded	0.4015	0.4532	0.2427	1.0891	0.4655	0	0.3798

Industry 331

